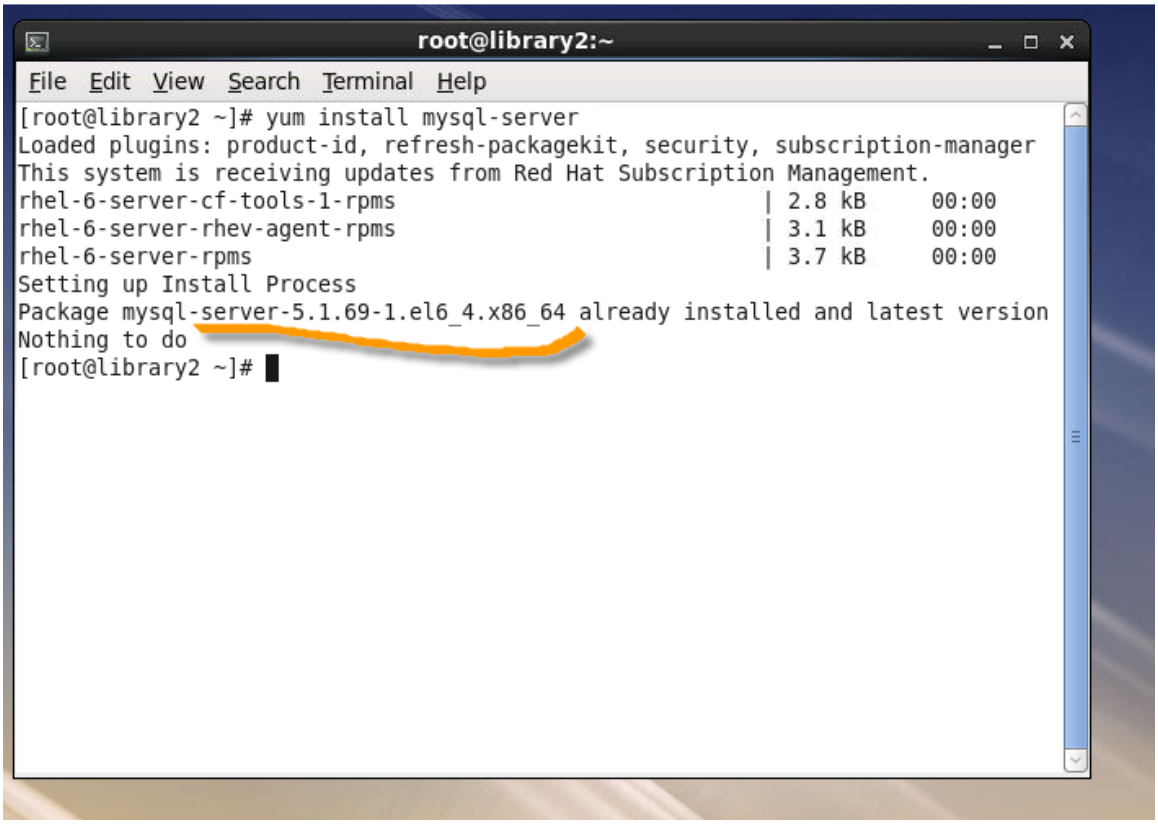
VuFind Installation on RedHat Linux 6.4 step by step

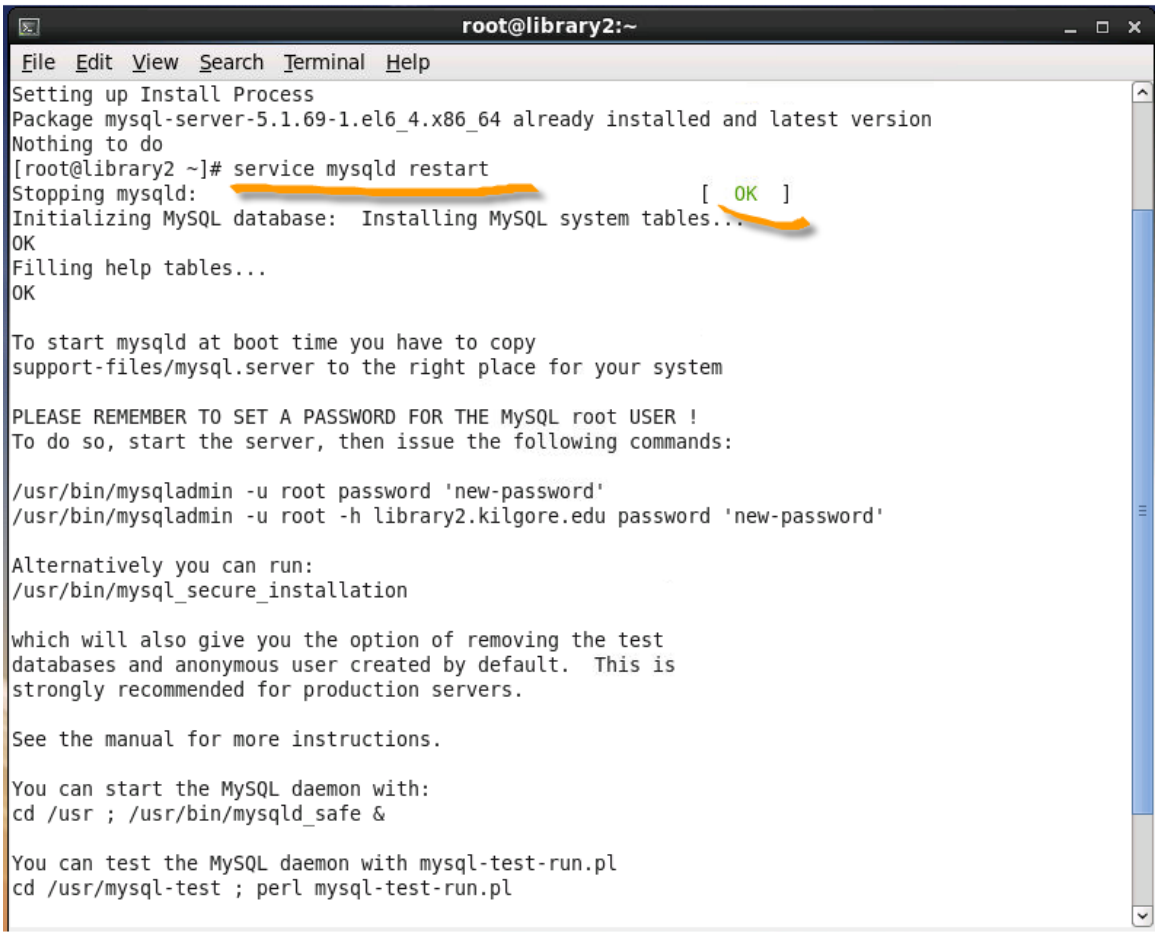
Andy Tang

Kilgore College

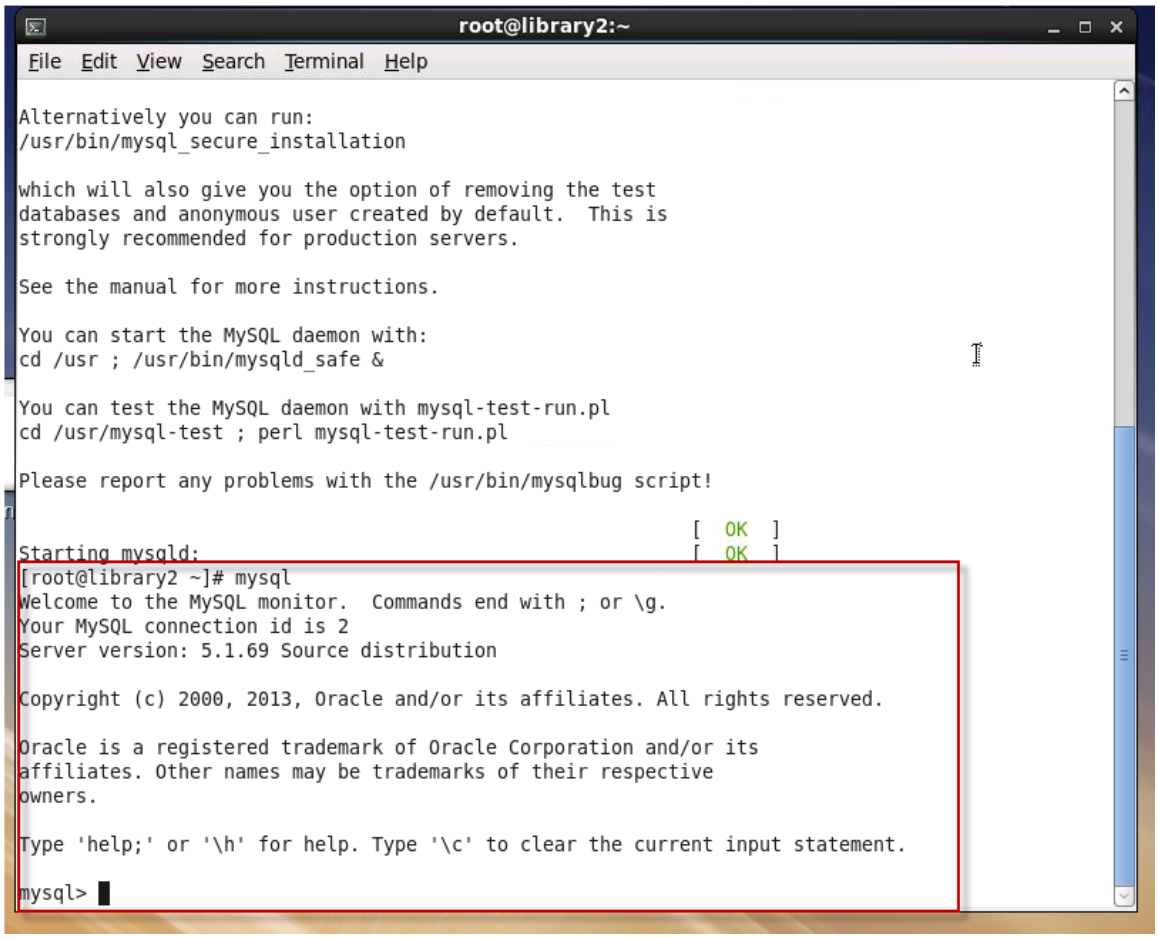
Install MySQL server



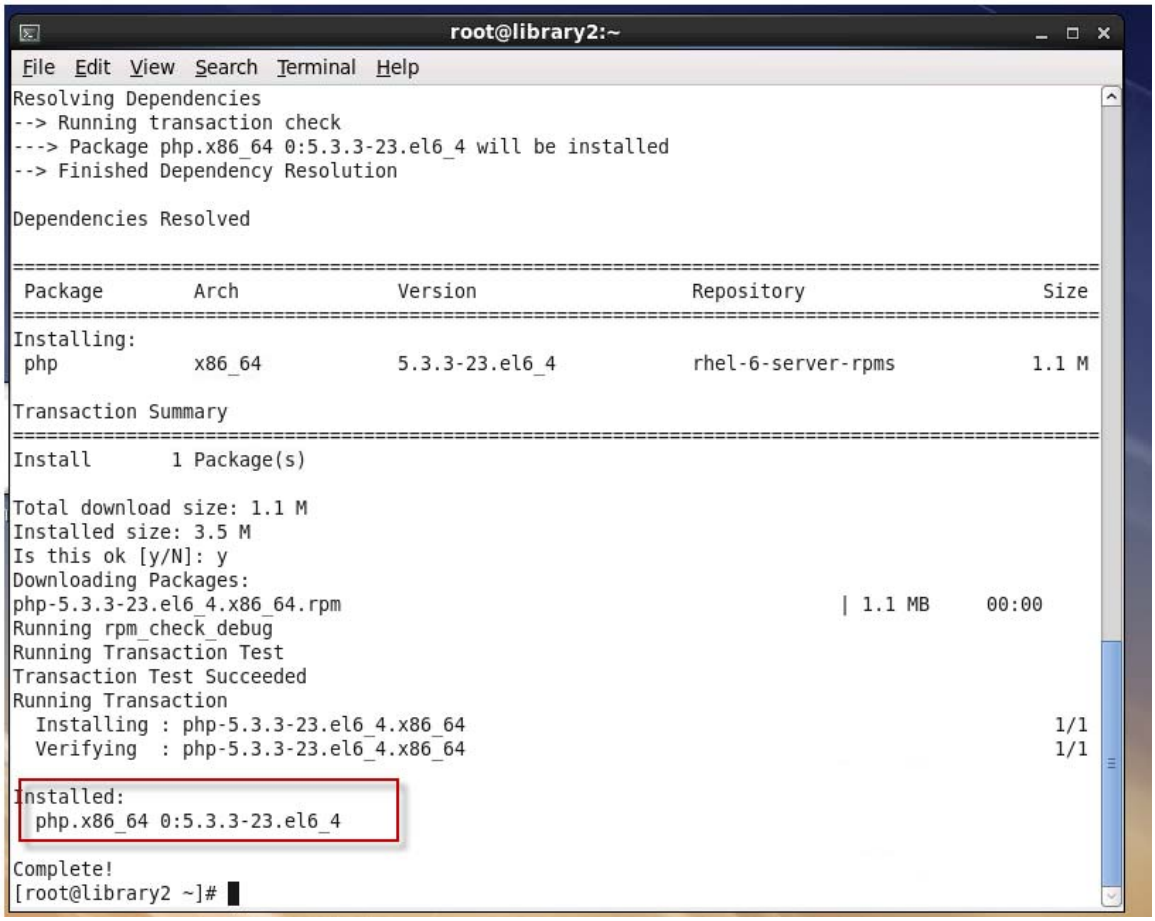
Restart MySQL server



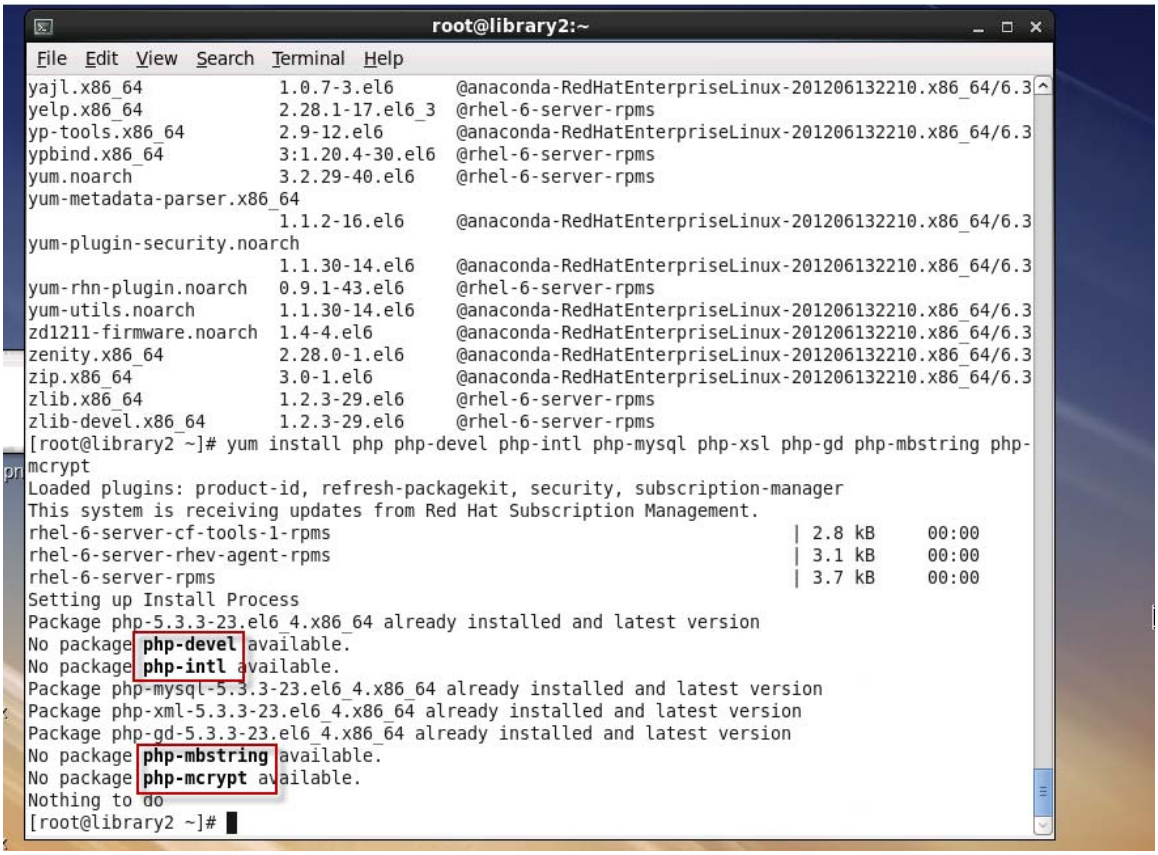
Check whether MySQL is installed successfully.



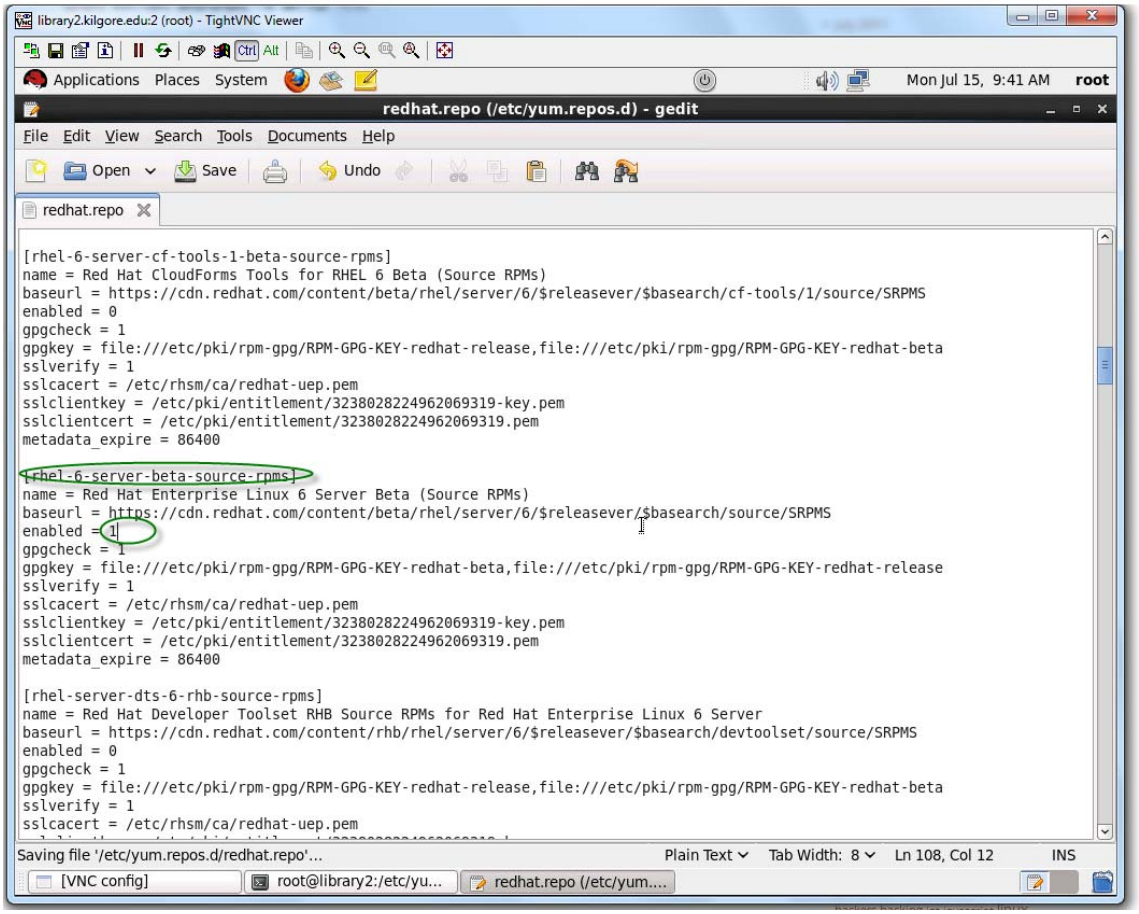
Install PHP


root@library2:~
File Edit View Search Terminal Help
Resolving Dependencies
--> Running transaction check
---> Package php.x86_64 0:5.3.3-23.el6_4 will be installed
--> Finished Dependency Resolution
Dependencies Resolved
Package Arch Version Repository Size
Installing:
php x86_64 5.3.3-23.el6_4 rhel-6-server-rpms 1.1 M
Transaction Summary
Install 1 Package(s)
Total download size: 1.1 M
Installed size: 3.5 M
Is this ok [y/N]: y
Downloading Packages:
php-5.3.3-23.el6_4.x86_64.rpm | 1.1 MB 00:00
Running rpm_check_debug
Running Transaction Test
Transaction Test Succeeded
Running Transaction
Installing : php-5.3.3-23.el6_4.x86_64 1/1
Verifying : php-5.3.3-23.el6_4.x86_64 1/1
Installed:
php.x86_64 0:5.3.3-23.el6_4
Complete!
[root@library2 ~]#

Find 4 PHP packages need to be installed.



Enable one of RedHat repositories.



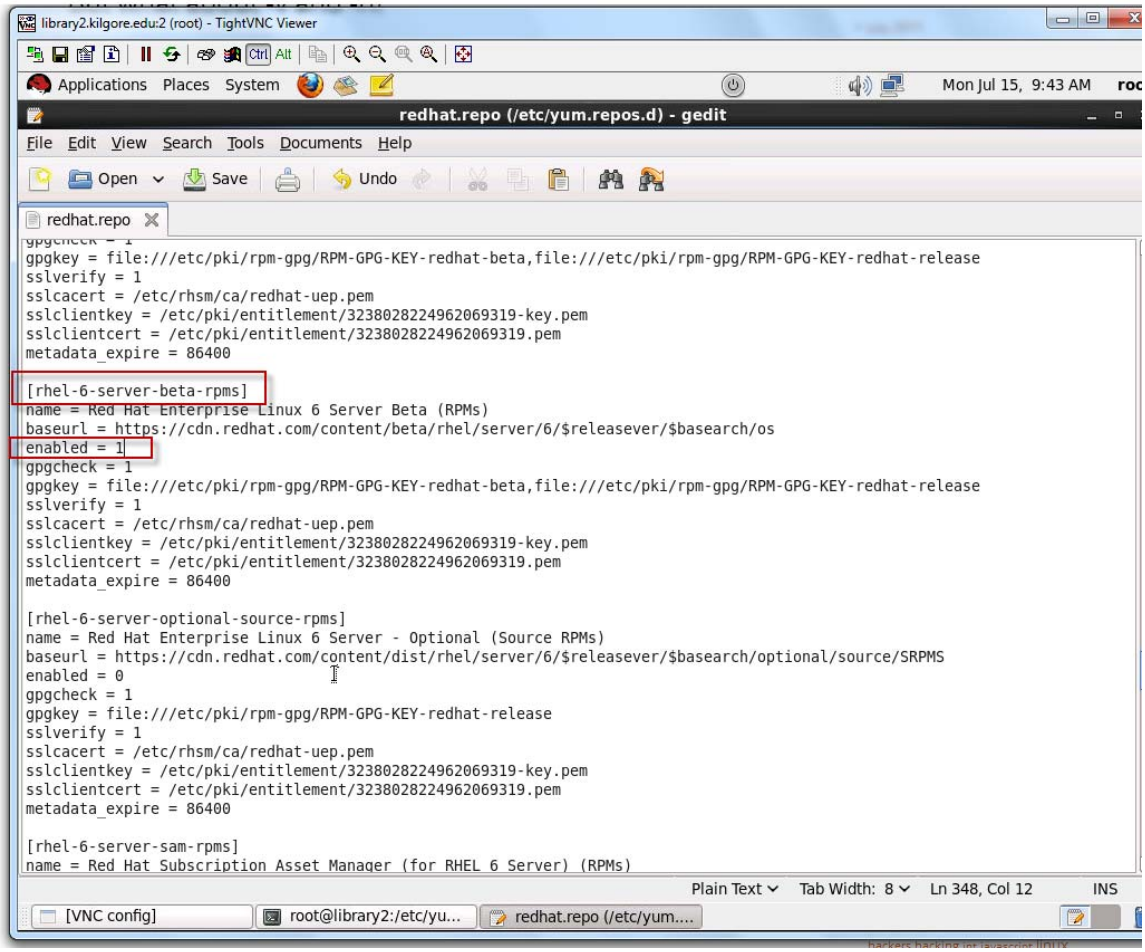
Enable one of RedHat repositories.



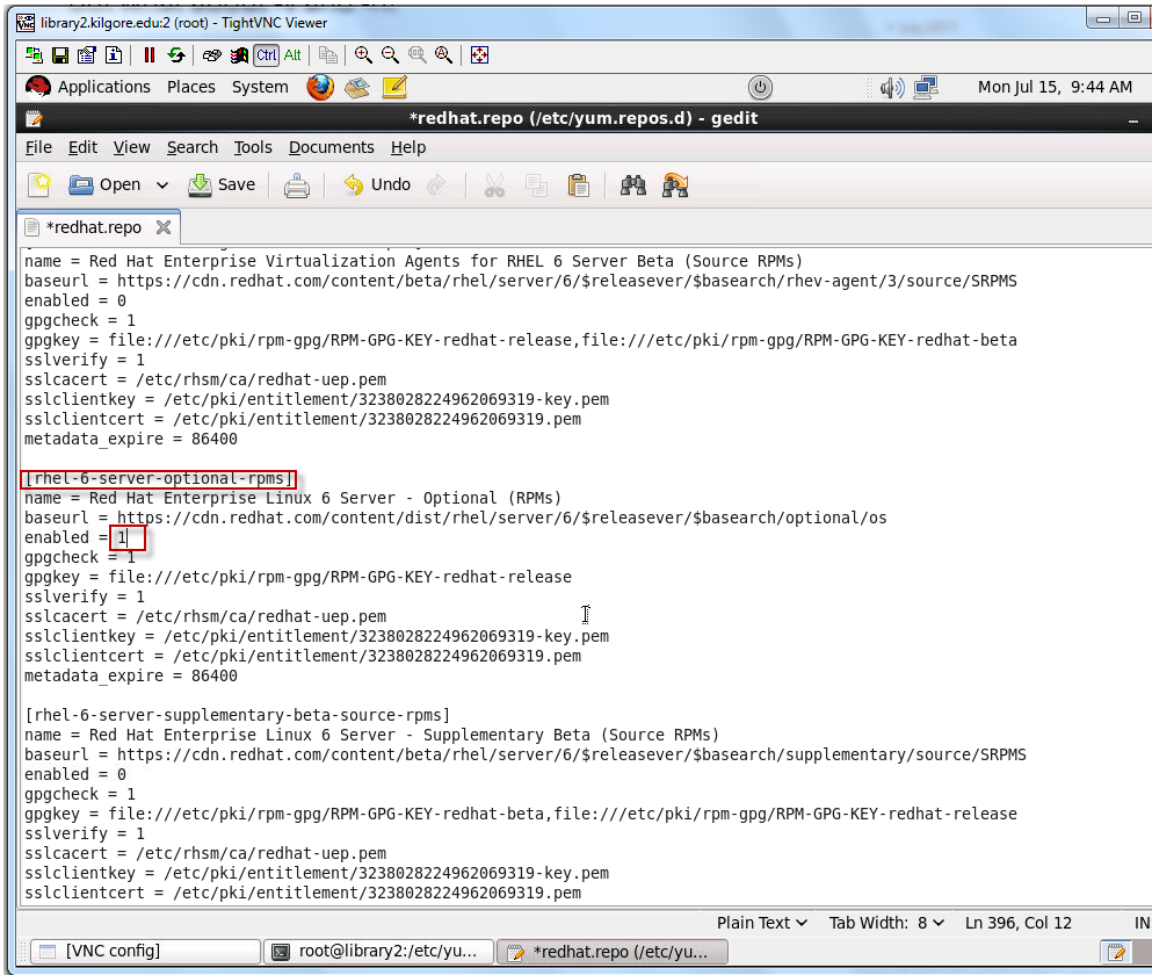
Enable one of RedHat repositories.



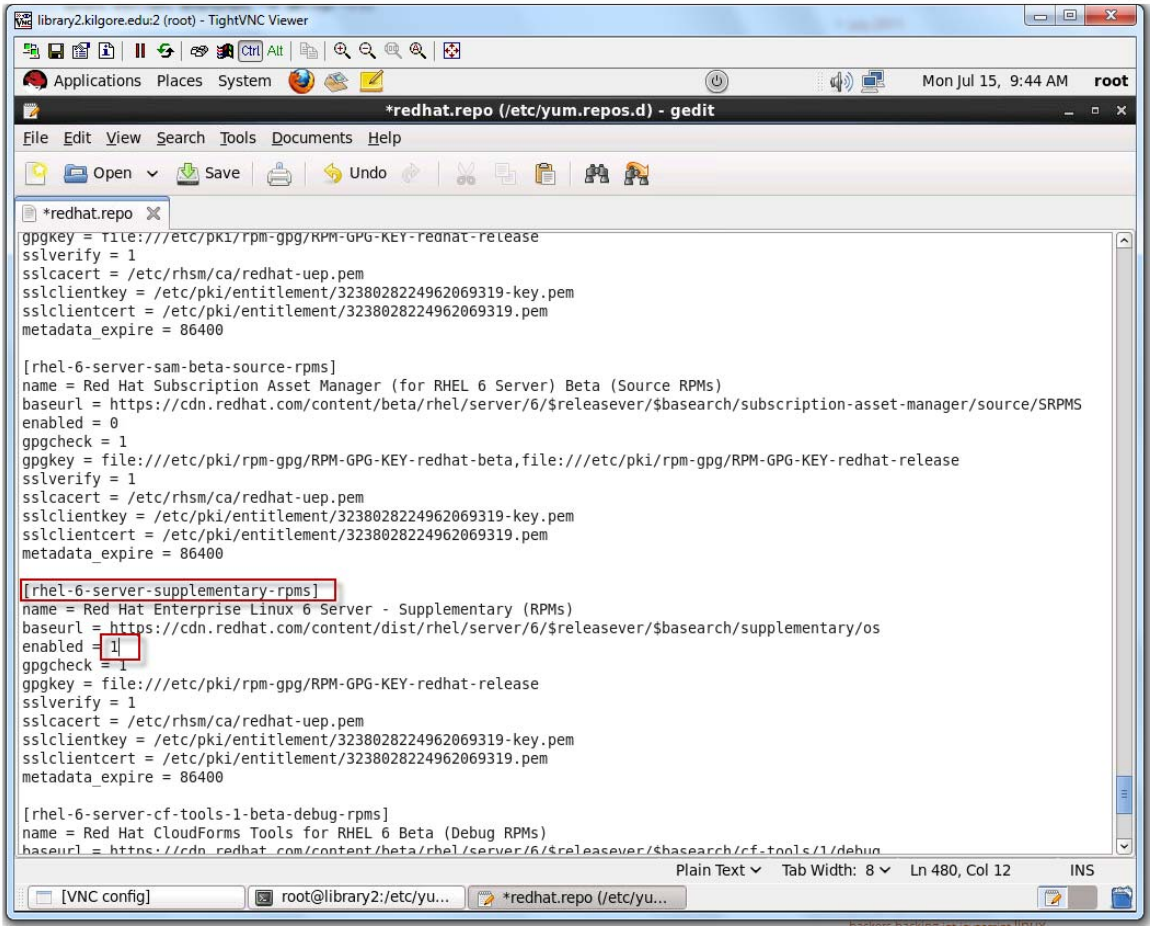
Enable one of RedHat repositories.



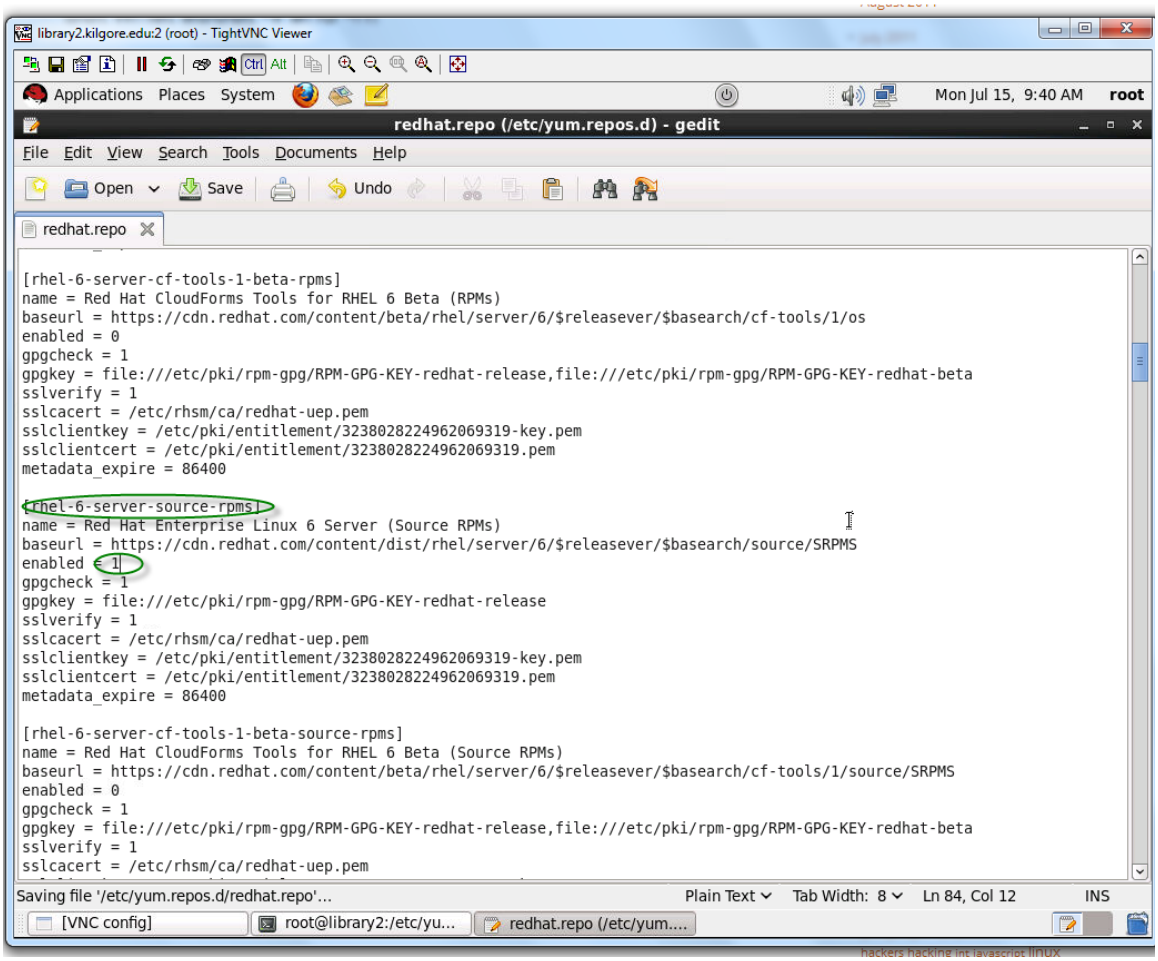
Enable one of RedHat repositories.



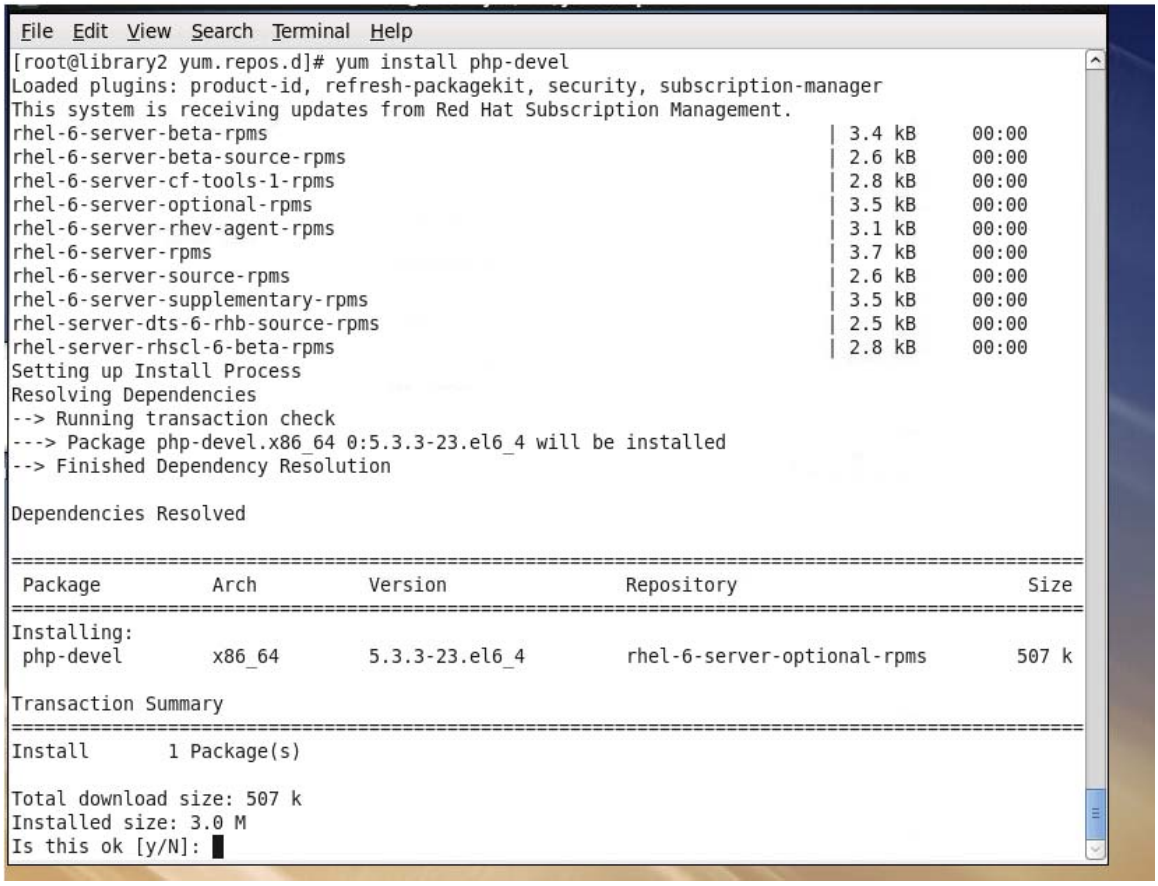
Enable one of RedHat repositories.



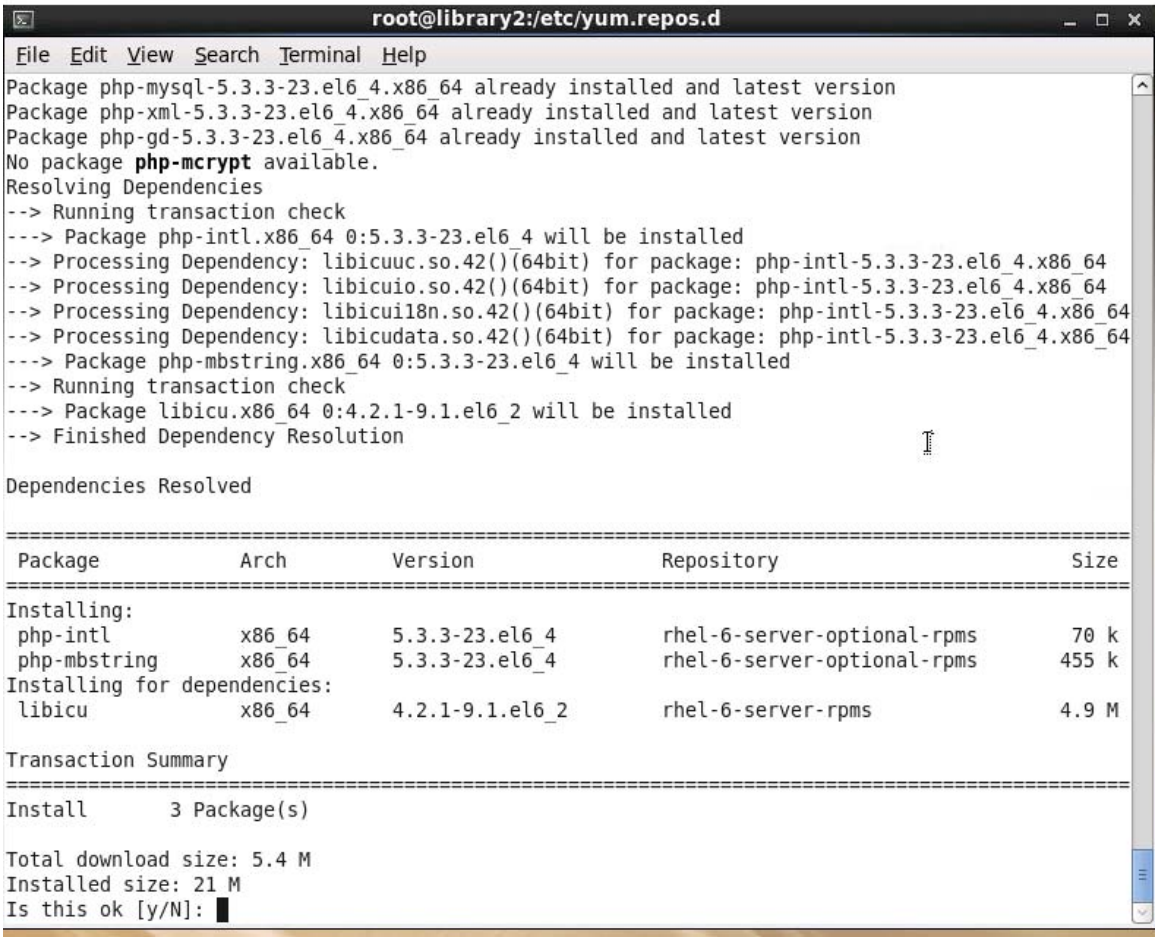
Enable several RedHat repositories and make some preparation to install PHP packages.



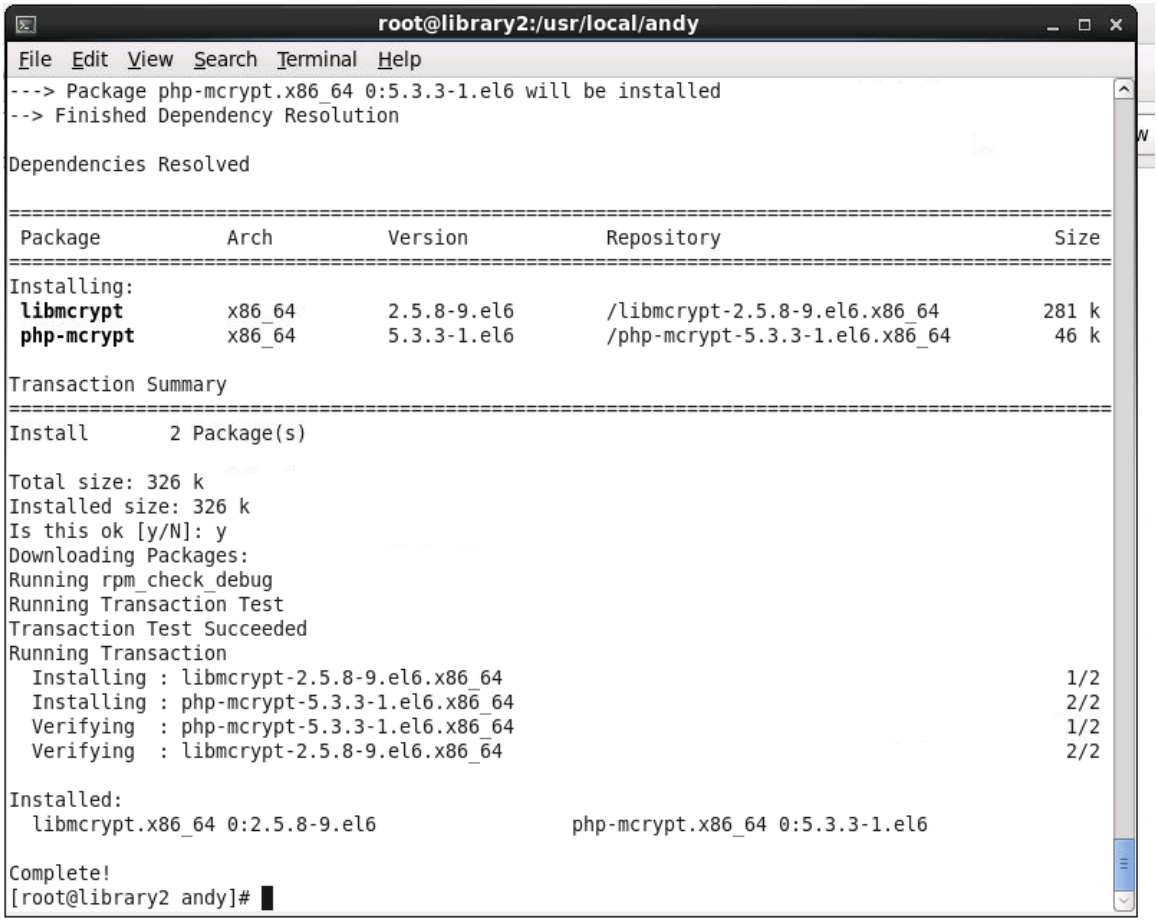
The package php-devel is installed.



The packages php-intl, php-mbstring, are installed. However, php-mcrypt is not available.



The packages libmcrypt and php-mcrypt are installed.



Test whether all important php packes are installed.

```
[root@library2 andy]# yum install php php-devel php-intl php-mysql php-xsl php-gd php-mbstring p
hp-mcrypt
Loaded plugins: product-id, refresh-packagekit, security, subscription-manager
This system is receiving updates from Red Hat Subscription Management.
rhel-6-server-beta-rpms                                                  | 3.4 kB     00:00
rhel-6-server-beta-source-rpms                                           | 2.6 kB     00:00
rhel-6-server-cf-tools-1-rpms                                            | 2.8 kB     00:00
rhel-6-server-optional-rpms                                              | 3.5 kB     00:00
rhel-6-server-rhev-agent-rpms                                            | 3.1 kB     00:00
rhel-6-server-rpms                                                       | 3.7 kB     00:00
rhel-6-server-source-rpms                                                | 2.6 kB     00:00
rhel-6-server-supplementary-rpms                                         | 3.5 kB     00:00
rhel-server-dts-6-rhb-source-rpms                                        | 2.5 kB     00:00
rhel-server-rhscl-6-beta-rpms                                            | 2.8 kB     00:00
Setting up Install Process
Package php-5.3.3-23.el6_4.x86_64 already installed and latest version
Package php-devel-5.3.3-23.el6_4.x86_64 already installed and latest version
Package php-intl-5.3.3-23.el6_4.x86_64 already installed and latest version
Package php-mysql-5.3.3-23.el6_4.x86_64 already installed and latest version
Package php-xml-5.3.3-23.el6_4.x86_64 already installed and latest version
Package php-gd-5.3.3-23.el6_4.x86_64 already installed and latest version
Package php-mbstring-5.3.3-23.el6_4.x86_64 already installed and latest version
Nothing to do
[root@library2 andy]#
```

Upgrade pear.

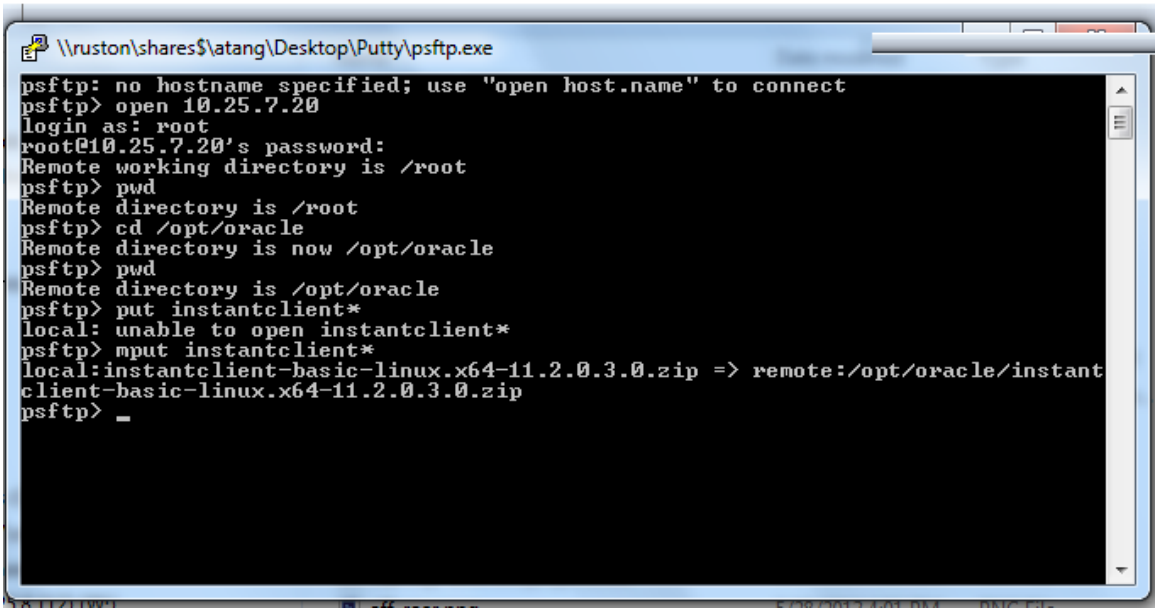
```
[root@library2 andy]# pear upgrade pear
Nothing to upgrade
[root@library2 andy]#
```

2. Extract the Oracle Instant Client

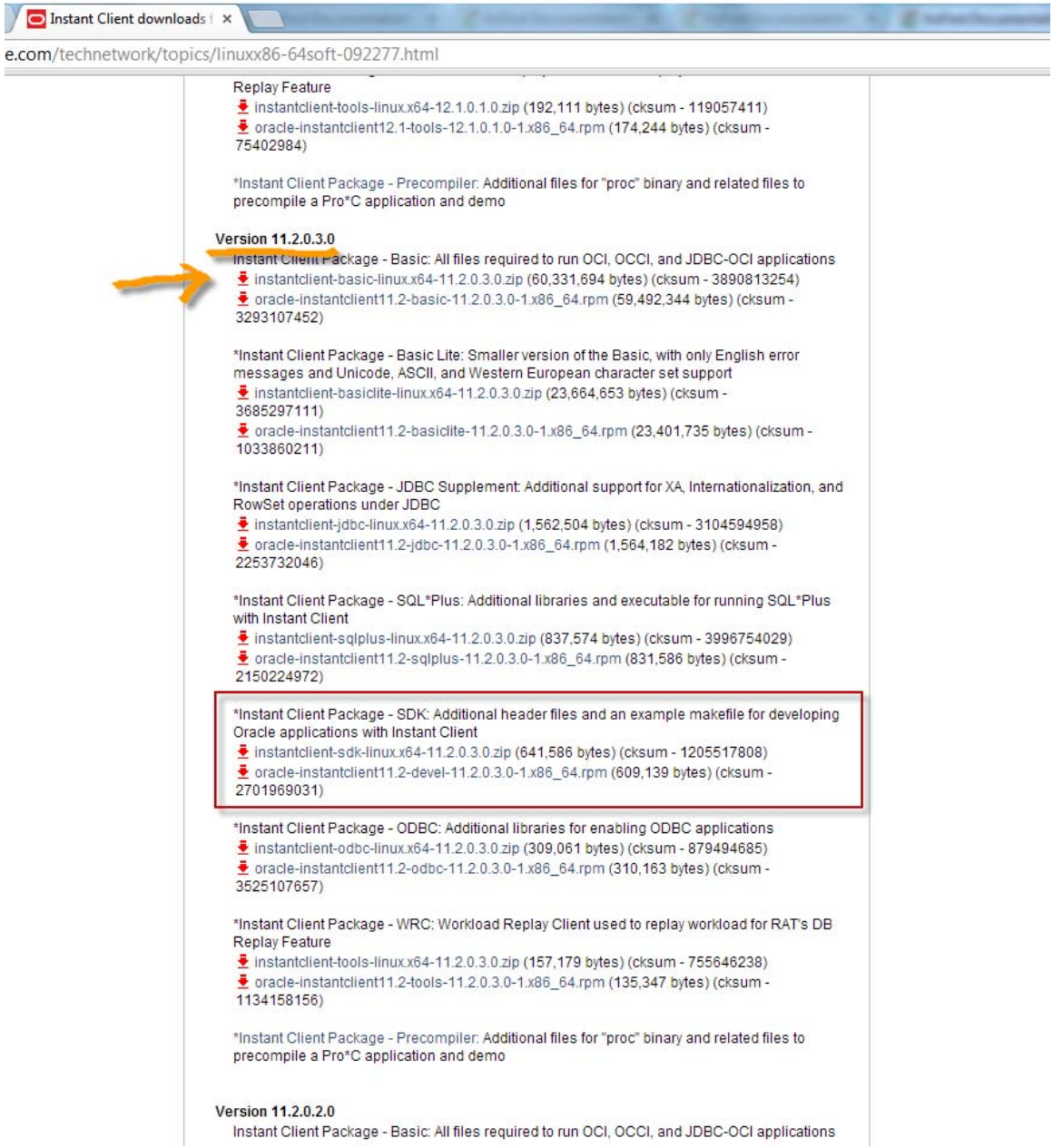
Create a directory to store the Oracle code and then switch to it:

```
mkdir -p /opt/oracle
cd /opt/oracle
```

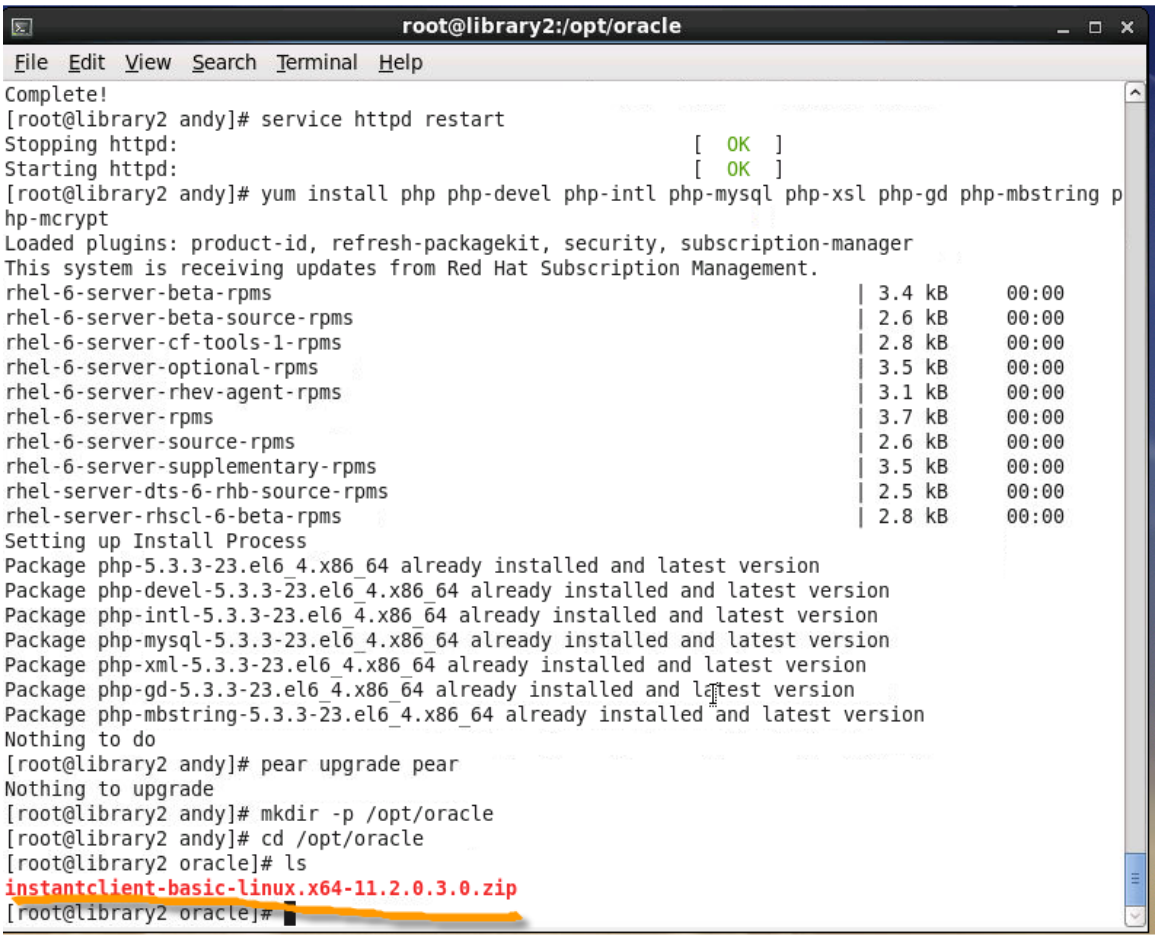
Upload zip file to Linux server.

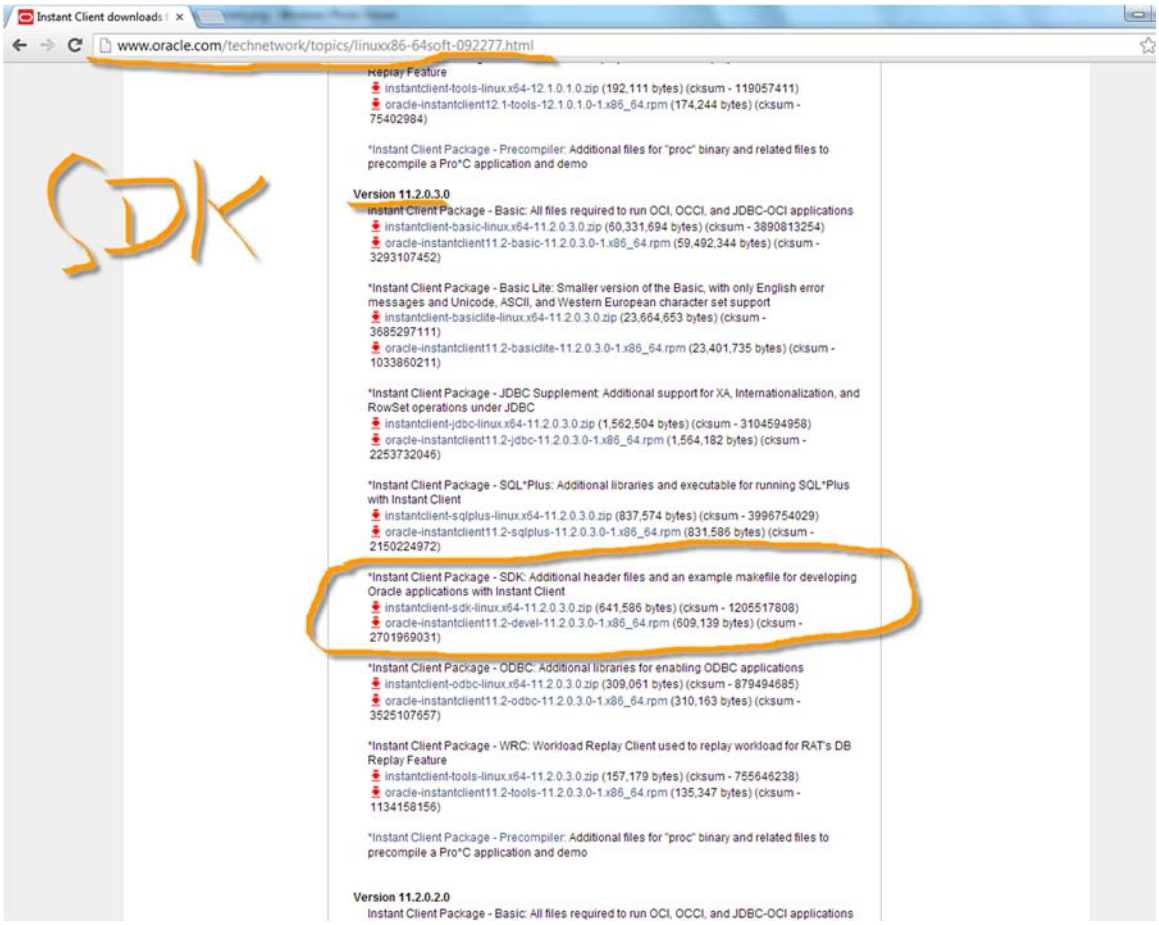


Download the newest Oracle client.

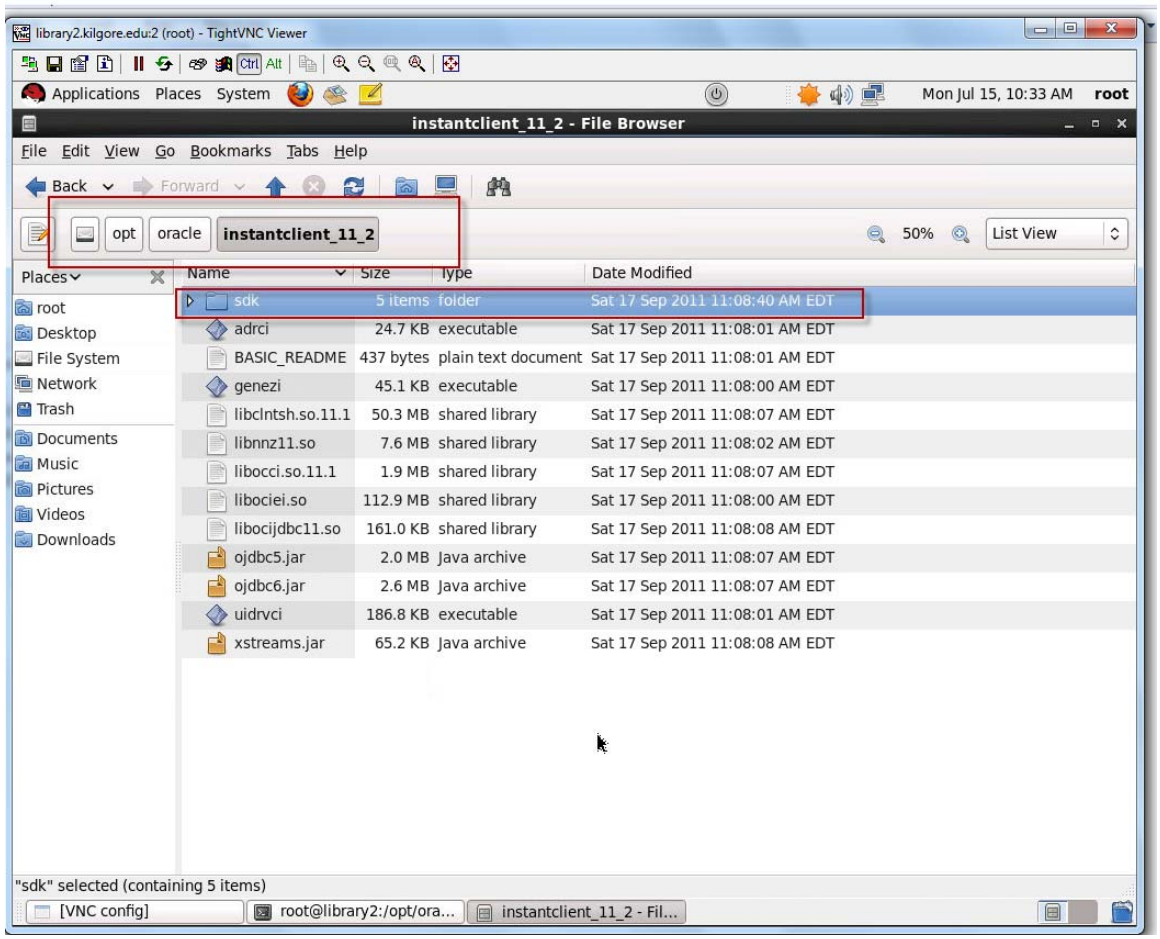


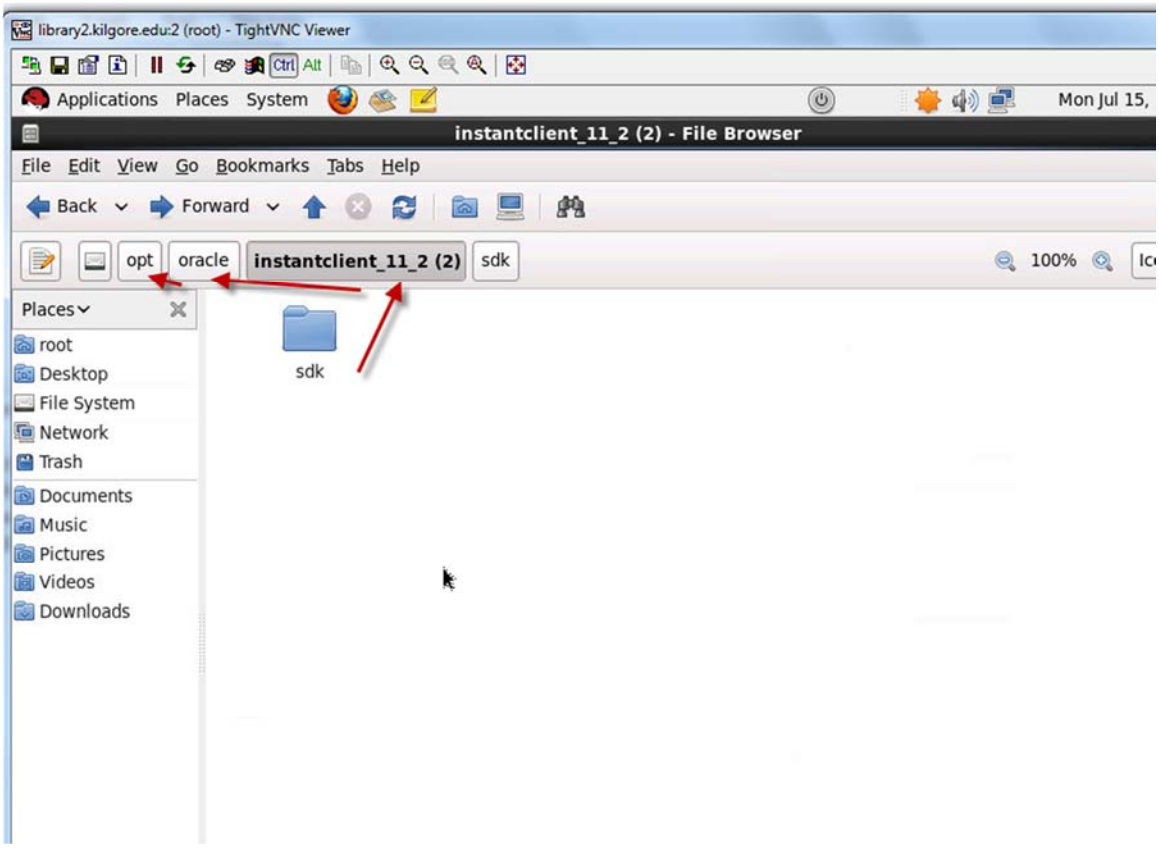
List the file and test the version of file.



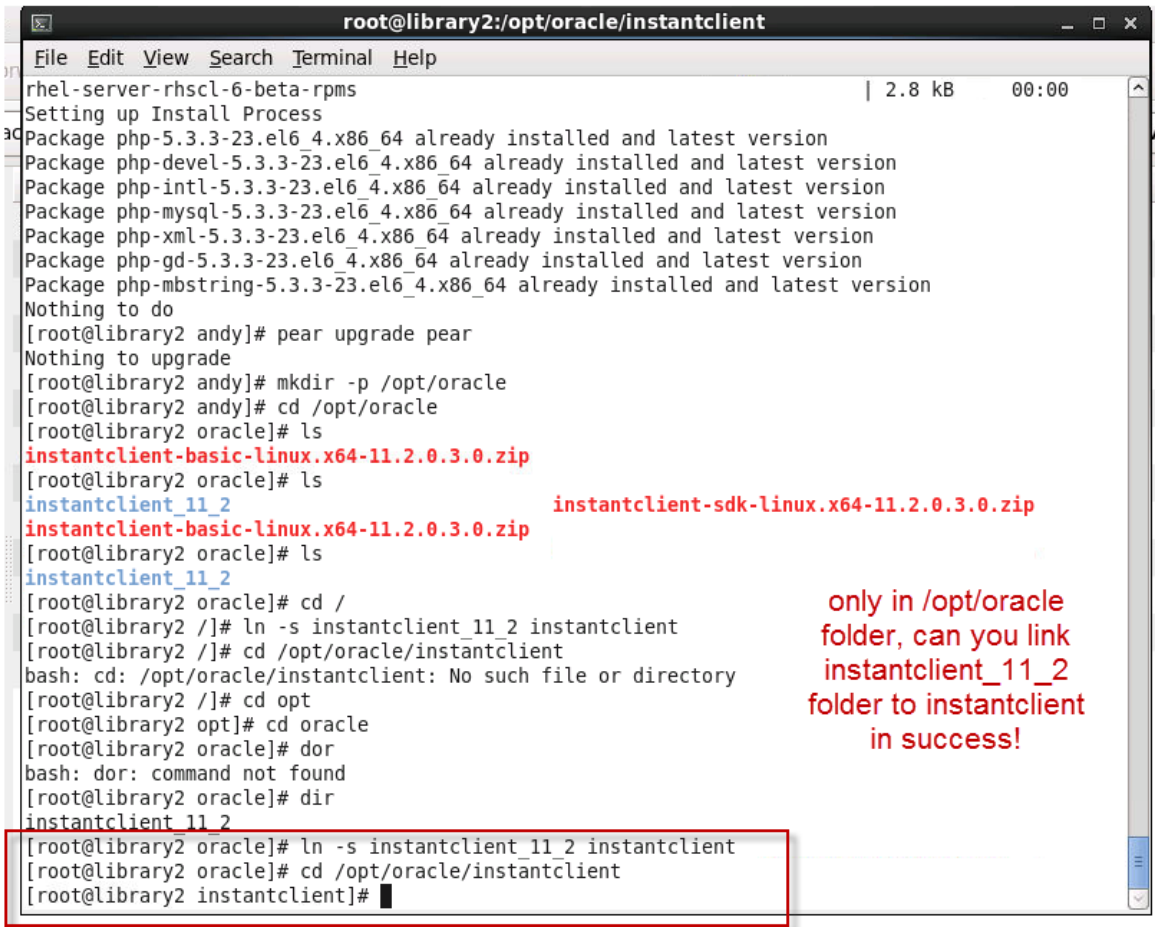

Instant Client downloads
www.oracle.com/technetwork/topics/linuxx86-64soft-092277.html
SDK
Replay Feature
instantclient-tools-linux.x64-12.1.0.1.0.zip (192,111 bytes) (cksum - 119057411)
oracle-instantclient12.1-tools-12.1.0.1.0-1.x86_64.rpm (174,244 bytes) (cksum - 75402984)
*Instant Client Package - Precompiler: Additional files for "proc" binary and related files to precompile a Pro*C application and demo
Version 11.2.0.3.0
Instant Client Package - Basic: All files required to run OCI, OCCI, and JDBC-OCI applications
instantclient-basic-linux.x64-11.2.0.3.0.zip (60,331,694 bytes) (cksum - 3890813254)
oracle-instantclient11.2-basic-11.2.0.3.0-1.x86_64.rpm (59,492,344 bytes) (cksum - 3293107452)
*Instant Client Package - Basic Lite: Smaller version of the Basic, with only English error messages and Unicode, ASCII, and Western European character set support
instantclient-basiclite-linux.x64-11.2.0.3.0.zip (23,664,653 bytes) (cksum - 3685297111)
oracle-instantclient11.2-basiclite-11.2.0.3.0-1.x86_64.rpm (23,401,735 bytes) (cksum - 1033860211)
*Instant Client Package - JDBC Supplement: Additional support for XA, Internationalization, and RowSet operations under JDBC
instantclient-jdbc-linux.x64-11.2.0.3.0.zip (1,562,504 bytes) (cksum - 3104594958)
oracle-instantclient11.2-jdbc-11.2.0.3.0-1.x86_64.rpm (1,564,182 bytes) (cksum - 2253732046)
*Instant Client Package - SQL*Plus: Additional libraries and executable for running SQL*Plus with Instant Client
instantclient-sqlplus-linux.x64-11.2.0.3.0.zip (837,574 bytes) (cksum - 3996754029)
oracle-instantclient11.2-sqlplus-11.2.0.3.0-1.x86_64.rpm (831,586 bytes) (cksum - 2150224972)
*Instant Client Package - SDK: Additional header files and an example makefile for developing Oracle applications with Instant Client
instantclient-sdk-linux.x64-11.2.0.3.0.zip (641,586 bytes) (cksum - 1205517808)
oracle-instantclient11.2-devel-11.2.0.3.0-1.x86_64.rpm (609,139 bytes) (cksum - 2701969031)
*Instant Client Package - ODBC: Additional libraries for enabling ODBC applications
instantclient-odbc-linux.x64-11.2.0.3.0.zip (309,061 bytes) (cksum - 879494685)
oracle-instantclient11.2-odbc-11.2.0.3.0-1.x86_64.rpm (310,163 bytes) (cksum - 3525107657)
*Instant Client Package - WRC: Workload Replay Client used to replay workload for RAT's DB Replay Feature
instantclient-tools-linux.x64-11.2.0.3.0.zip (157,179 bytes) (cksum - 755646238)
oracle-instantclient11.2-tools-11.2.0.3.0-1.x86_64.rpm (135,347 bytes) (cksum - 1134158156)
*Instant Client Package - Precompiler: Additional files for "proc" binary and related files to precompile a Pro*C application and demo
Version 11.2.0.2.0
Instant Client Package - Basic: All files required to run OCI, OCCI, and JDBC-OCI applications

Check the folder contents after unzipping the file.

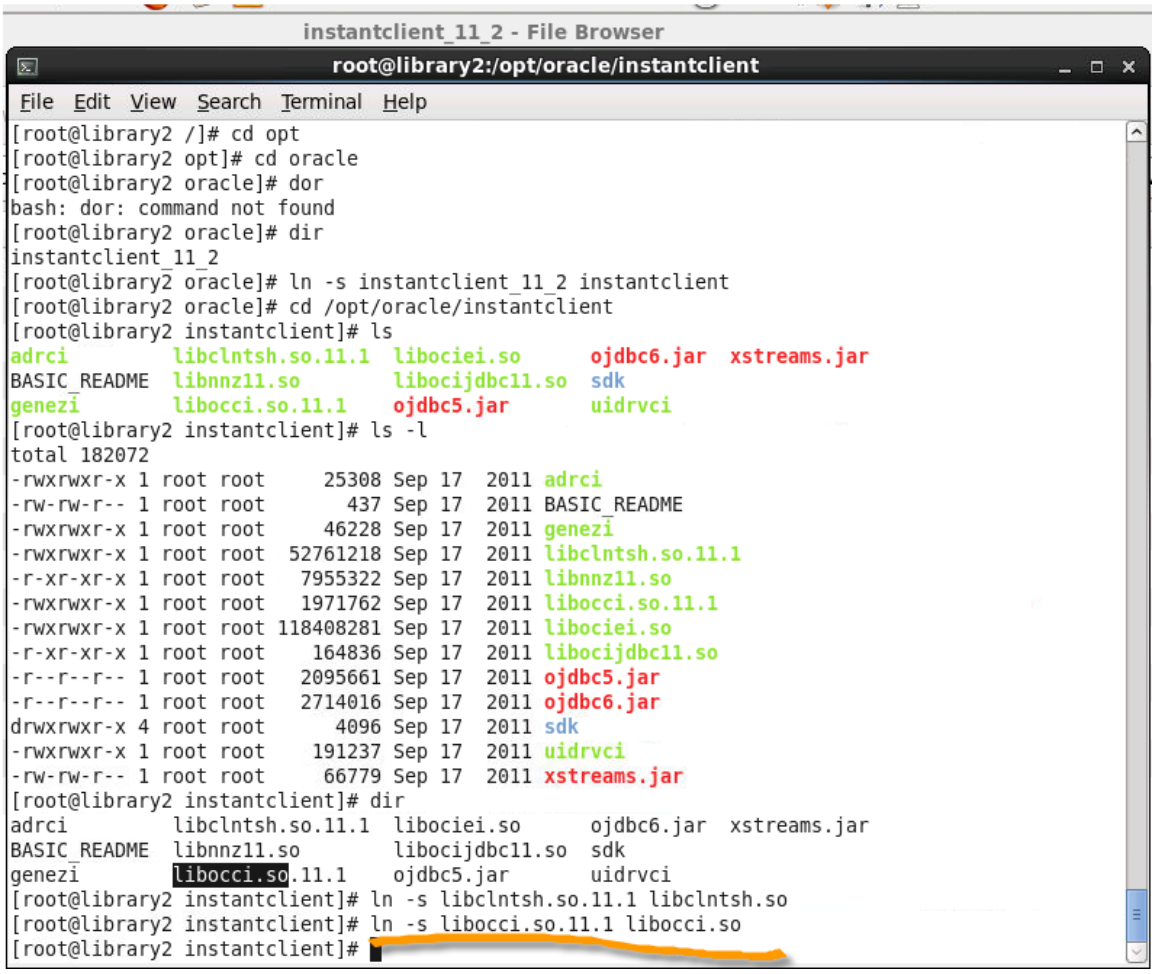



library2.kilgore.edu:2 (root) - TightVNC Viewer
Applications Places System
Mon Jul 15,
instantclient_11_2 (2) - File Browser
File Edit View Go Bookmarks Tabs Help
Back Forward
opt
oracle
instantclient_11_2 (2)
sdk
100%
Places
root
Desktop
File System
Network
Trash
Documents
Music
Pictures
Videos
Downloads
sdk

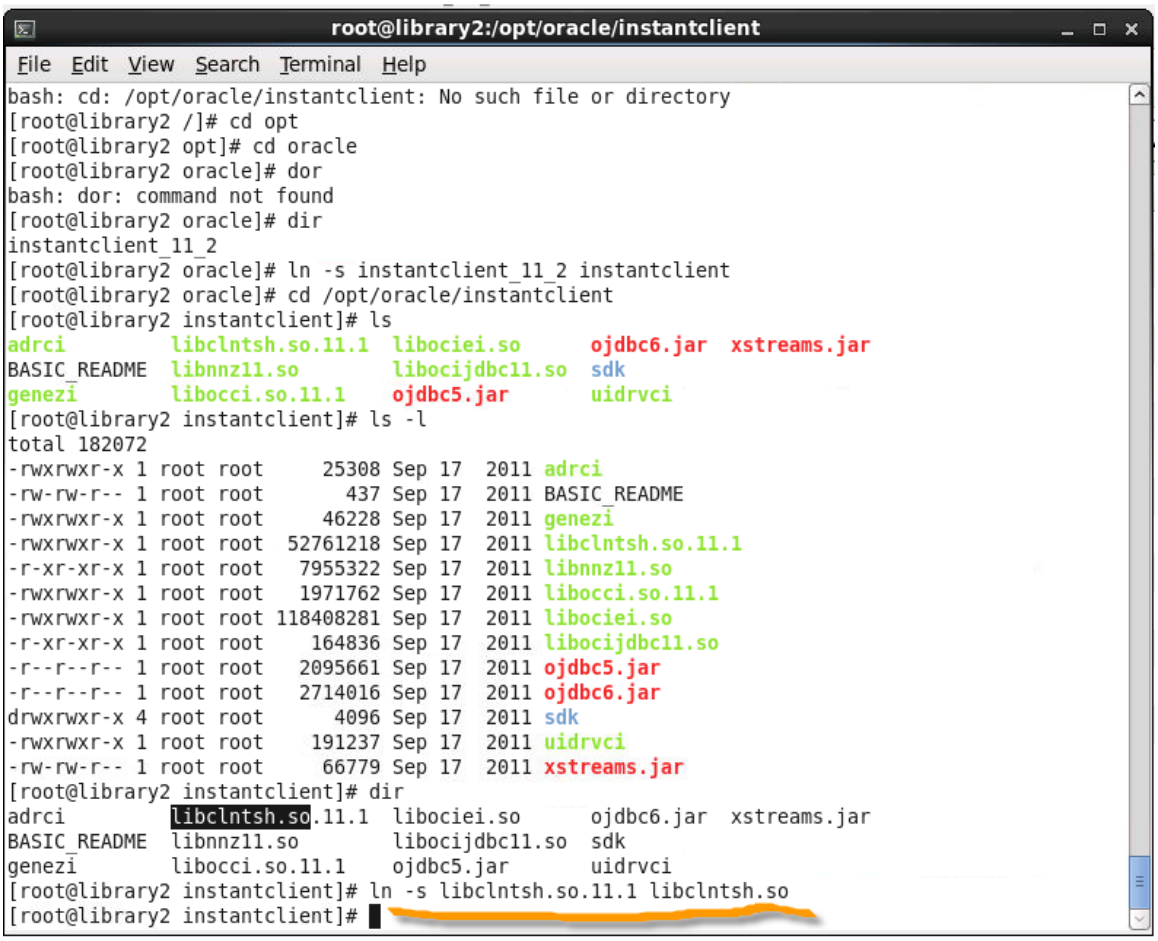
Link instant_client file.



Link libocci.so file.



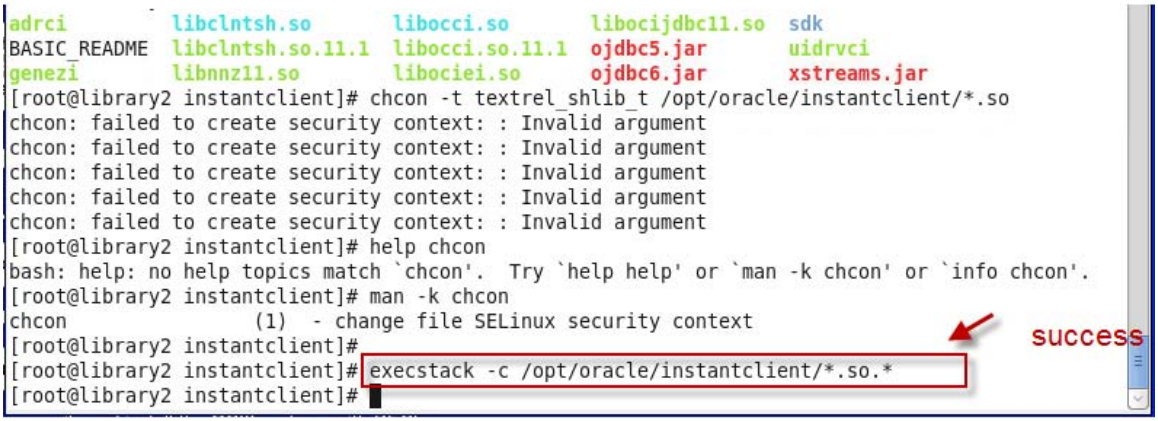
Link libclntsh.so file.



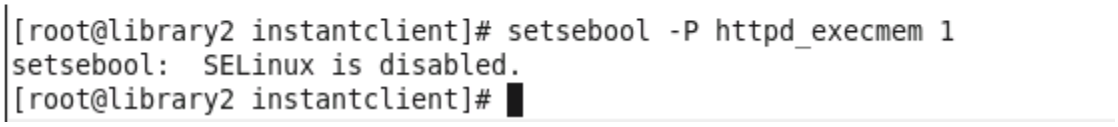
Chcon failed to create security context

```
[root@library2 instantclient]# chcon -t textrel_shlib_t /opt/oracle/instantclient/*.so
chcon: failed to create security context: : Invalid argument
chcon: failed to create security context: : Invalid argument
chcon: failed to create security context: : Invalid argument
chcon: failed to create security context: : Invalid argument
chcon: failed to create security context: : Invalid argument
```

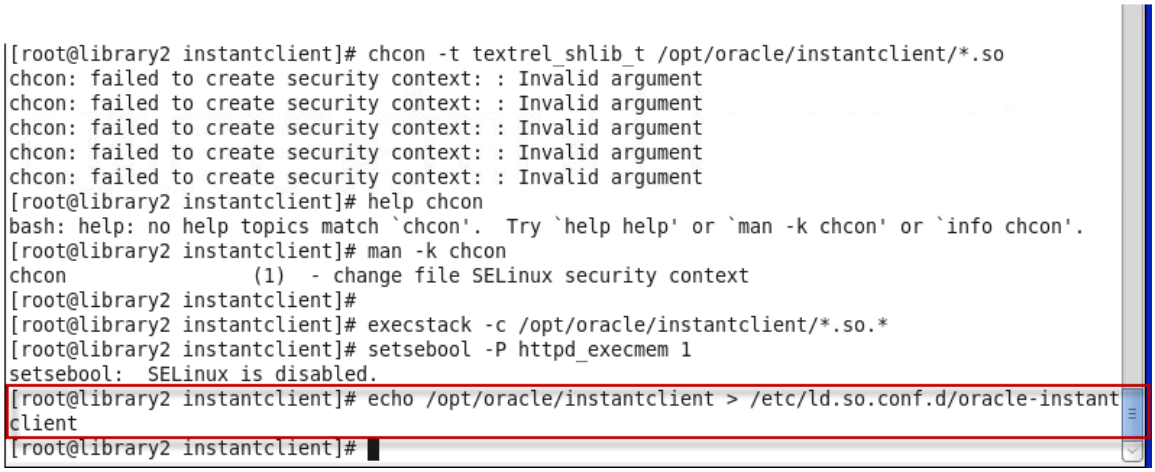
By using execstack command, it is successful.



SELinux is disabled be default.



Add the Instant Client to the system dynamic library loader



Install gcc

```
[root@library2 instantclient]# yum install gcc
Loaded plugins: product-id, refresh-packagekit, security, subscription-manager
This system is receiving updates from Red Hat Subscription Management.
rhel-6-server-beta-rpms                                          | 3.4 kB     00:00
rhel-6-server-beta-source-rpms                                   | 2.6 kB     00:00
rhel-6-server-cf-tools-1-rpms                                    | 2.8 kB     00:00
rhel-6-server-optional-rpms                                      | 3.5 kB     00:00
rhel-6-server-rhev-agent-rpms                                    | 3.1 kB     00:00
rhel-6-server-rpms                                               | 3.7 kB     00:00
rhel-6-server-source-rpms                                        | 2.6 kB     00:00
rhel-6-server-supplementary-rpms                                 | 3.5 kB     00:00
rhel-server-dts-6-rhb-source-rpms                                | 2.5 kB     00:00
rhel-server-rhscl-6-beta-rpms                                    | 2.8 kB     00:00
Setting up Install Process
Package gcc-4.4.7-3.el6.x86_64 already installed and latest version
Nothing to do
[root@library2 instantclient]#
```

Install oci8 in success by using PECL.

```
root@library2:/opt/oracle/instantclient
File  Edit  View  Search  Terminal  Help
    during linking
  - use the `-Wl,-rpath -Wl,LIBDIR' linker flag
  - have your system administrator add LIBDIR to `/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.
----------------------------------------------------------------------

Build complete.
Don't forget to run 'make test'.

running: make INSTALL_ROOT="/var/tmp/pear-build-rootyDNDhr/install-oci8-1.4.10" install
Installing shared extensions:     /var/tmp/pear-build-rootyDNDhr/install-oci8-1.4.10/usr/lib64/p
hp/modules/
running: find "/var/tmp/pear-build-rootyDNDhr/install-oci8-1.4.10" | xargs ls -dils
2491073   4 drwxr-xr-x 3 root root   4096 Jul 15 10:57 /var/tmp/pear-build-rootyDNDhr/install-oc
i8-1.4.10
2491102   4 drwxr-xr-x 3 root root   4096 Jul 15 10:57 /var/tmp/pear-build-rootyDNDhr/install-oc
i8-1.4.10/usr
2491103   4 drwxr-xr-x 3 root root   4096 Jul 15 10:57 /var/tmp/pear-build-rootyDNDhr/install-oc
i8-1.4.10/usr/lib64
2491104   4 drwxr-xr-x 3 root root   4096 Jul 15 10:57 /var/tmp/pear-build-rootyDNDhr/install-oc
i8-1.4.10/usr/lib64/php
2491105   4 drwxr-xr-x 2 root root   4096 Jul 15 10:57 /var/tmp/pear-build-rootyDNDhr/install-oc
i8-1.4.10/usr/lib64/php/modules
2491101 492 -rwxr-xr-x 1 root root 502499 Jul 15 10:57 /var/tmp/pear-build-rootyDNDhr/install-oc
i8-1.4.10/usr/lib64/php/modules/oci8.so

Build process completed successfully
Installing '/usr/lib64/php/modules/oci8.so'
install ok: channel://pecl.php.net/oci8-1.4.10
configuration option "php_ini" is not set to php.ini location
You should add "extension=oci8.so" to php.ini
[root@library2 instantclient]#
```

Install the PHP extension

Now install the PHP extension and restart Apache

```
chcon system_u:object_r:textrel_shlib_t:s0 /usr/lib/php/modules/oci8.so
chmod +x /usr/lib/php/modules/oci8.so
echo extension=oci8.so > /etc/php.d/oci8.ini
apachectl restart
```

pay attention, in redhat linux 64, it should be lib64

Give oci8.so executable authority.

```
chcon: failed to change context of /usr/lib64/php/modules/oci8.so' to system_u:obje
l_shlib_t:s0': Operation not supported
[root@library2 instantclient]# chmod +x /usr/lib/php/modules/oci8.so
chmod: cannot access `/usr/lib/php/modules/oci8.so': No such file or directory
[root@library2 instantclient]# chmod +x /usr/lib64/php/modules/oci8.so
[root@library2 instantclient]#
```

Create oci8.ini file.

```
2491101 492 -rwxr-xr-x 1 root root 502499 Jul 15 10:57 /var/tmp/pear-build-rootyDNDhr/inst
i8-1.4.10/usr/lib64/php/modules/oci8.so

Build process completed successfully
Installing '/usr/lib64/php/modules/oci8.so'
install ok: channel://pecl.php.net/oci8-1.4.10
configuration option "php_ini" is not set to php.ini location
You should add "extension=oci8.so" to php.ini
[root@library2 instantclient]# chcon system_u:object_r:textrel_shlib_t:s0 /usr/lib/php/mod
ci8.so
chcon: cannot access `/usr/lib/php/modules/oci8.so': No such file or directory
[root@library2 instantclient]# chcon system_u:object_r:textrel_shlib_t:s0 /usr/lib64/php/m
/oci8.so
chcon: failed to change context of `/usr/lib64/php/modules/oci8.so' to `system_u:object_r:
l_shlib_t:s0': Operation not supported
[root@library2 instantclient]# chmod +x /usr/lib/php/modules/oci8.so
chmod: cannot access `/usr/lib/php/modules/oci8.so': No such file or directory
[root@library2 instantclient]# chmod +x /usr/lib64/php/modules/oci8.so
[root@library2 instantclient]# pwd
/opt/oracle/instantclient
[root@library2 instantclient]# ls /etc/php.d/oci8.ini
ls: cannot access /etc/php.d/oci8.ini: No such file or directory
[root@library2 instantclient]# echo extension=oci8.so > /etc/php.d/oci8.ini
[root@library2 instantclient]# ls /etc/php.d/oci8.ini
/etc/php.d/oci8.ini
[root@library2 instantclient]#
```

Restart Apache server.

```
chmod: cannot access `/usr/lib/php/modules/oci8.so': No such file or di
[root@library2 instantclient]# chmod +x /usr/lib64/php/modules/oci8.so
[root@library2 instantclient]# pwd
/opt/oracle/instantclient
[root@library2 instantclient]# ls /etc/php.d/oci8.ini
ls: cannot access /etc/php.d/oci8.ini: No such file or directory
[root@library2 instantclient]# echo extension=oci8.so > /etc/php.d/oci8
[root@library2 instantclient]# ls /etc/php.d/oci8.ini
/etc/php.d/oci8.ini
[root@library2 instantclient]# apachectl restart
[root@library2 instantclient]#
```

Create a temporary folder for pear download.

```
Build process completed successfully
Installing '/usr/lib64/php/modules/oci8.so'
install ok: channel://pecl.php.net/oci8-1.4.10
configuration option "php_ini" is not set to php.ini location
You should add "extension=oci8.so" to php.ini
[root@library2 instantclient]# chcon system_u:object_r:textrel_shlib_t:s0 /usr/lib
ci8.so
chcon: cannot access `/usr/lib/php/modules/oci8.so': No such file or directory
[root@library2 instantclient]# chcon system_u:object_r:textrel_shlib_t:s0 /usr/lib
/oci8.so
chcon: failed to change context of `/usr/lib64/php/modules/oci8.so' to `system_u:o
l_shlib_t:s0': Operation not supported
[root@library2 instantclient]# chmod +x /usr/lib/php/modules/oci8.so
chmod: cannot access `/usr/lib/php/modules/oci8.so': No such file or directory
[root@library2 instantclient]# chmod +x /usr/lib64/php/modules/oci8.so
[root@library2 instantclient]# pwd
/opt/oracle/instantclient
[root@library2 instantclient]# ls /etc/php.d/oci8.ini
ls: cannot access /etc/php.d/oci8.ini: No such file or directory
[root@library2 instantclient]# echo extension=oci8.so > /etc/php.d/oci8.ini
[root@library2 instantclient]# ls /etc/php.d/oci8.ini
/etc/php.d/oci8.ini
[root@library2 instantclient]# apachectl restart
[root@library2 instantclient]# mkdir -p /tmp/pear/download/
[root@library2 instantclient]#
```

Download pdo_oci

```
[root@library2 instantclient]# chmod +x /usr/lib64/php/modules/oci8.so
[root@library2 instantclient]# pwd
/opt/oracle/instantclient
[root@library2 instantclient]# ls /etc/php.d/oci8.ini
ls: cannot access /etc/php.d/oci8.ini: No such file or directory
[root@library2 instantclient]# echo extension=oci8.so > /etc/php.d/oci8.ini
[root@library2 instantclient]# ls /etc/php.d/oci8.ini
/etc/php.d/oci8.ini
[root@library2 instantclient]# apachectl restart
[root@library2 instantclient]# mkdir -p /tmp/pear/download/
[root@library2 instantclient]# cd /tmp/pear/download/
[root@library2 download]# pecl download pdo_oci
WARNING: "pear/PDO_OCI" is deprecated in favor of "channel://http://www.php.net/pd
oci"
WARNING: channel "pear.php.net" has updated its protocols, use "pecl channel-updat
" to update
downloading PDO_OCI-1.0.tgz ...
Starting to download PDO_OCI-1.0.tgz (13,679 bytes)
.....done: 13,679 bytes
File /tmp/pear/download/PDO_OCI-1.0.tgz downloaded
[root@library2 download]#
```

Uncompress the downloaded file.

```
[root@library2 download]# tar xvf PDO_OCI-1.0.tgz
package.xml
PDO_OCI-1.0/config.m4
PDO_OCI-1.0/config.w32
PDO_OCI-1.0/pdo_oci.c
PDO_OCI-1.0/oci_driver.c
PDO_OCI-1.0/oci_statement.c
PDO_OCI-1.0/php_pdo_oci.h
PDO_OCI-1.0/php_pdo_oci_int.h
PDO_OCI-1.0/CREDITS
[root@library2 download]#
```

Go into the folder PDO_OCI-1.0

```
[root@library2 download]# ls -l
total 196
-rw-r--r-- 1 root root    1402 Jul 15 11:06 channel.xml
-rw-r--r-- 1 root root  169248 Jul 15 10:56 oci8-1.4.10.tgz
-rw-r--r-- 1 1000 users   1861 Nov 26  2005 package.xml
drwxr-xr-x 2 root root    4096 Jul 15 11:07 PDO_OCI-1.0
-rw-r--r-- 1 root root   13679 Jul 15 11:06 PDO_OCI-1.0.tgz
[root@library2 download]# cd PDO_OCI-1.0
[root@library2 PDO_OCI-1.0]#
```

Modify the file config.m4.

```
[root@library2 download]# ls -l
total 196
-rw-r--r-- 1 root root    1402 Jul 15 11:06 channel.xml
-rw-r--r-- 1 root root  169248 Jul 15 10:56 oci8-1.4.10.tgz
-rw-r--r-- 1 1000 users   1861 Nov 26  2005 package.xml
drwxr-xr-x 2 root root    4096 Jul 15 11:07 PDO_OCI-1.0
-rw-r--r-- 1 root root   13679 Jul 15 11:06 PDO_OCI-1.0.tgz
[root@library2 download]# cd PDO_OCI-1.0
[root@library2 PDO_OCI-1.0]# ls
config.m4   CREDITS       oci_statement.c  php_pdo_oci.h
config.w32  oci_driver.c  pdo_oci.c        php_pdo_oci_int.h
[root@library2 PDO_OCI-1.0]# gedit config.m4
[root@library2 PDO_OCI-1.0]#
```

Add the below contents to config.m4 file

Edit the config.m4 file...

Add these lines:

```
    elif test -f $PDO_OCI_DIR/lib/libclntsh.$SHLIB_SUFFIX_NAME.11.1; then
      PDO_OCI_VERSION=11.1
```

above these lines:

```
    elif test -f $PDO_OCI_DIR/lib/libclntsh.$SHLIB_SUFFIX_NAME.10.1; then
      PDO_OCI_VERSION=10.1
```

Also add these lines:

```
      11.1)
        PHP_ADD_LIBRARY(clntsh, 1, PDO_OCI_SHARED_LIBADD)
        ;;
```

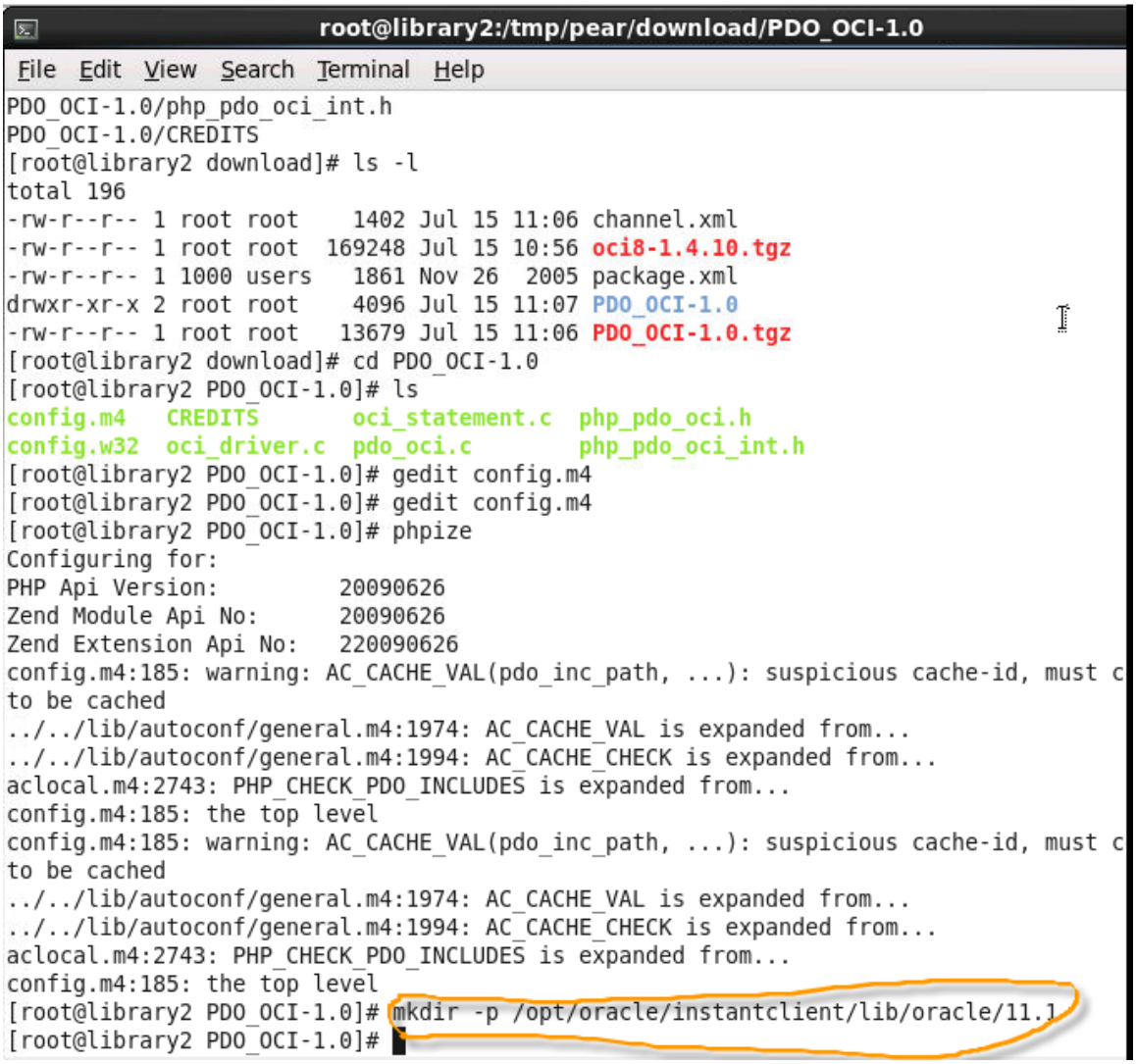
above these lines:

```
      *)
        AC_MSG_ERROR(Unsupported Oracle version! $PDO_OCI_VERSION)
        ;;
```

Run phpize.

```
[root@library2 PDO_OCI-1.0]# ls
config.m4   CREDITS       oci_statement.c  php_pdo_oci.h
config.w32  oci_driver.c  pdo_oci.c        php_pdo_oci_int.h
[root@library2 PDO_OCI-1.0]# gedit config.m4
[root@library2 PDO_OCI-1.0]# gedit config.m4
[root@library2 PDO_OCI-1.0]# phpize
Configuring for:
PHP Api Version:         20090626
Zend Module Api No:      20090626
Zend Extension Api No:   220090626
config.m4:185: warning: AC_CACHE_VAL(pdo_inc_path, ...): suspicious cache-i
to be cached
../../lib/autoconf/general.m4:1974: AC_CACHE_VAL is expanded from...
../../lib/autoconf/general.m4:1994: AC_CACHE_CHECK is expanded from...
aclocal.m4:2743: PHP_CHECK_PDO_INCLUDES is expanded from...
config.m4:185: the top level
config.m4:185: warning: AC_CACHE_VAL(pdo_inc_path, ...): suspicious cache-i
to be cached
../../lib/autoconf/general.m4:1974: AC_CACHE_VAL is expanded from...
../../lib/autoconf/general.m4:1994: AC_CACHE_CHECK is expanded from...
aclocal.m4:2743: PHP_CHECK_PDO_INCLUDES is expanded from...
config.m4:185: the top level
[root@library2 PDO_OCI-1.0]#
```

Create the directory as below.



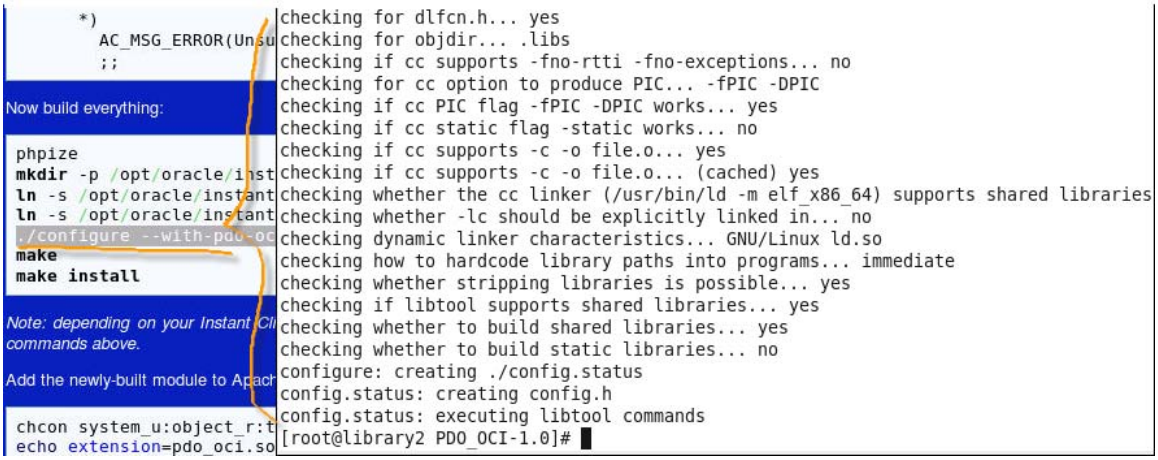
Create softlink for /opt/oracle/instantclient/lib/oracle/11.1/client

```
[root@library2 PDO_OCI-1.0]# phpize
Configuring for:
PHP Api Version:         20090626
Zend Module Api No:      20090626
Zend Extension Api No:   220090626
config.m4:185: warning: AC_CACHE_VAL(pdo_inc_path, ...): suspicious cache-id, must c
to be cached
../../lib/autoconf/general.m4:1974: AC_CACHE_VAL is expanded from...
../../lib/autoconf/general.m4:1994: AC_CACHE_CHECK is expanded from...
aclocal.m4:2743: PHP_CHECK_PDO_INCLUDES is expanded from...
config.m4:185: the top level
config.m4:185: warning: AC_CACHE_VAL(pdo_inc_path, ...): suspicious cache-id, must c
to be cached
../../lib/autoconf/general.m4:1974: AC_CACHE_VAL is expanded from...
../../lib/autoconf/general.m4:1994: AC_CACHE_CHECK is expanded from...
aclocal.m4:2743: PHP_CHECK_PDO_INCLUDES is expanded from...
config.m4:185: the top level
[root@library2 PDO_OCI-1.0]# mkdir -p /opt/oracle/instantclient/lib/oracle/11.1
[root@library2 PDO_OCI-1.0]# ln -s /opt/oracle/instantclient/sdk /opt/oracle/instant
racle/11.1/client
[root@library2 PDO_OCI-1.0]#
```

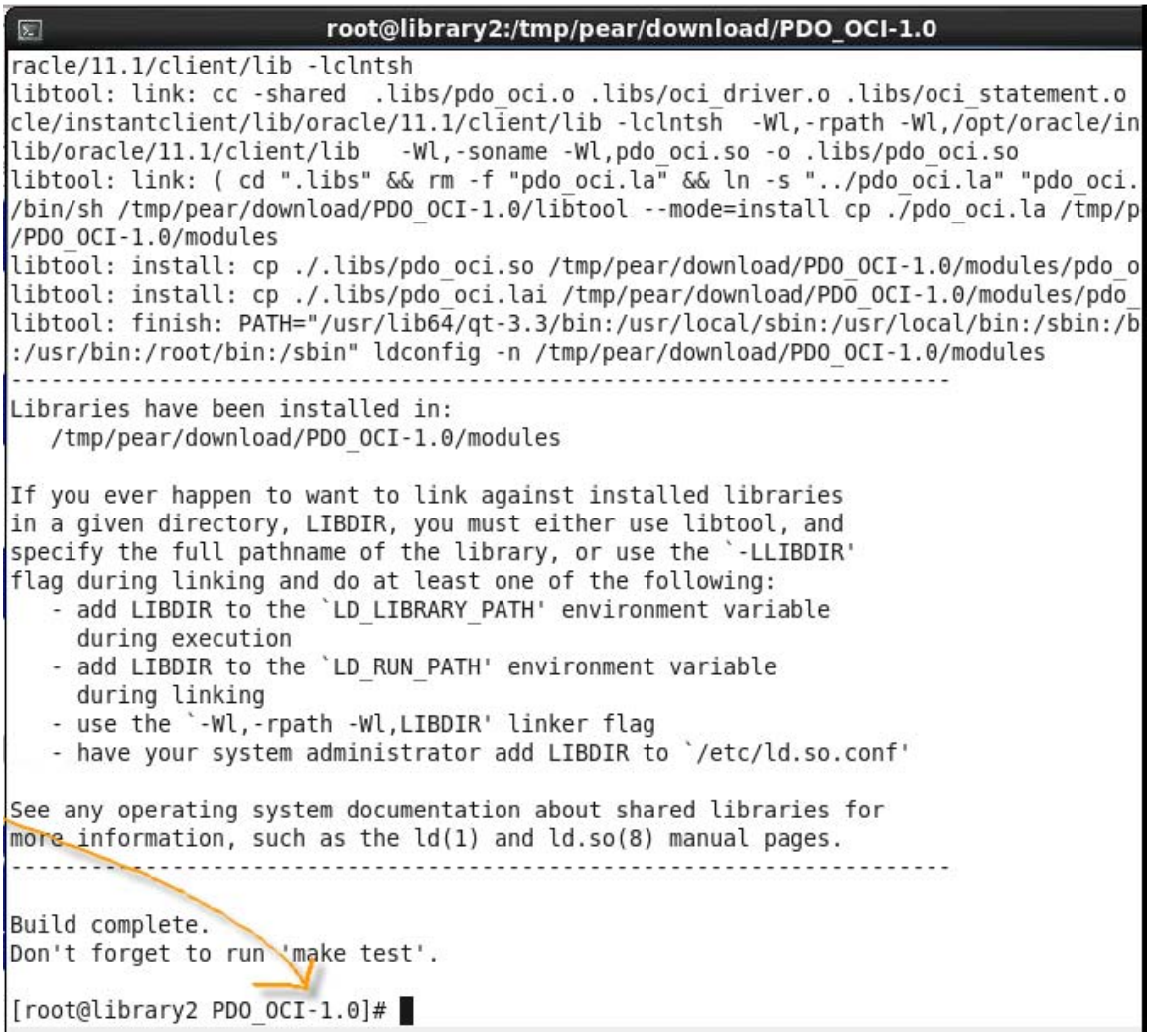
Create softlink for /opt/oracle/instantclient/lib/oracle/11.1/client

```
root@library2:/tmp/pear/download/PDO_OCI-1.0
-rw-r--r-- 1 root root    1402 Jul 15 11:06 channel.xml
-rw-r--r-- 1 root root  169248 Jul 15 10:56 oci8-1.4.10.tgz
-rw-r--r-- 1 1000 users   1861 Nov 26  2005 package.xml
drwxr-xr-x 2 root root    4096 Jul 15 11:07 PDO_OCI-1.0
-rw-r--r-- 1 root root   13679 Jul 15 11:06 PDO_OCI-1.0.tgz
[root@library2 download]# cd PDO_OCI-1.0
[root@library2 PDO_OCI-1.0]# ls
config.m4   CREDITS       oci_statement.c  php_pdo_oci.h
config.w32  oci_driver.c  pdo_oci.c        php_pdo_oci_int.h
[root@library2 PDO_OCI-1.0]# gedit config.m4
[root@library2 PDO_OCI-1.0]# gedit config.m4
[root@library2 PDO_OCI-1.0]# phpize
Configuring for:
PHP Api Version:         20090626
Zend Module Api No:      20090626
Zend Extension Api No:   220090626
config.m4:185: warning: AC_CACHE_VAL(pdo_inc_path, ...): suspicious cache-id, must c
to be cached
../../lib/autoconf/general.m4:1974: AC_CACHE_VAL is expanded from...
../../lib/autoconf/general.m4:1994: AC_CACHE_CHECK is expanded from...
aclocal.m4:2743: PHP_CHECK_PDO_INCLUDES is expanded from...
config.m4:185: the top level
config.m4:185: warning: AC_CACHE_VAL(pdo_inc_path, ...): suspicious cache-id, must c
to be cached
../../lib/autoconf/general.m4:1974: AC_CACHE_VAL is expanded from...
../../lib/autoconf/general.m4:1994: AC_CACHE_CHECK is expanded from...
aclocal.m4:2743: PHP_CHECK_PDO_INCLUDES is expanded from...
config.m4:185: the top level
[root@library2 PDO_OCI-1.0]# mkdir -p /opt/oracle/instantclient/lib/oracle/11.1
[root@library2 PDO_OCI-1.0]# ln -s /opt/oracle/instantclient/sdk /opt/oracle/instant
racle/11.1/client
[root@library2 PDO_OCI-1.0]# ln -s /opt/oracle/instantclient /opt/oracle/instantclie
e/11.1/client/lib
[root@library2 PDO_OCI-1.0]#
```

php-oci configure



Make command is in success.



Make install is succsessful.

```
If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the `-LLIBDIR'
flag during linking and do at least one of the following:
   - add LIBDIR to the `LD_LIBRARY_PATH' environment variable
     during execution
   - add LIBDIR to the `LD_RUN_PATH' environment variable
     during linking
   - use the `-Wl,-rpath -Wl,LIBDIR' linker flag
   - have your system administrator add LIBDIR to `/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.
----------------------------------------------------------------------

Build complete.
Don't forget to run 'make test'.

[root@library2 PDO_OCI-1.0]# make install
Installing shared extensions:     /usr/lib64/php/modules/
[root@library2 PDO_OCI-1.0]#
```

Add newly-built module to Apache PHP configuration

```
root@library2:/tmp/pear/download/PDO_OCI-1.0
/PDO_OCI-1.0/modules
libtool: install: cp ./.libs/pdo_oci.so /tmp/pear/download/PDO_OCI-1.0/modules/pdo_o
libtool: install: cp ./.libs/pdo_oci.lai /tmp/pear/download/PDO_OCI-1.0/modules/pdo_
libtool: finish: PATH="/usr/lib64/qt-3.3/bin:/usr/local/sbin:/usr/local/bin:/sbin:/b
:/usr/bin:/root/bin:/sbin" ldconfig -n /tmp/pear/download/PDO_OCI-1.0/modules
----------------------------------------------------------------------
Libraries have been installed in:
   /tmp/pear/download/PDO_OCI-1.0/modules

If you ever happen to want to link against installed libraries
in a given directory, LIBDIR, you must either use libtool, and
specify the full pathname of the library, or use the `-LLIBDIR'
flag during linking and do at least one of the following:
   - add LIBDIR to the `LD_LIBRARY_PATH' environment variable
     during execution
   - add LIBDIR to the `LD_RUN_PATH' environment variable
     during linking
   - use the `-Wl,-rpath -Wl,LIBDIR' linker flag
   - have your system administrator add LIBDIR to `/etc/ld.so.conf'

See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.
----------------------------------------------------------------------

Build complete.
Don't forget to run 'make test'.

[root@library2 PDO_OCI-1.0]# make install
Installing shared extensions:     /usr/lib64/php/modules/
[root@library2 PDO_OCI-1.0]# chcon system_u:object_r:textrel_shlib_t:s0 /usr/lib/php
_oci.so
chcon: cannot access `/usr/lib/php/modules/pdo_oci.so': No such file or directory
[root@library2 PDO_OCI-1.0]# echo extension=pdo_oci.so > /etc/php.d/pdo_oci.ini
[root@library2 PDO_OCI-1.0]#
```

Restart Apache again

```
[root@library2 PDO_OCI-1.0]# make install
Installing shared extensions:     /usr/lib64/php/modules/
[root@library2 PDO_OCI-1.0]# chcon system_u:object_r:textrel_s
_oci.so
chcon: cannot access `/usr/lib/php/modules/pdo_oci.so': No suc
[root@library2 PDO_OCI-1.0]# echo extension=pdo_oci.so > /etc/
[root@library2 PDO_OCI-1.0]# apachectl restart
[root@library2 PDO_OCI-1.0]#
```

SeLinux is disabled.

```
See any operating system documentation about shared libraries for
more information, such as the ld(1) and ld.so(8) manual pages.
----------------------------------------------------------------------

Build complete.
Don't forget to run 'make test'.

[root@library2 PDO_OCI-1.0]# make install
Installing shared extensions:     /usr/lib64/php/modules/
[root@library2 PDO_OCI-1.0]# chcon system_u:object_r:textrel_shlib_t:s0 /usr/li
_oci.so
chcon: cannot access `/usr/lib/php/modules/pdo_oci.so': No such file or directo
[root@library2 PDO_OCI-1.0]# echo extension=pdo_oci.so > /etc/php.d/pdo_oci.ini
[root@library2 PDO_OCI-1.0]# apachectl restart
[root@library2 PDO_OCI-1.0]# setsebool -P httpd_can_network_relay=1
setsebool:  SELinux is disabled.
[root@library2 PDO_OCI-1.0]#
```

Install JAVA

```
================================================================================
 Package                    Arch      Version                      Repository
================================================================================
Installing:
 java-1.7.0-openjdk-devel   x86_64    1:1.7.0.25-2.3.10.4.el6_4    rhel-6-server-rp
Installing for dependencies:
 java-1.7.0-openjdk         x86_64    1:1.7.0.25-2.3.10.4.el6_4    rhel-6-server-rp

Transaction Summary
================================================================================
Install       2 Package(s)

Total download size: 35 M
Installed size: 125 M
Is this ok [y/N]:
```

Download and uncompress VuFind

```
root@library2:/tmp
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/S3/_file
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/S3/_file
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/S3/_file
tml
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/SimpleDb
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/SimpleDb
php
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/Sqs/
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/Sqs/Onli
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/_files/
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/_files/a
se-invalid.xml
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/_files/a
se-valid.xml
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/_files/m
.xml
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/ZendService/Amazon/_files/o
ames.xml
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/_autoload.php
vufind-2.0/vendor/zendframework/zendservice-amazon/tests/phpunit.xml.dist
vufind-2.0/vufind.sh
[root@library2 tmp]# ls vu*
vufind-2.0.tar.gz

vufind-2.0:
build.xml       import-marc-auth.bat          languages  run_vufind.bat
composer.json   import-marc-auth.sh           LICENSE    solr
composer.lock   import-marc.bat               local      tests
config          import-marc.sh                module     themes
data            index-alphabetic-browse.bat   packages   util
harvest         index-alphabetic-browse.sh    public     vendor
import          install.php                   README.md  vufind.sh
[root@library2 tmp]# mv vufind-2.0 /usr/local/vufind2
[root@library2 tmp]#
```

VuFind Folder

```
[root@library2 tmp]# ls vu*
vufind-2.0.tar.gz

vufind-2.0:
build.xml       import-marc-auth.bat          languages  run_vufind.bat
composer.json   import-marc-auth.sh           LICENSE    solr
composer.lock   import-marc.bat               local      tests
config          import-marc.sh                module     themes
data            index-alphabetic-browse.bat   packages   util
harvest         index-alphabetic-browse.sh    public     vendor
import          install.php                   README.md  vufind.sh
[root@library2 tmp]# mv vufind-2.0 /usr/local/vufind2
[root@library2 tmp]# cd /usr/local/vufind2
[root@library2 vufind2]# dir
build.xml       import-marc-auth.bat          languages  run_vufind.bat
composer.json   import-marc-auth.sh           LICENSE    solr
composer.lock   import-marc.bat               local      tests
config          import-marc.sh                module     themes
data            index-alphabetic-browse.bat   packages   util
harvest         index-alphabetic-browse.sh    public     vendor
import          install.php                   README.md  vufind.sh
[root@library2 vufind2]#
```

Install VuFind

```
vufind-2.0:
build.xml       import-marc-auth.bat          languages  run_vufind.bat
composer.json   import-marc-auth.sh           LICENSE    solr
composer.lock   import-marc.bat               local      tests
config          import-marc.sh                module     themes
data            index-alphabetic-browse.bat   packages   util
harvest         index-alphabetic-browse.sh    public     vendor
import          install.php                   README.md  vufind.sh
[root@library2 tmp]# mv vufind-2.0 /usr/local/vufind2
[root@library2 tmp]# cd /usr/local/vufind2
[root@library2 vufind2]# dir
build.xml       import-marc-auth.bat          languages  run_vufind.bat
composer.json   import-marc-auth.sh           LICENSE    solr
composer.lock   import-marc.bat               local      tests
config          import-marc.sh                module     themes
data            index-alphabetic-browse.bat   packages   util
harvest         index-alphabetic-browse.sh    public     vendor
import          install.php                   README.md  vufind.sh
[root@library2 vufind2]# php install.php
VuFind has been found in /usr/local/vufind2.

Where would you like to store your local settings? [/usr/local/vufind2/local]
```

The module we will use in VuFind

```
vufind-2.0/vufind.sh
[root@library2 tmp]# ls vu*
vufind-2.0.tar.gz

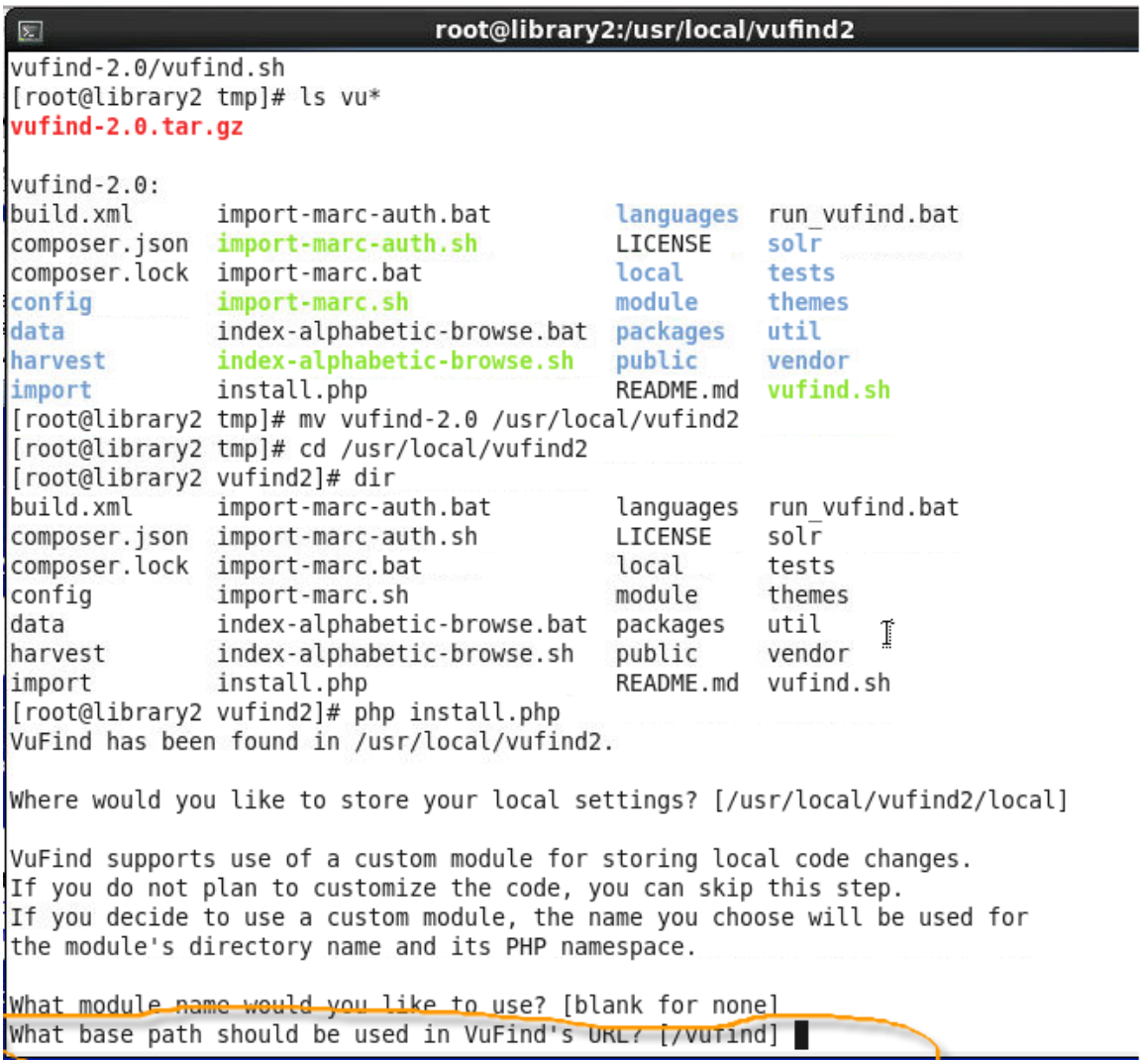
vufind-2.0:
build.xml       import-marc-auth.bat          languages  run_vufind.bat
composer.json   import-marc-auth.sh           LICENSE    solr
composer.lock   import-marc.bat               local      tests
config          import-marc.sh                module     themes
data            index-alphabetic-browse.bat   packages   util
harvest         index-alphabetic-browse.sh    public     vendor
import          install.php                   README.md  vufind.sh
[root@library2 tmp]# mv vufind-2.0 /usr/local/vufind2
[root@library2 tmp]# cd /usr/local/vufind2
[root@library2 vufind2]# dir
build.xml       import-marc-auth.bat          languages  run_vufind.bat
composer.json   import-marc-auth.sh           LICENSE    solr
composer.lock   import-marc.bat               local      tests
config          import-marc.sh                module     themes
data            index-alphabetic-browse.bat   packages   util
harvest         index-alphabetic-browse.sh    public     vendor
import          install.php                   README.md  vufind.sh
[root@library2 vufind2]# php install.php
VuFind has been found in /usr/local/vufind2.

Where would you like to store your local settings? [/usr/local/vufind2/local]

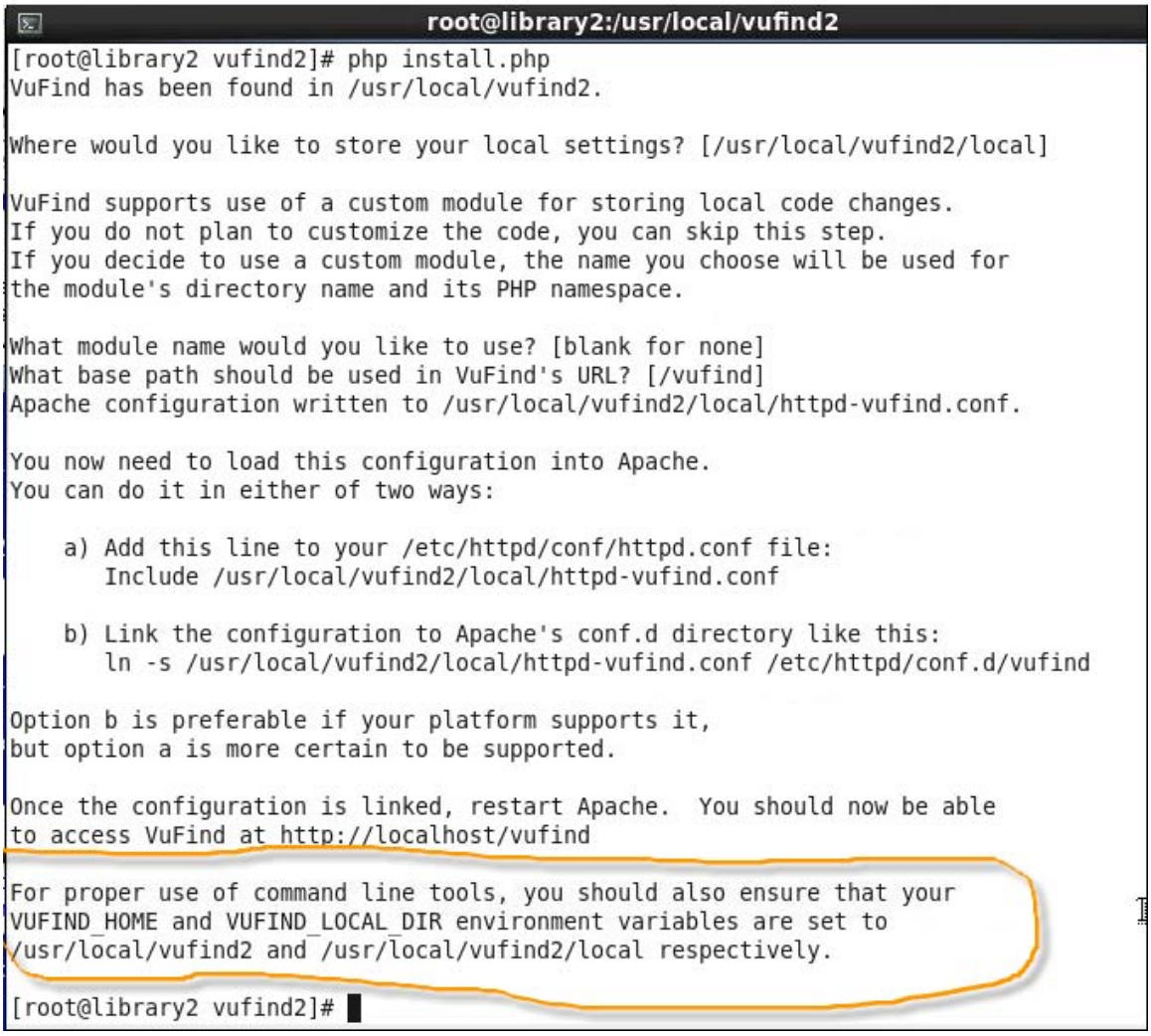
VuFind supports use of a custom module for storing local code changes.
If you do not plan to customize the code, you can skip this step.
If you decide to use a custom module, the name you choose will be used for
the module's directory name and its PHP namespace.

What module name would you like to use? [blank for none]
```

Base URL of VuFind



Vufind_home Vufind_Local_dir reminding



Setup Cache and config folder authority

```
[root@library2 vufind2]# chown -R apache:apache /usr/local/vufind2/local/cache
[root@library2 vufind2]# chcon -R unconfined_u:object_r:httpd_sys_rw_content_t:s0 /u
ind2/local/cache
chcon: failed to change context of `.gitignore' to `unconfined_u:object_r:httpd_sys_
:s0': Operation not supported
chcon: failed to change context of `/usr/local/vufind2/local/cache' to `unconfined_u
tpd_sys_rw_content_t:s0': Operation not supported
[root@library2 vufind2]# chown -R apache:apache /usr/local/vufind2/local/config
[root@library2 vufind2]# chcon -R unconfined_u:object_r:httpd_sys_rw_content_t:s0 /u
ind2/local/config
chcon: failed to change context of `README.txt' to `unconfined_u:object_r:httpd_sys_
:s0': Operation not supported
chcon: failed to change context of `vufind' to `unconfined_u:object_r:httpd_sys_rw_c
: Operation not supported
chcon: failed to change context of `/usr/local/vufind2/local/config' to `unconfined_
ttpd_sys_rw_content_t:s0': Operation not supported
[root@library2 vufind2]#
[root@library2 vufind2]#
```

Link VuFind to Apache

```
chcon: failed to change context of `/usr/local/vufind2/local/cache' to `unconfined_u
tpd_sys_rw_content_t:s0': Operation not supported
[root@library2 vufind2]# chown -R apache:apache /usr/local/vufind2/local/config
[root@library2 vufind2]# chcon -R unconfined_u:object_r:httpd_sys_rw_content_t:s0 /u
ind2/local/config
chcon: failed to change context of `README.txt' to `unconfined_u:object_r:httpd_sys_
:s0': Operation not supported
chcon: failed to change context of `vufind' to `unconfined_u:object_r:httpd_sys_rw_c
: Operation not supported
chcon: failed to change context of `/usr/local/vufind2/local/config' to `unconfined_
ttpd_sys_rw_content_t:s0': Operation not supported
[root@library2 vufind2]#
[root@library2 vufind2]# chcon system_u:object_r:httpd_config_t:s0 /usr/local/vufind
d-vufind.conf
chcon: failed to change context of `/usr/local/vufind2/local/httpd-vufind.conf' to `
ect_r:httpd_config_t:s0': Operation not supported
[root@library2 vufind2]# ln -s /usr/local/vufind2/local/httpd-vufind.conf /etc/httpd
nd2.conf
[root@library2 vufind2]#
```

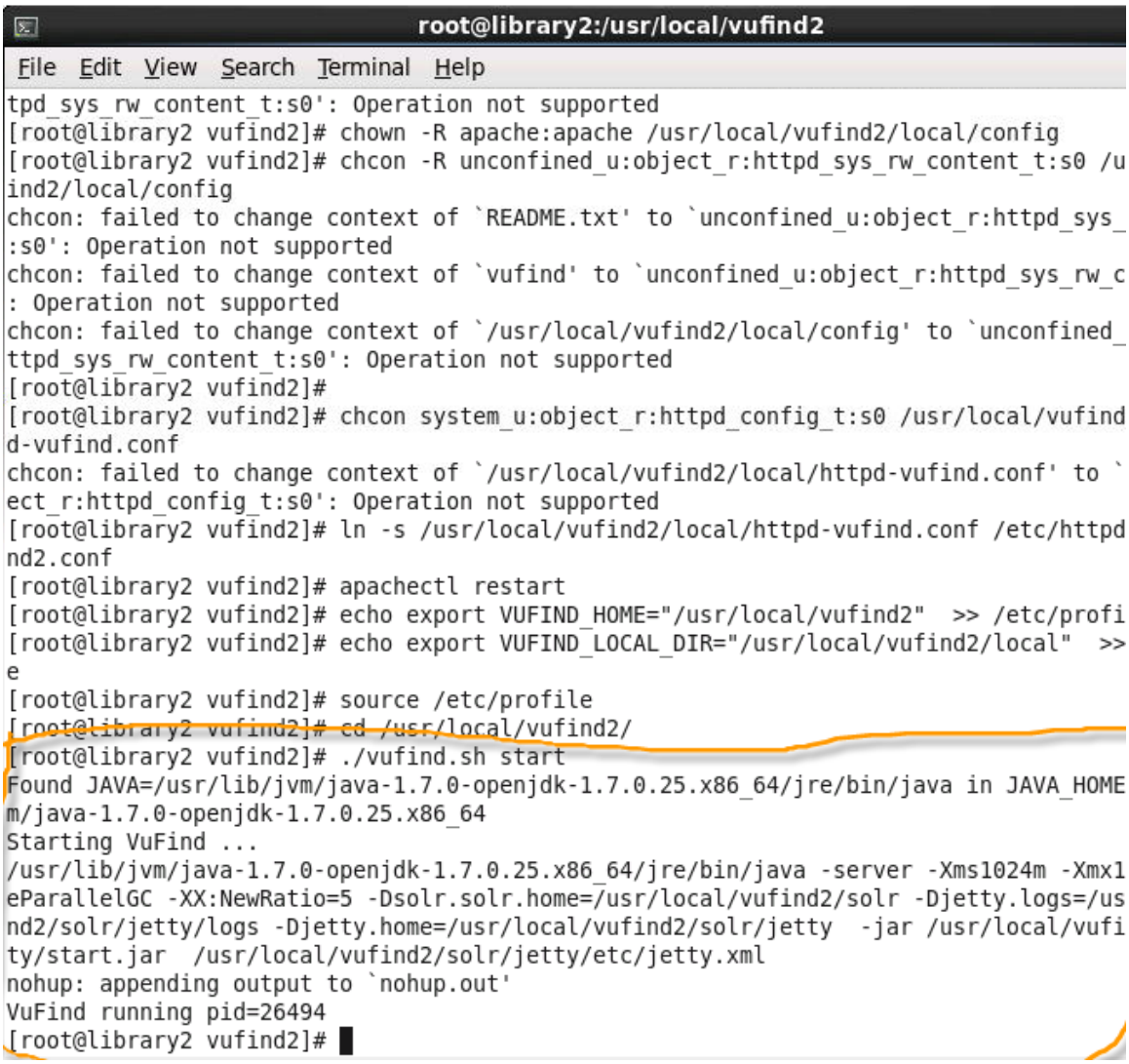
Export VuFind home and VuFind local

```
tpd_sys_rw_content_t:s0': Operation not supported
[root@library2 vufind2]# chown -R apache:apache /usr/local/vufind2/local/config
[root@library2 vufind2]# chcon -R unconfined_u:object_r:httpd_sys_rw_content_t:s0 /u
ind2/local/config
chcon: failed to change context of `README.txt' to `unconfined_u:object_r:httpd_sys_
:s0': Operation not supported
chcon: failed to change context of `vufind' to `unconfined_u:object_r:httpd_sys_rw_c
: Operation not supported
chcon: failed to change context of `/usr/local/vufind2/local/config' to `unconfined_
ttpd_sys_rw_content_t:s0': Operation not supported
[root@library2 vufind2]#
[root@library2 vufind2]# chcon system_u:object_r:httpd_config_t:s0 /usr/local/vufind
d-vufind.conf
chcon: failed to change context of `/usr/local/vufind2/local/httpd-vufind.conf' to `
ect_r:httpd_config_t:s0': Operation not supported
[root@library2 vufind2]# ln -s /usr/local/vufind2/local/httpd-vufind.conf /etc/httpd
nd2.conf
[root@library2 vufind2]# apachectl restart
[root@library2 vufind2]# echo export VUFIND_HOME="/usr/local/vufind2"  >> /etc/profi
[root@library2 vufind2]# echo export VUFIND_LOCAL_DIR="/usr/local/vufind2/local"  >>
e
[root@library2 vufind2]#
```

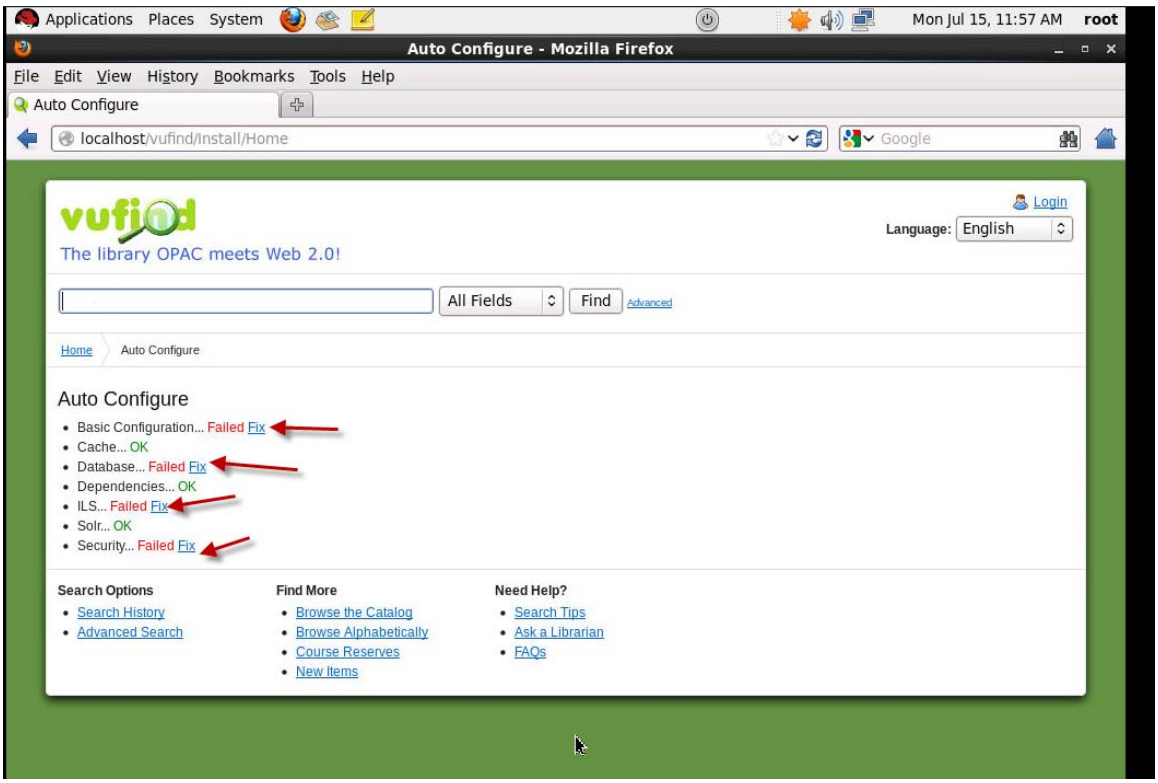
Run command in /etc/profile



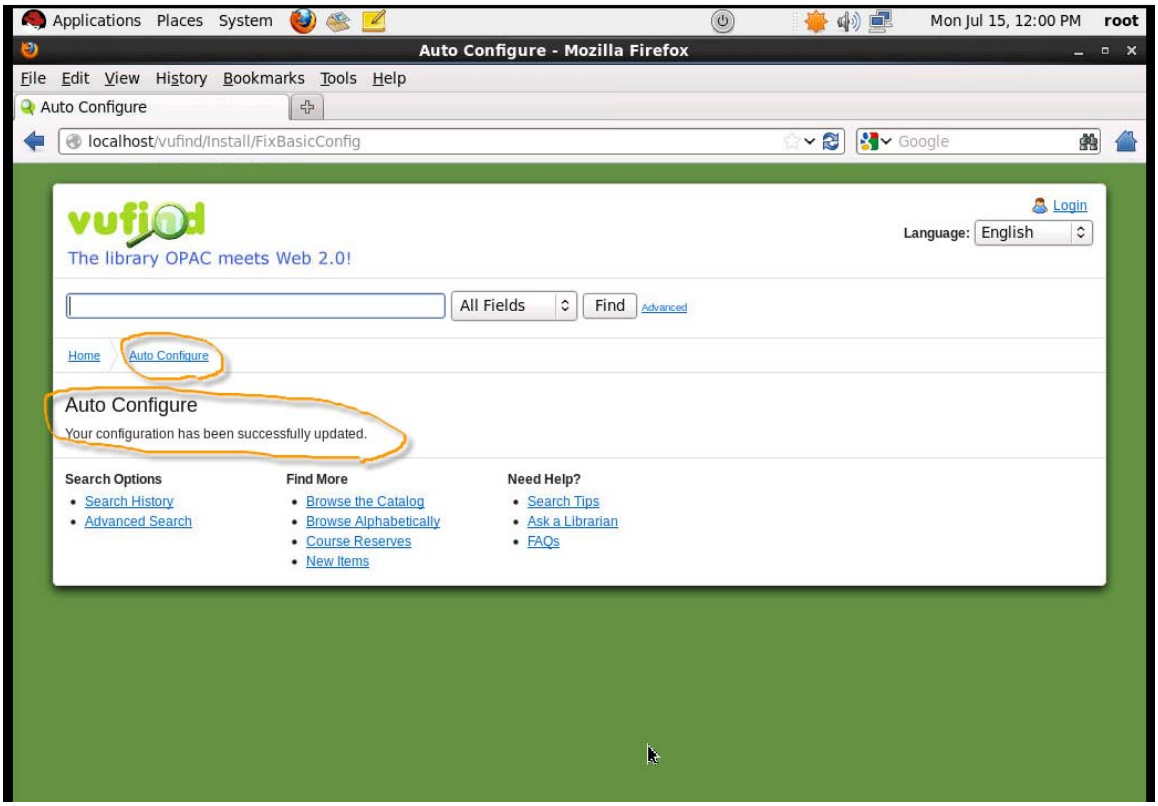
Restart VuFind



Vufind is up, but some modules need to be configured!



Auto configure is completed



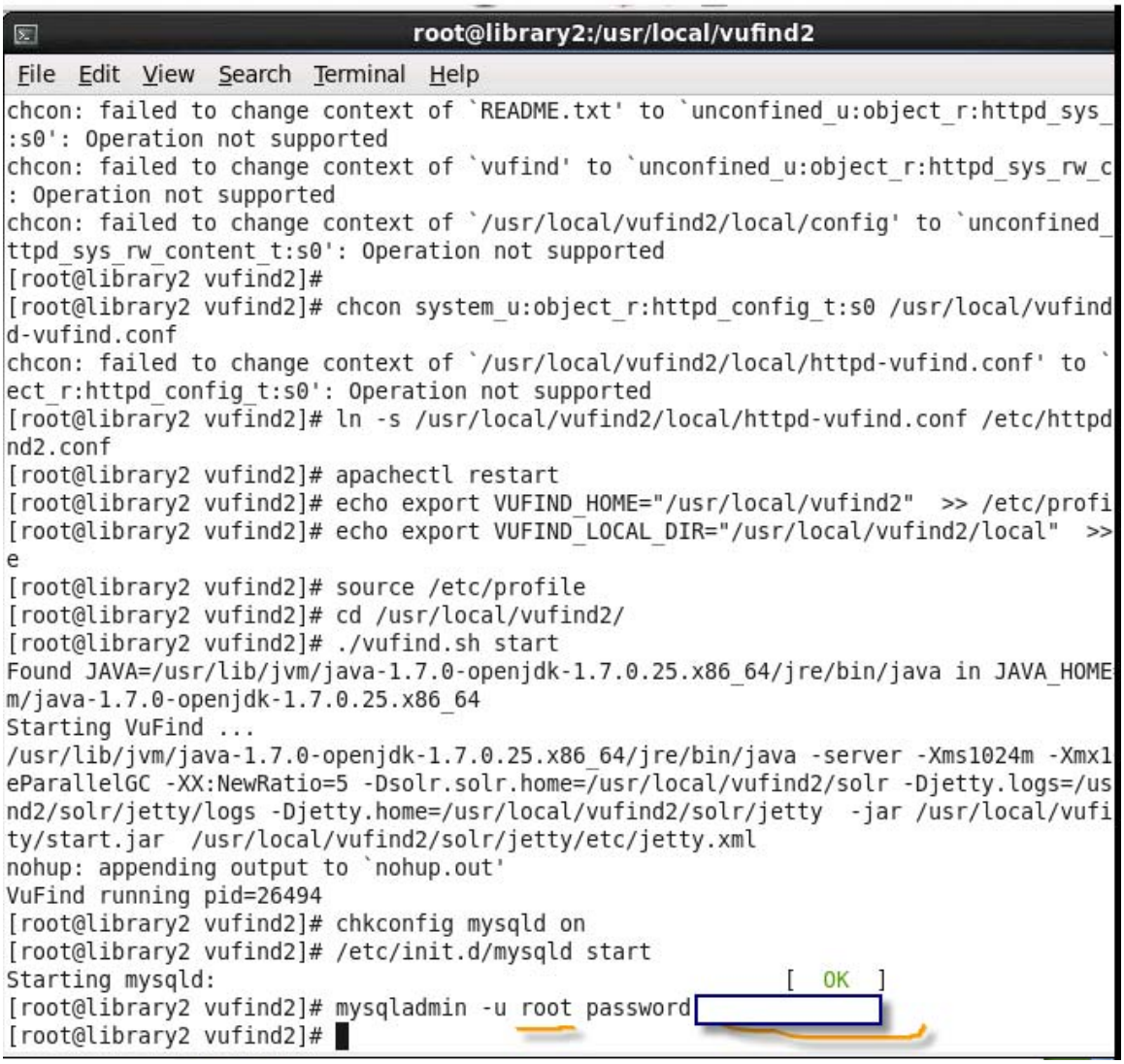
Add MySQL service to system and make it start automatically

```
root@library2:/usr/local/vufind2
File  Edit  View  Search  Terminal  Help
[root@library2 vufind2]# chown -R apache:apache /usr/local/vufind2/local/config
[root@library2 vufind2]# chcon -R unconfined_u:object_r:httpd_sys_rw_content_t:s0 /u
ind2/local/config
chcon: failed to change context of `README.txt' to `unconfined_u:object_r:httpd_sys_
:s0': Operation not supported
chcon: failed to change context of `vufind' to `unconfined_u:object_r:httpd_sys_rw_c
: Operation not supported
chcon: failed to change context of `/usr/local/vufind2/local/config' to `unconfined_
ttpd_sys_rw_content_t:s0': Operation not supported
[root@library2 vufind2]#
[root@library2 vufind2]# chcon system_u:object_r:httpd_config_t:s0 /usr/local/vufind
d-vufind.conf
chcon: failed to change context of `/usr/local/vufind2/local/httpd-vufind.conf' to `
ect_r:httpd_config_t:s0': Operation not supported
[root@library2 vufind2]# ln -s /usr/local/vufind2/local/httpd-vufind.conf /etc/httpd
nd2.conf
[root@library2 vufind2]# apachectl restart
[root@library2 vufind2]# echo export VUFIND_HOME="/usr/local/vufind2"  >> /etc/profi
[root@library2 vufind2]# echo export VUFIND_LOCAL_DIR="/usr/local/vufind2/local"  >>
e
[root@library2 vufind2]# source /etc/profile
[root@library2 vufind2]# cd /usr/local/vufind2/
[root@library2 vufind2]# ./vufind.sh start
Found JAVA=/usr/lib/jvm/java-1.7.0-openjdk-1.7.0.25.x86_64/jre/bin/java in JAVA_HOME
m/java-1.7.0-openjdk-1.7.0.25.x86_64
Starting VuFind ...
/usr/lib/jvm/java-1.7.0-openjdk-1.7.0.25.x86_64/jre/bin/java -server -Xms1024m -Xmx1
eParallelGC -XX:NewRatio=5 -Dsolr.solr.home=/usr/local/vufind2/solr -Djetty.logs=/us
nd2/solr/jetty/logs -Djetty.home=/usr/local/vufind2/solr/jetty  -jar /usr/local/vufi
ty/start.jar  /usr/local/vufind2/solr/jetty/etc/jetty.xml
nohup: appending output to `nohup.out'
VuFind running pid=26494
[root@library2 vufind2]# chkconfig mysqld on
[root@library2 vufind2]#
```

Manually start MySQL

```
[root@library2 vufind2]# chkconfig mysqld on
[root@library2 vufind2]# /etc/init.d/mysqld start
Starting mysqld:                                           [  OK  ]
[root@library2 vufind2]#
```

Create MySQL root user and password



Login MySQL

```
[root@library2 vufind2]# chkconfig mysqld on
[root@library2 vufind2]# /etc/init.d/mysqld start
Starting mysqld:                                           [  OK  ]
[root@library2 vufind2]# mysqladmin -u root password
[root@library2 vufind2]# mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g
Your MySQL connection id is 7
Server version: 5.1.69 Source distribution

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owners.

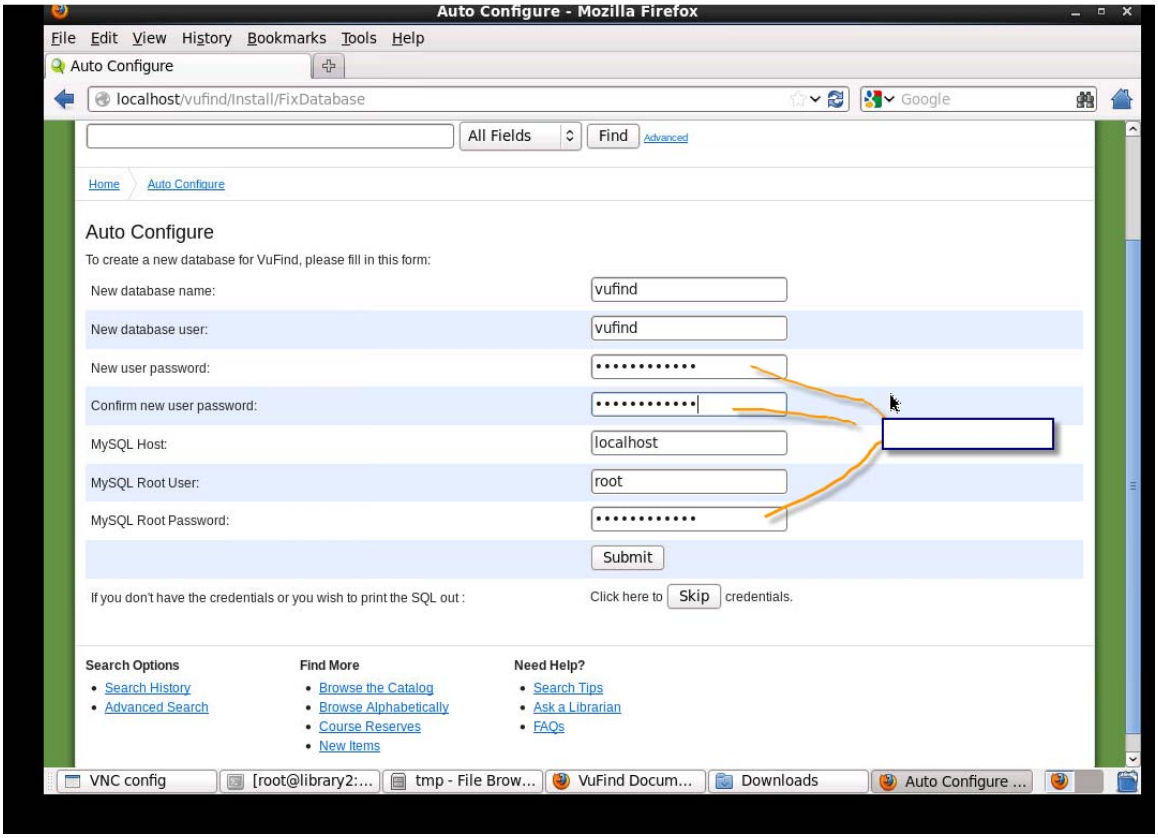
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

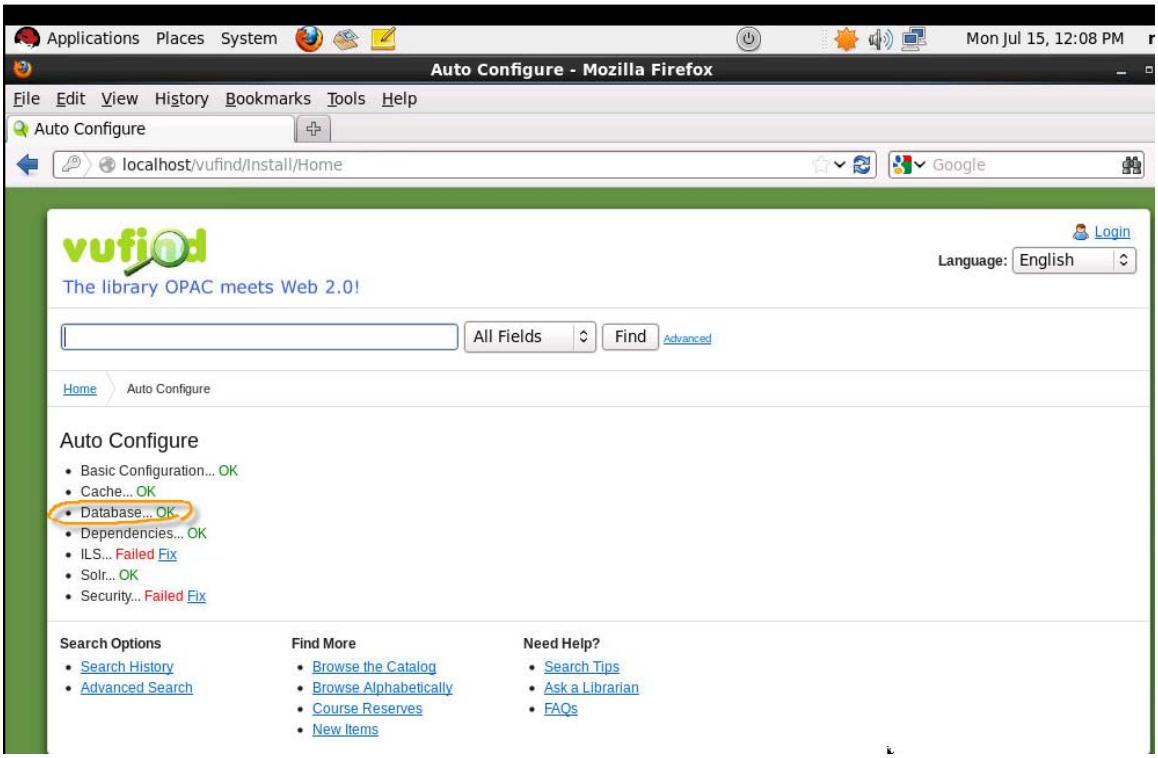
Modify MySQL configuration

```
mysql> quit
Bye
[root@library2 vufind2]# gedit /etc/my.cnf
[root@library2 vufind2]#
```

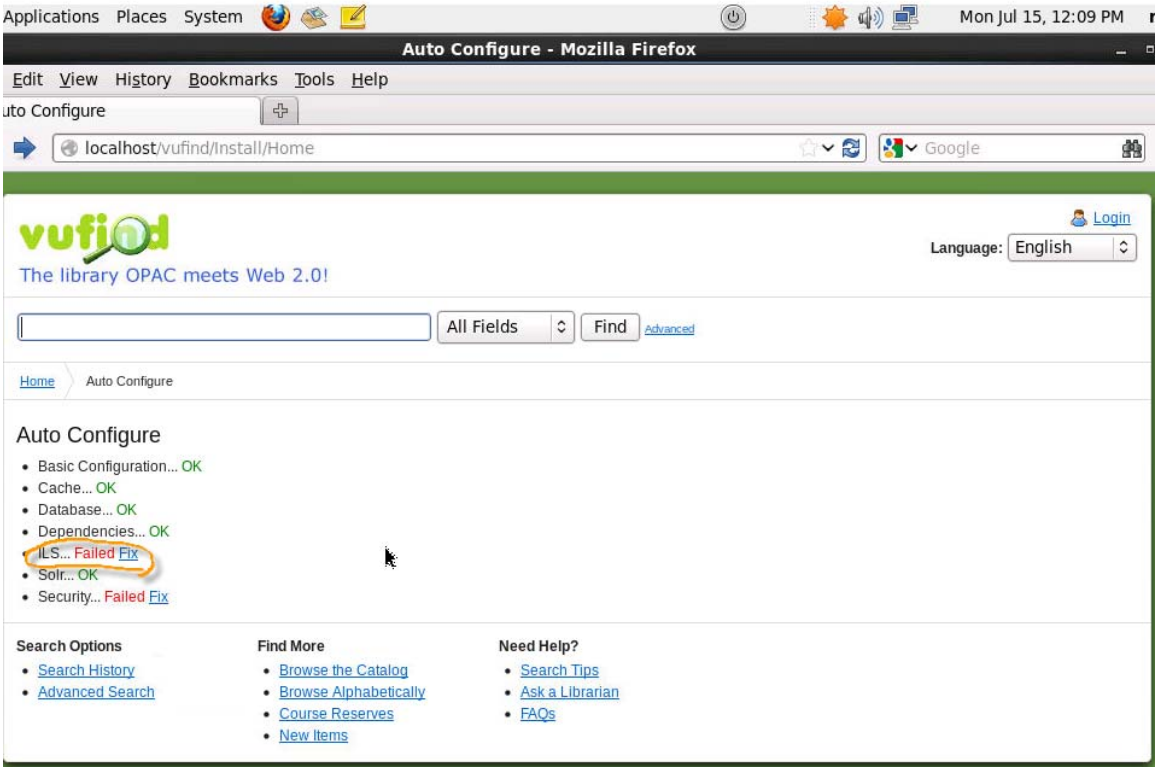
Configure MySQL database for VuFind



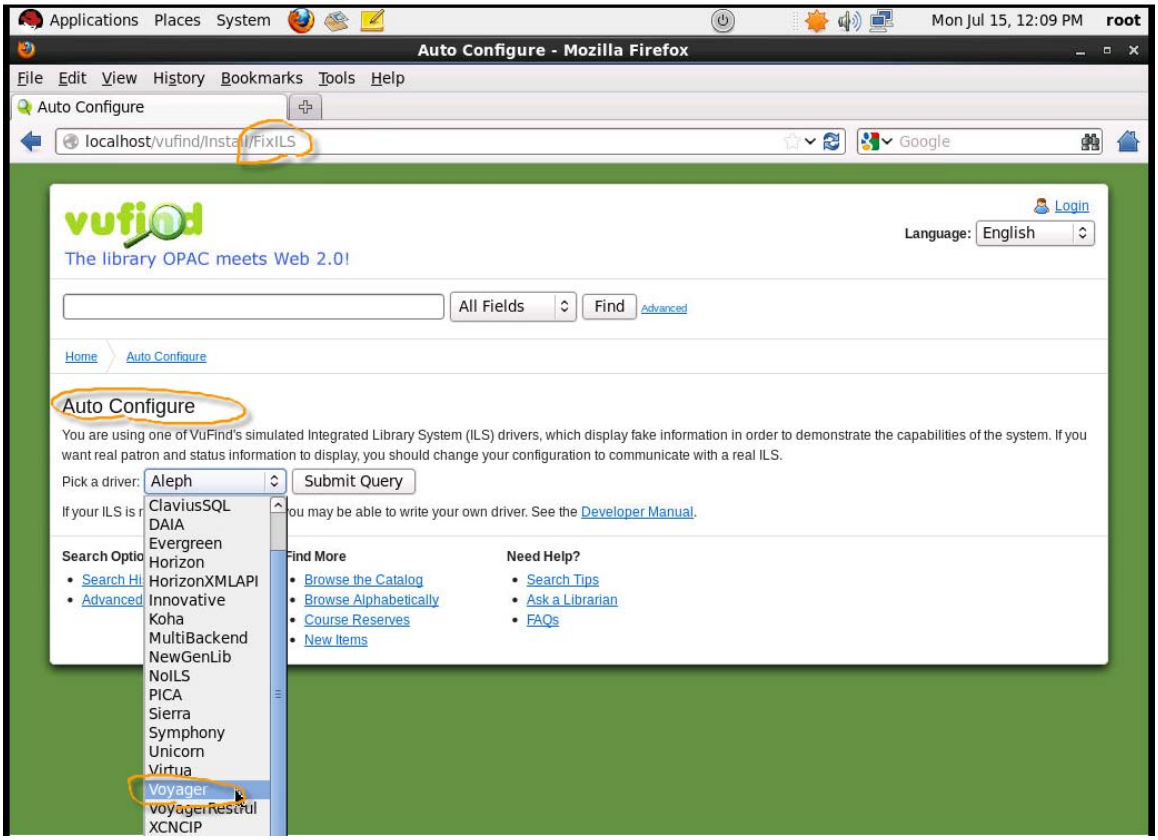
Database configuration is completed



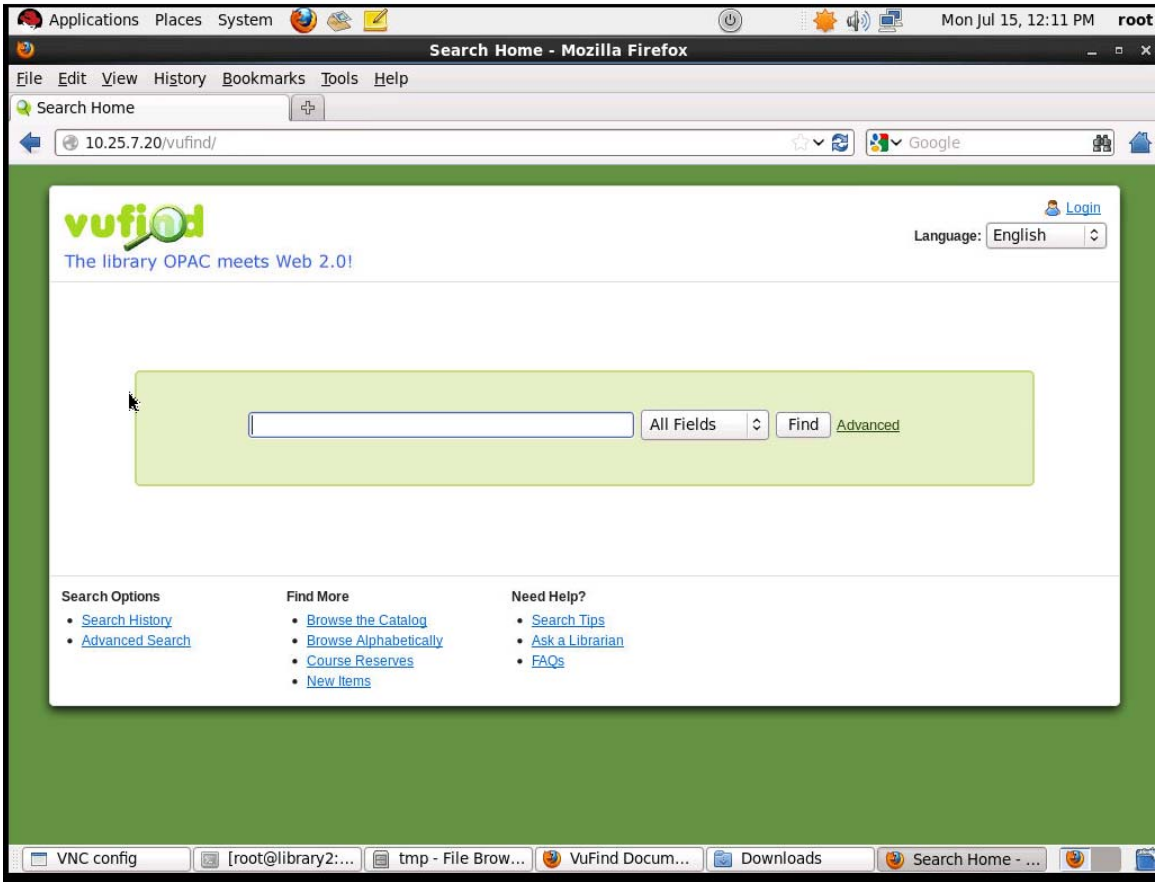
ILS configuration needs to be initialized.



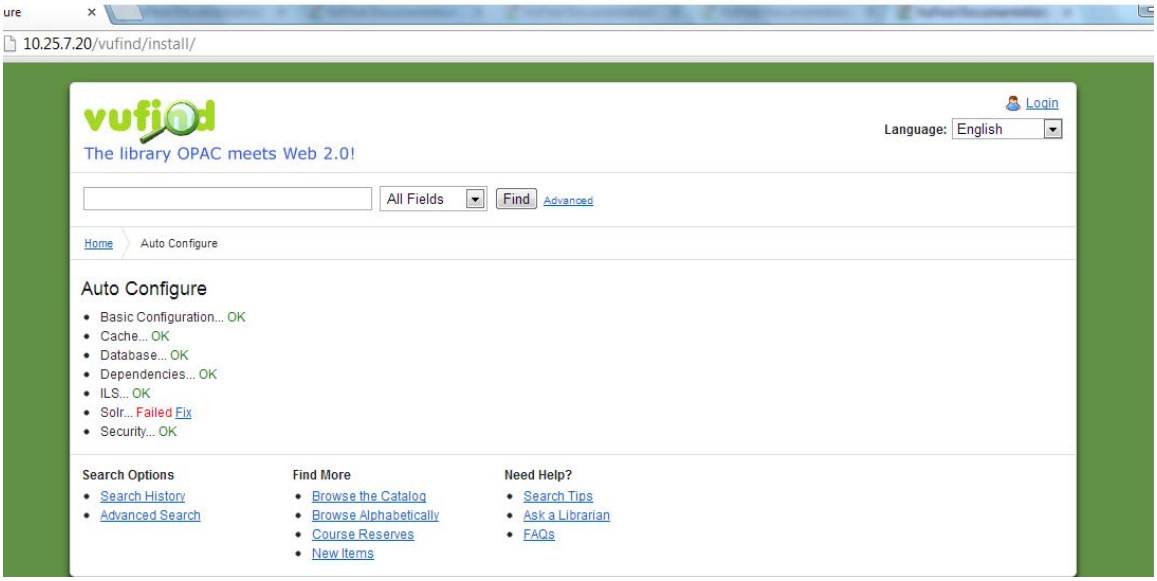
Choose voyager for ILS engine configuration



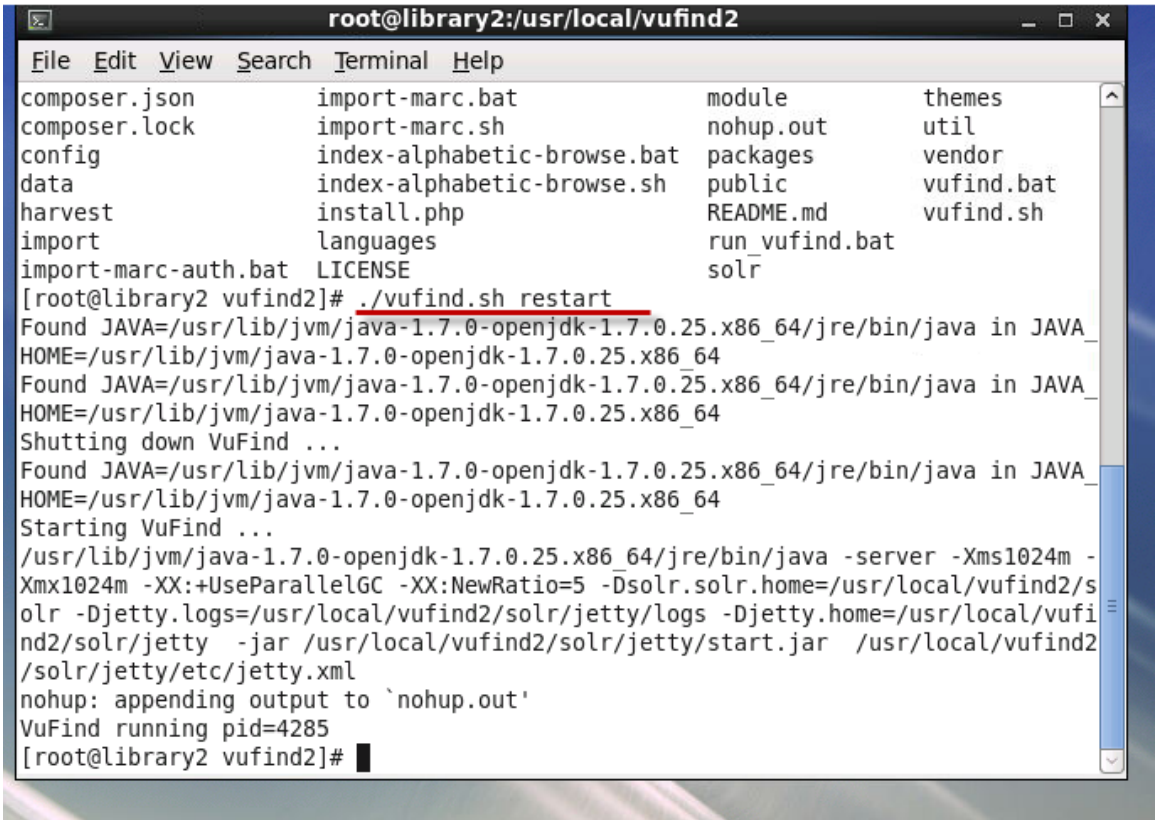
VuFind configuration is completed and OPAC is up



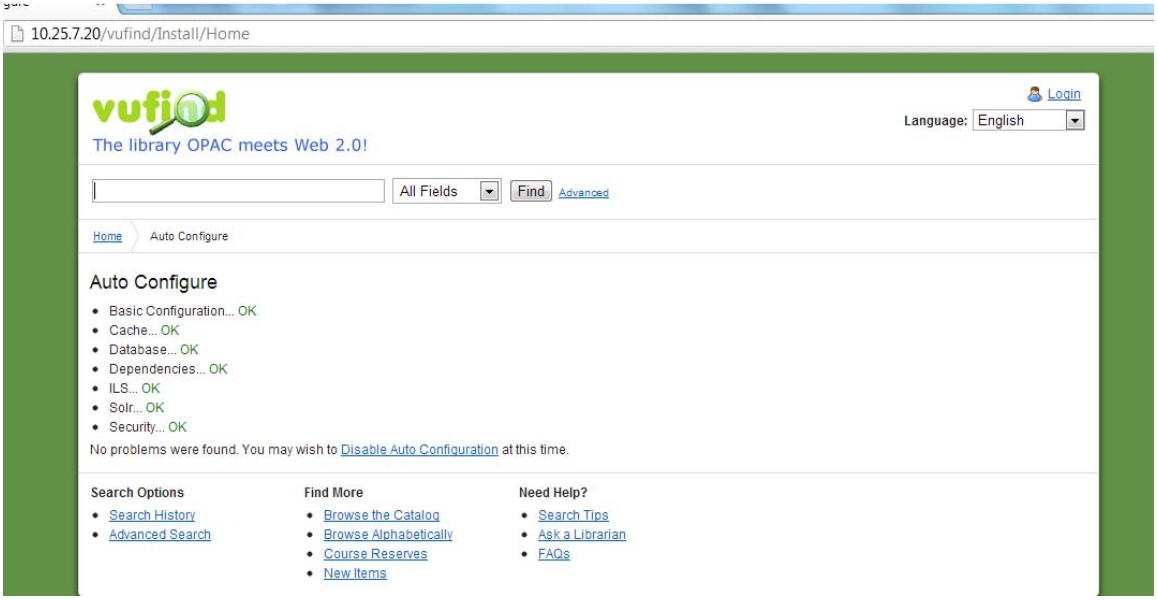
Solr has some problem, and we need to fix it.



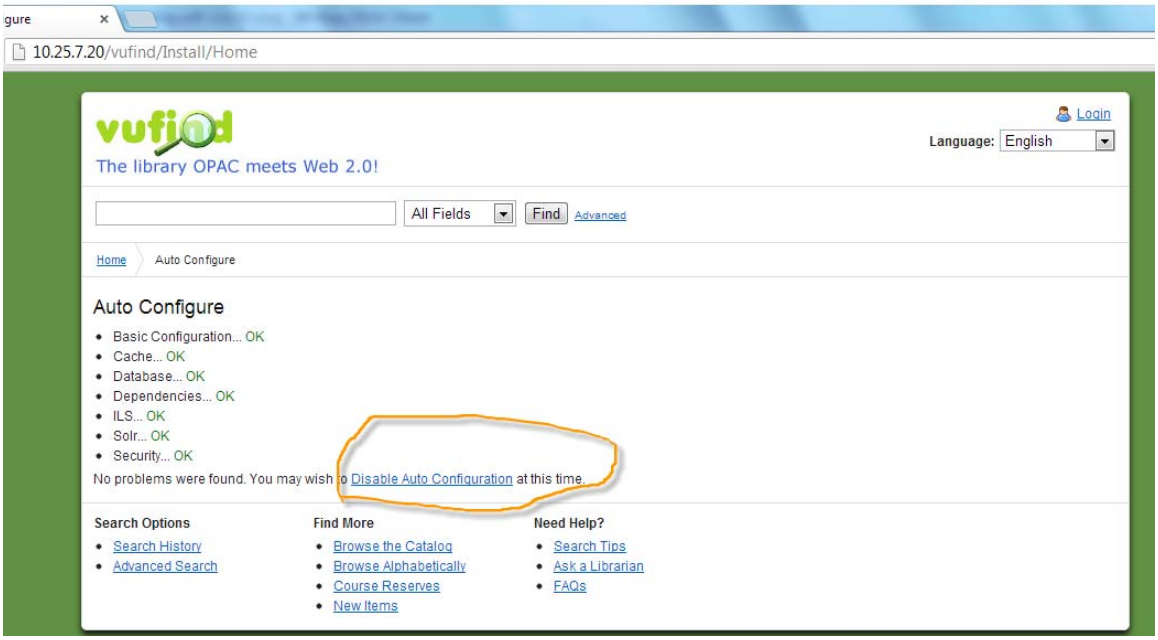
Restart VuFind again



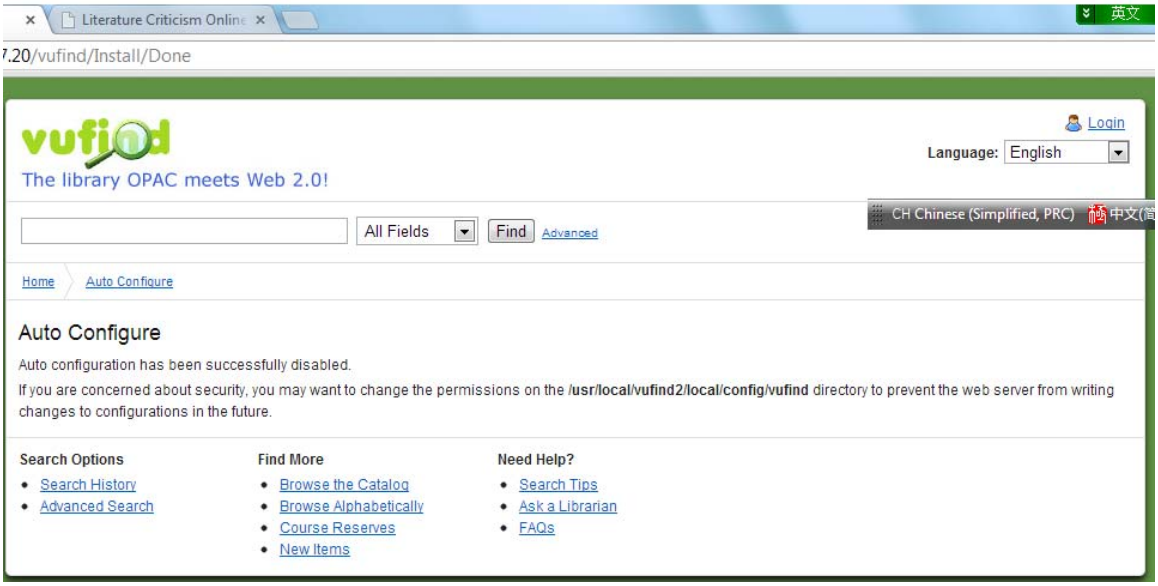
Finally, all modules are perfectly configured.



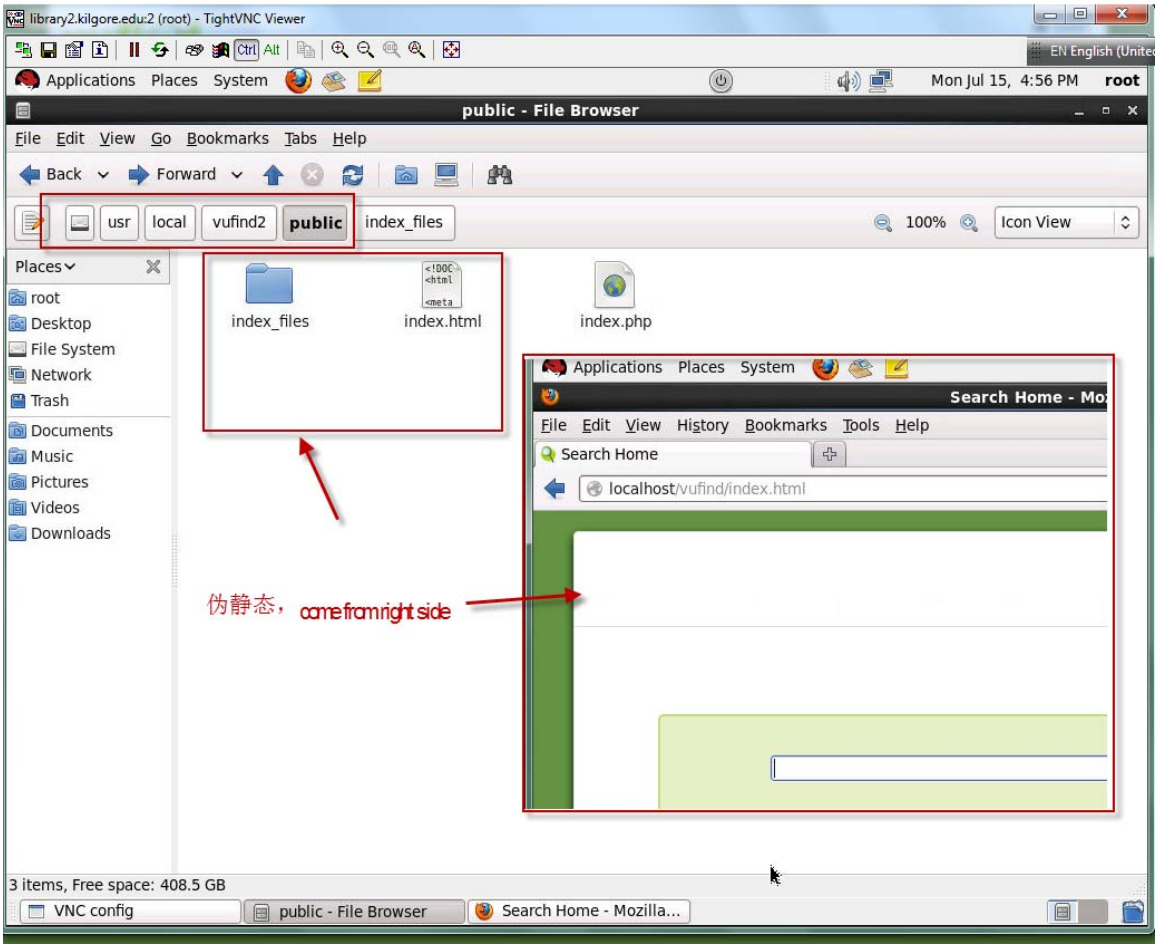
Auto configuration will be disabled after its duty is completed.



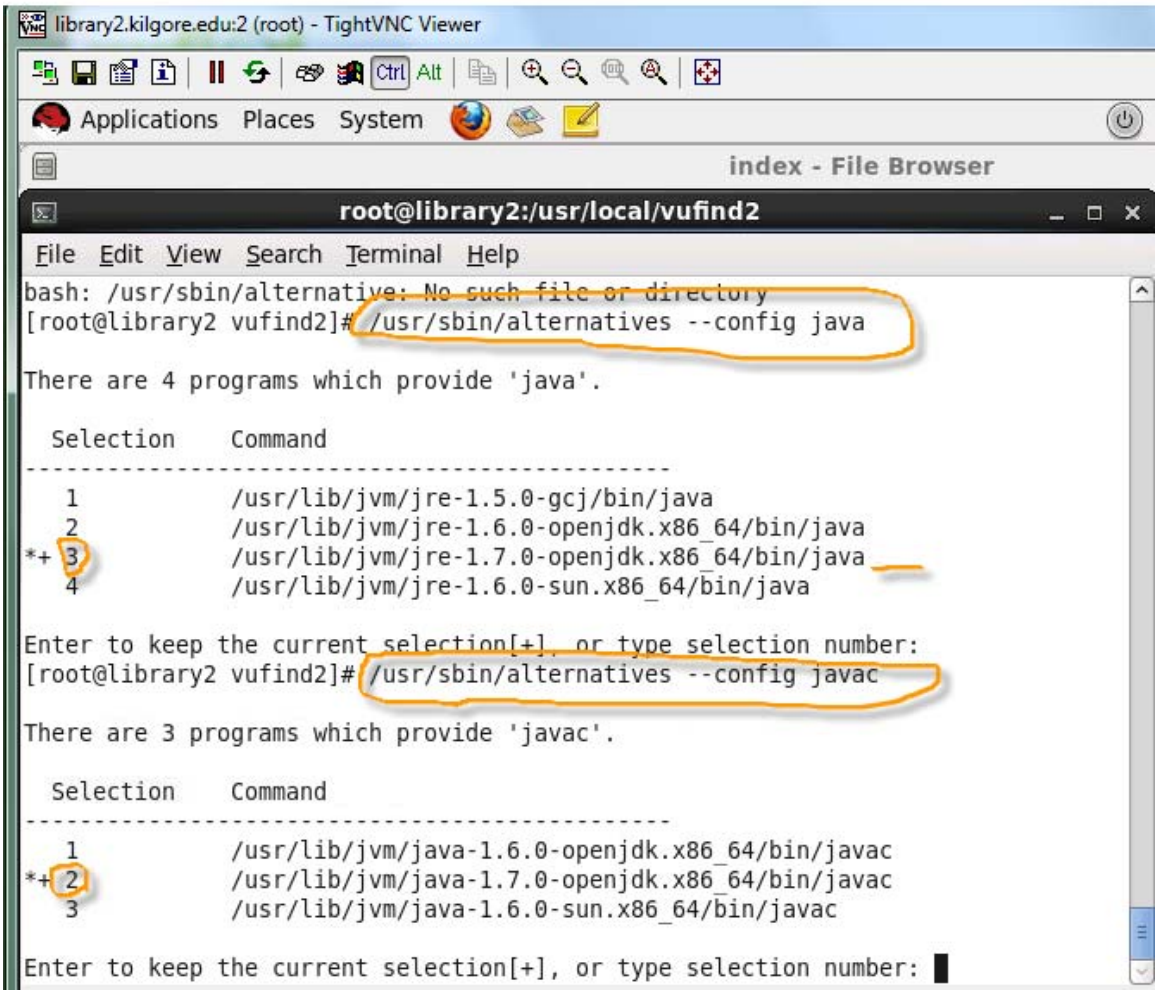
Auto Configure is successfuly disabled.

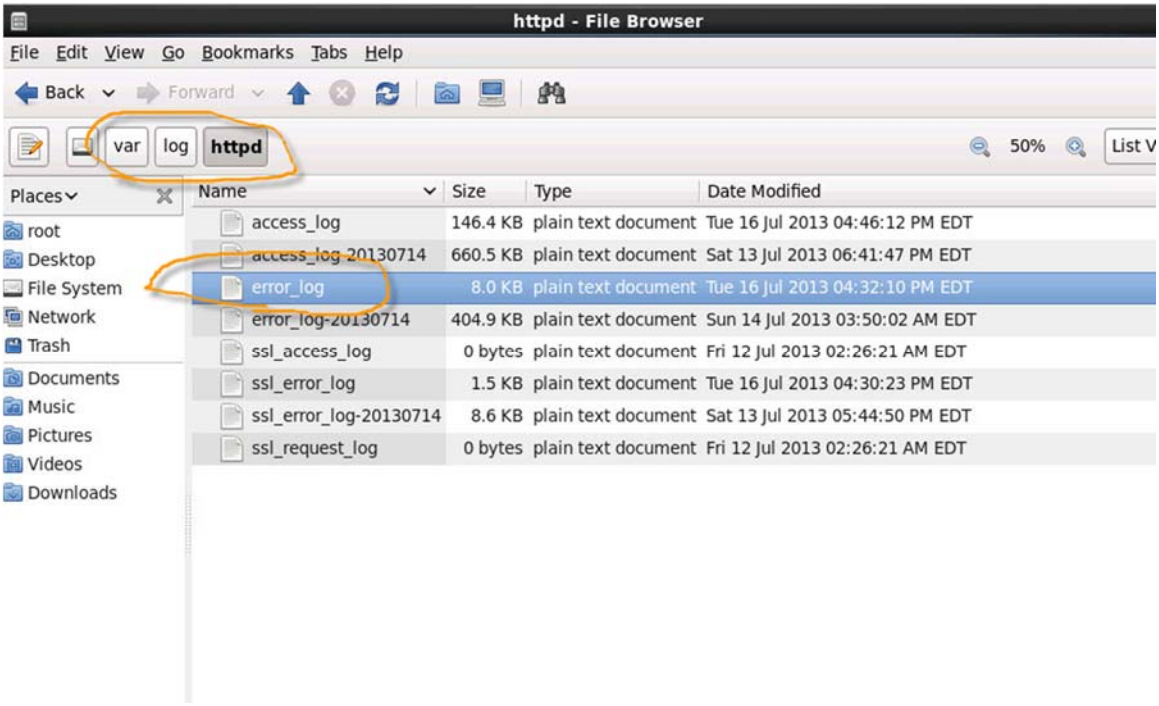


Pseudo-static front page will be created.

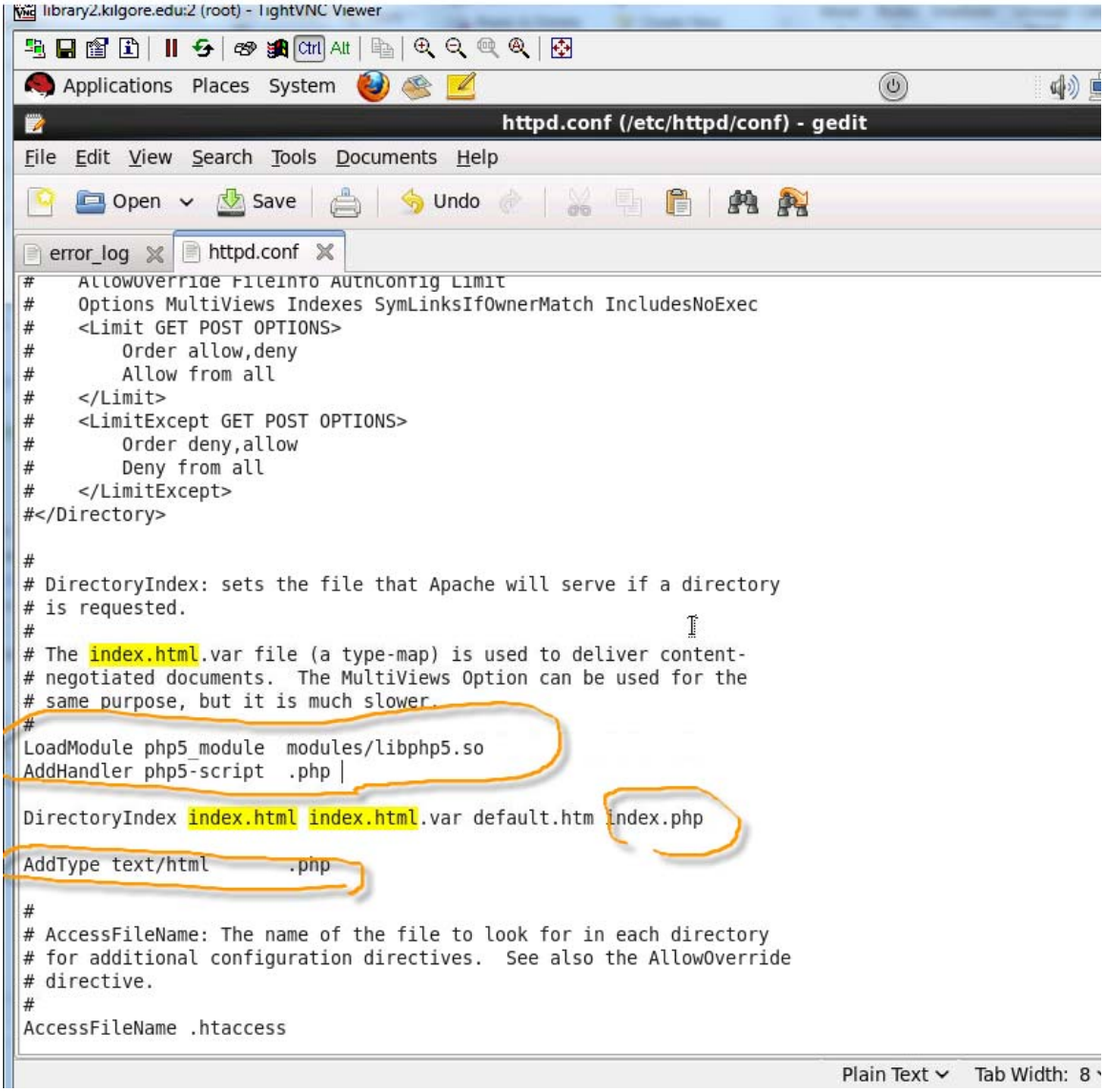


Currently openjdk is the default java. I will change it.

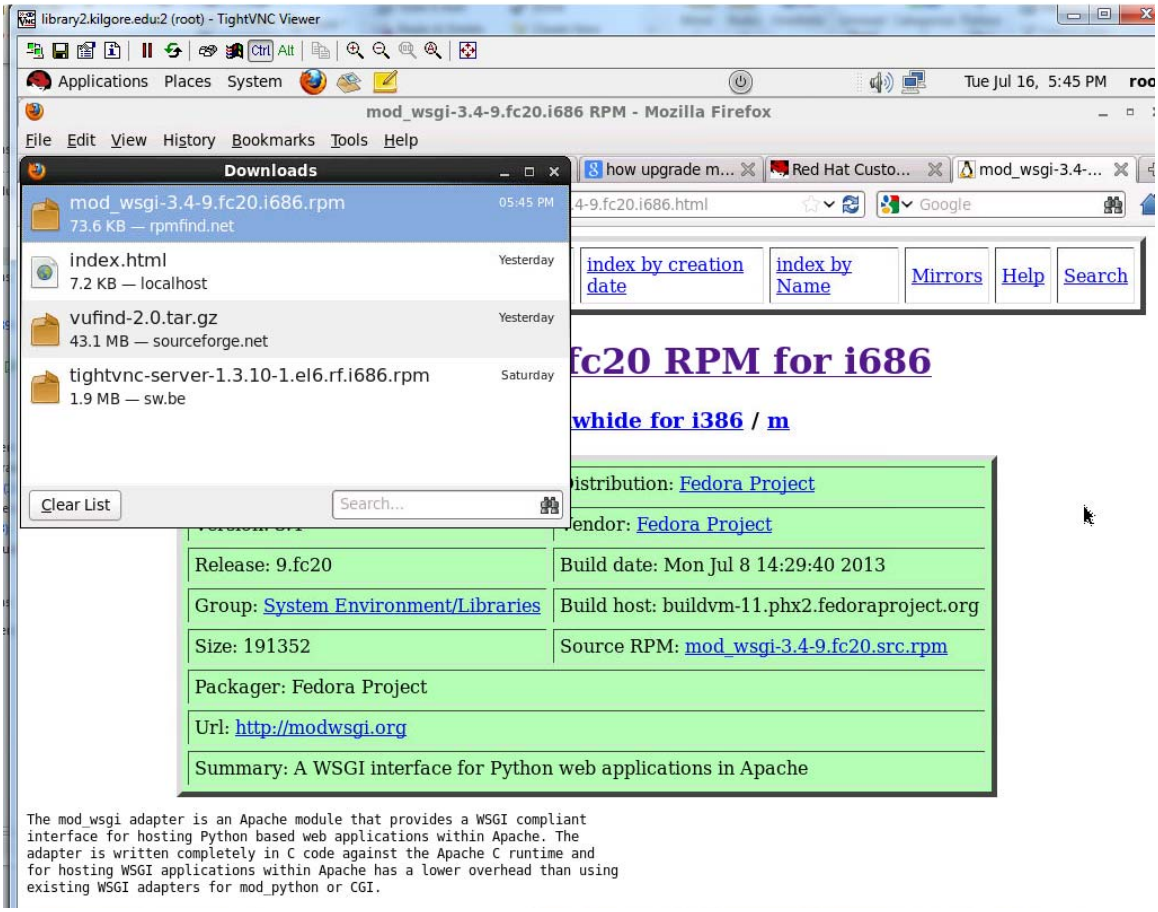



httpd - File Browser
File Edit View Go Bookmarks Tabs Help
Back Forward
var log httpd
50%
Places
root
Desktop
File System
Network
Trash
Documents
Music
Pictures
Videos
Downloads
Name Size Type Date Modified
access_log 146.4 KB plain text document Tue 16 Jul 2013 04:46:12 PM EDT
access_log-20130714 660.5 KB plain text document Sat 13 Jul 2013 06:41:47 PM EDT
error_log 8.0 KB plain text document Tue 16 Jul 2013 04:32:10 PM EDT
error_log-20130714 404.9 KB plain text document Sun 14 Jul 2013 03:50:02 AM EDT
ssl_access_log 0 bytes plain text document Fri 12 Jul 2013 02:26:21 AM EDT
ssl_error_log 1.5 KB plain text document Tue 16 Jul 2013 04:30:23 PM EDT
ssl_error_log-20130714 8.6 KB plain text document Sat 13 Jul 2013 05:44:50 PM EDT
ssl_request_log 0 bytes plain text document Fri 12 Jul 2013 02:26:21 AM EDT

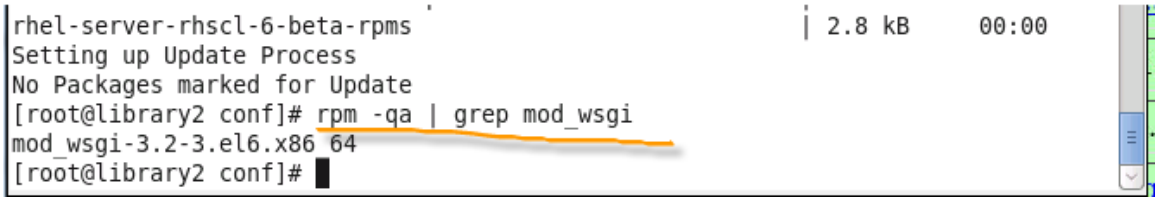
Configure Apache to make it better service VuFind



Upgrade mod_wsgi from 3.2 to 3.49



Before installing new version mod_swgi, old version mod_wsgi will be uninstalled.



Find old version glibc, uninstall it. Then, install a new version glibc.

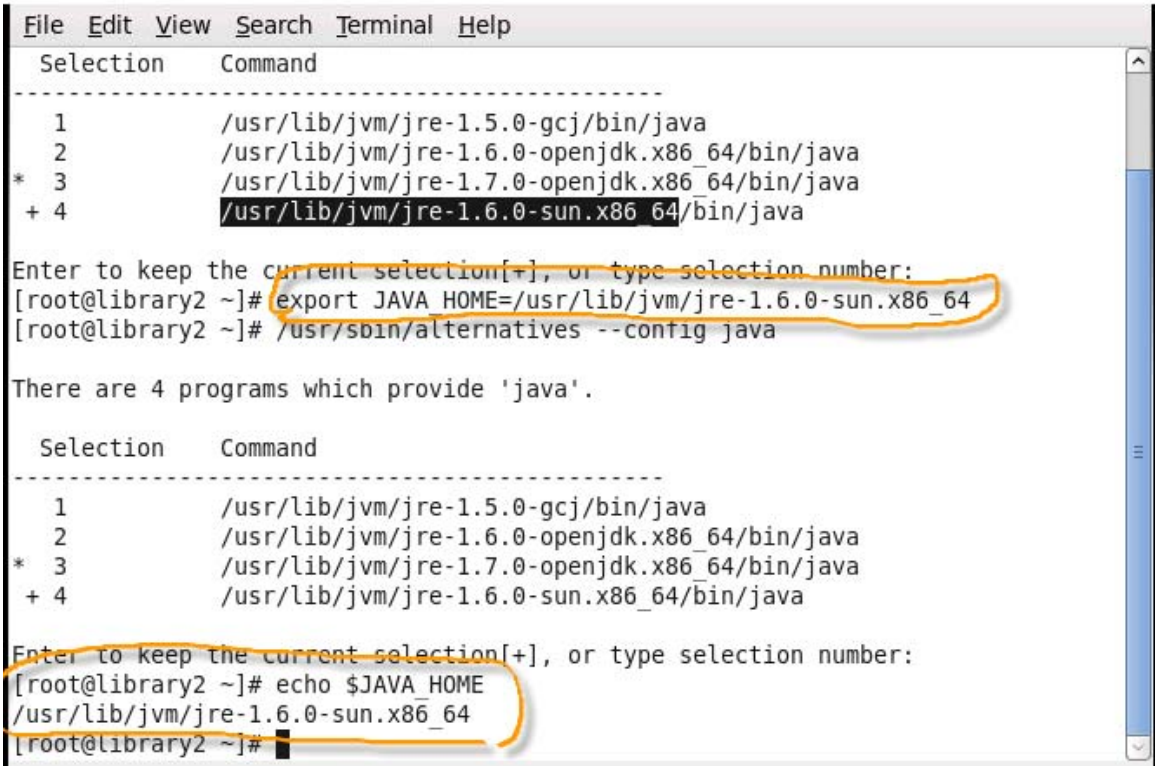
```
[root@library2 conf]# service httpd restart
Stopping httpd:                                            [  O
Starting httpd: [Tue Jul 16 17:51:40 2013] [warn] module php5_m
loaded, skipping
                                                           [  O
[root@library2 conf]# rpm -qa | grep glibc
glibc-2.12-1.107.el6_4.2.x86_64
glibc-devel-2.12-1.107.el6_4.2.x86_64
glibc-common-2.12-1.107.el6_4.2.x86_64
glibc-utils-2.12-1.107.el6_4.2.x86_64
glibc-headers-2.12-1.107.el6_4.2.x86_64
[root@library2 conf]#
```

Test

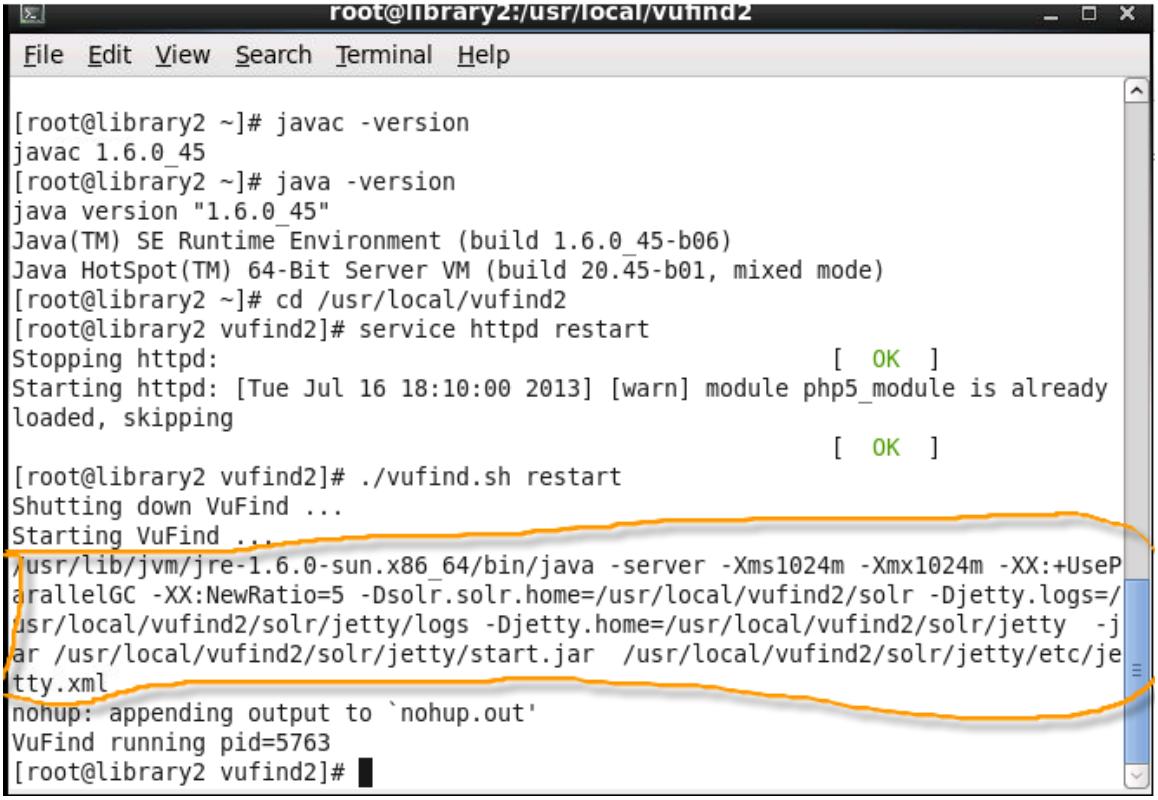
```
  -J<flag>                  Pass <flag> directly to the runtime system

[root@library2 ~]# javac -version
javac 1.6.0_45
[root@library2 ~]# java -version
java version "1.6.0_45"
Java(TM) SE Runtime Environment (build 1.6.0_45-b06)
Java HotSpot(TM) 64-Bit Server VM (build 20.45-b01, mixed mode)
[root@library2 ~]#
```

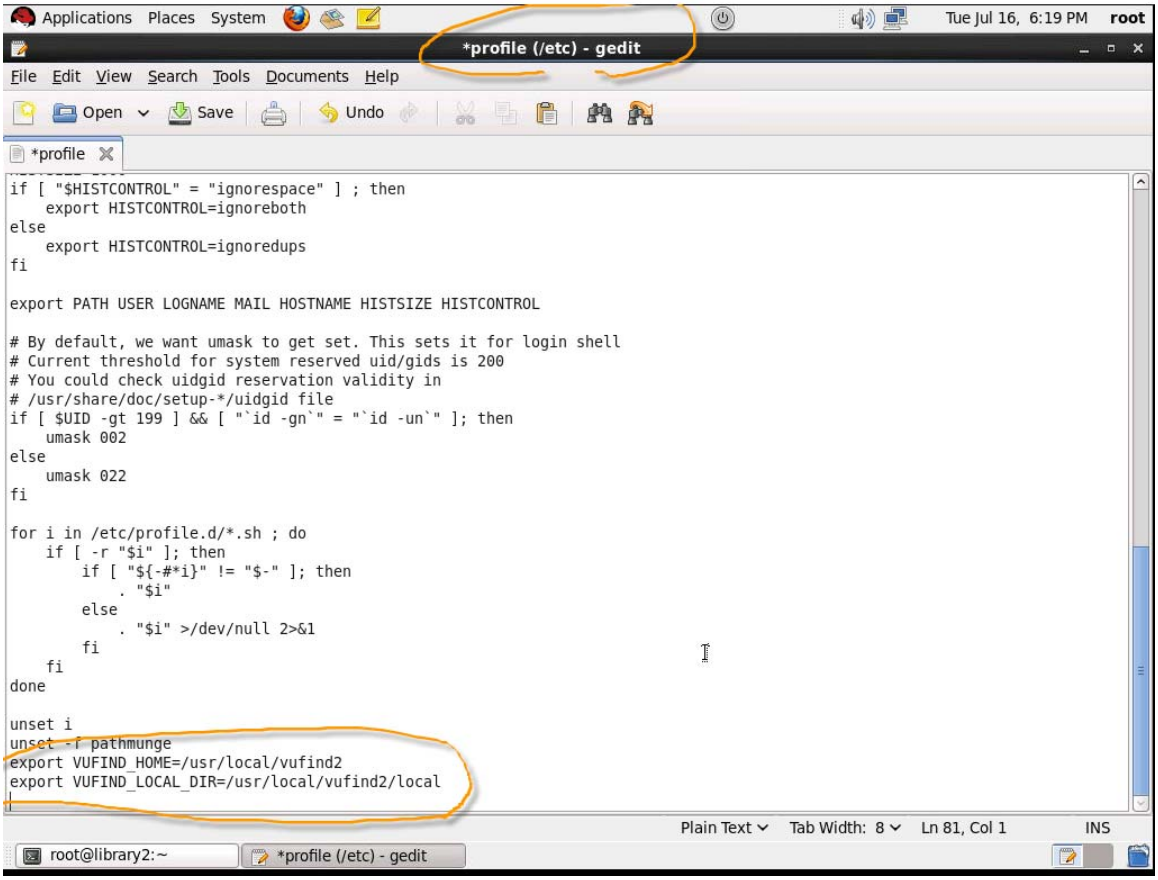
Use Java from Sun as the default java.



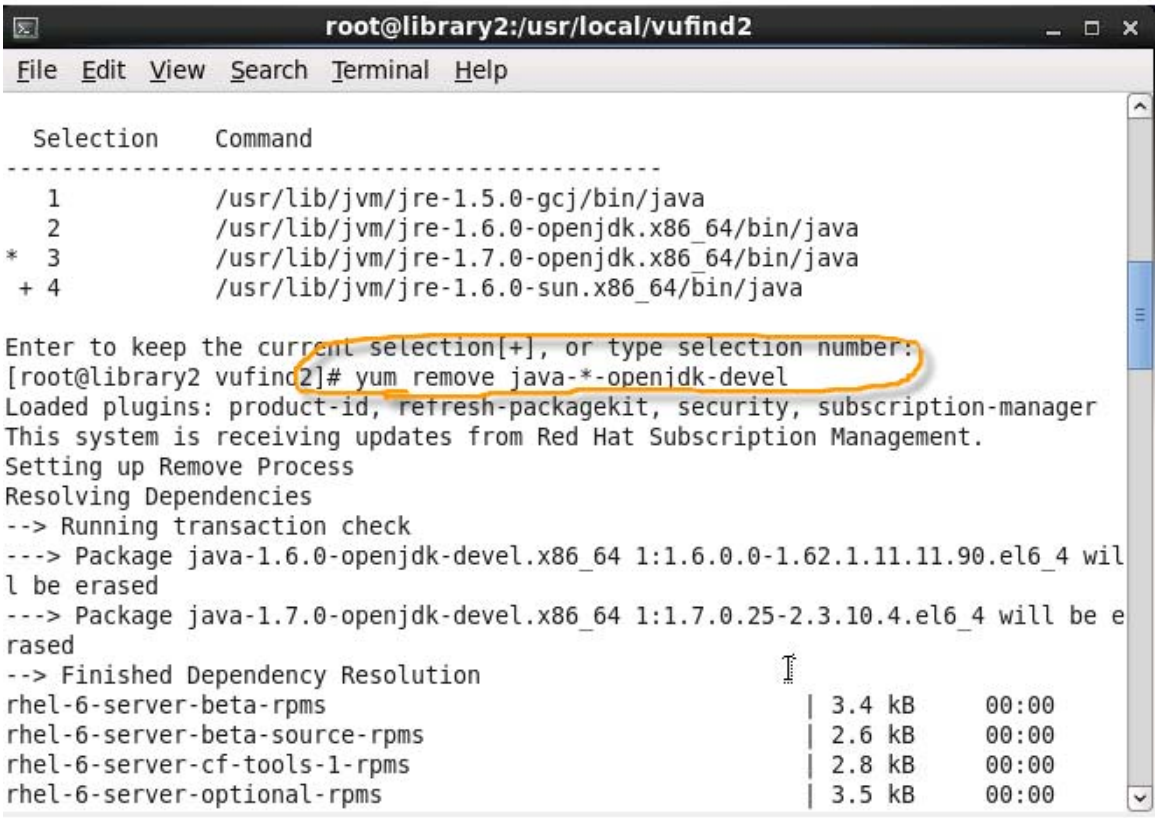
Run VuFind and find java is to setup sun version one.



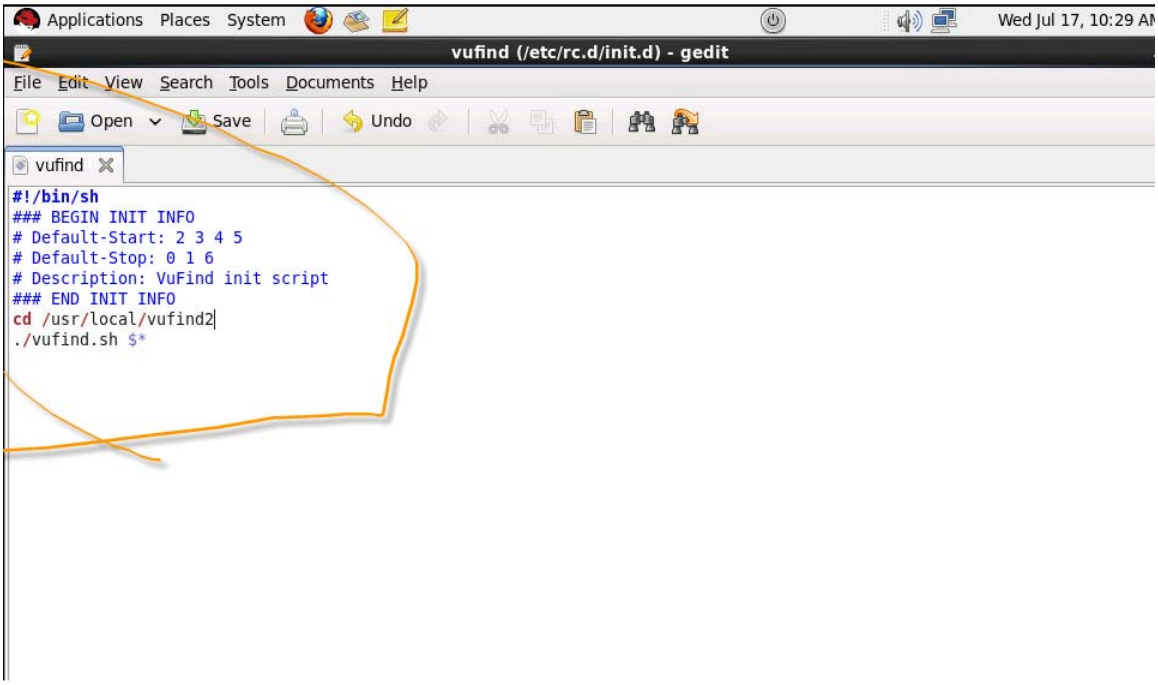
Edit vufind profile and add correct VuFind_Home and VuFind_local_dir



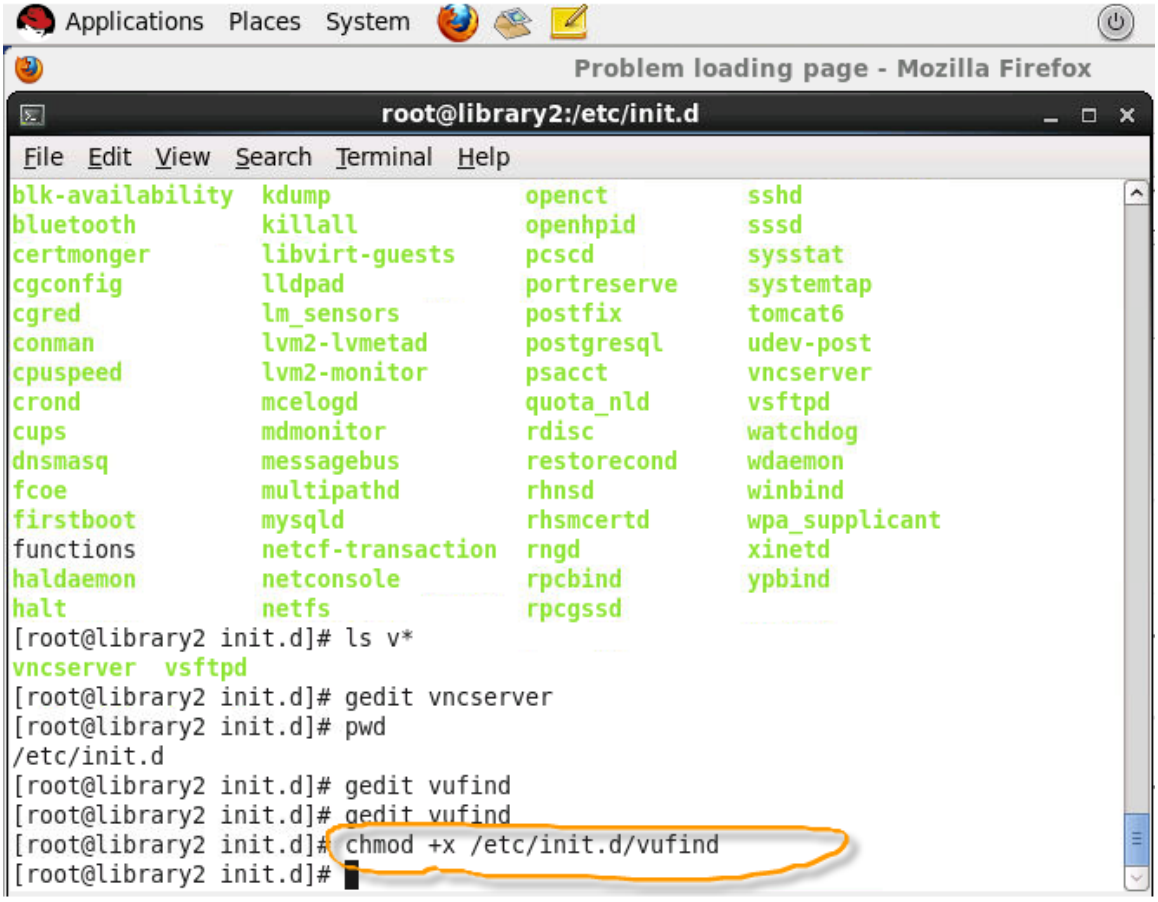
Remove the openjdk from system



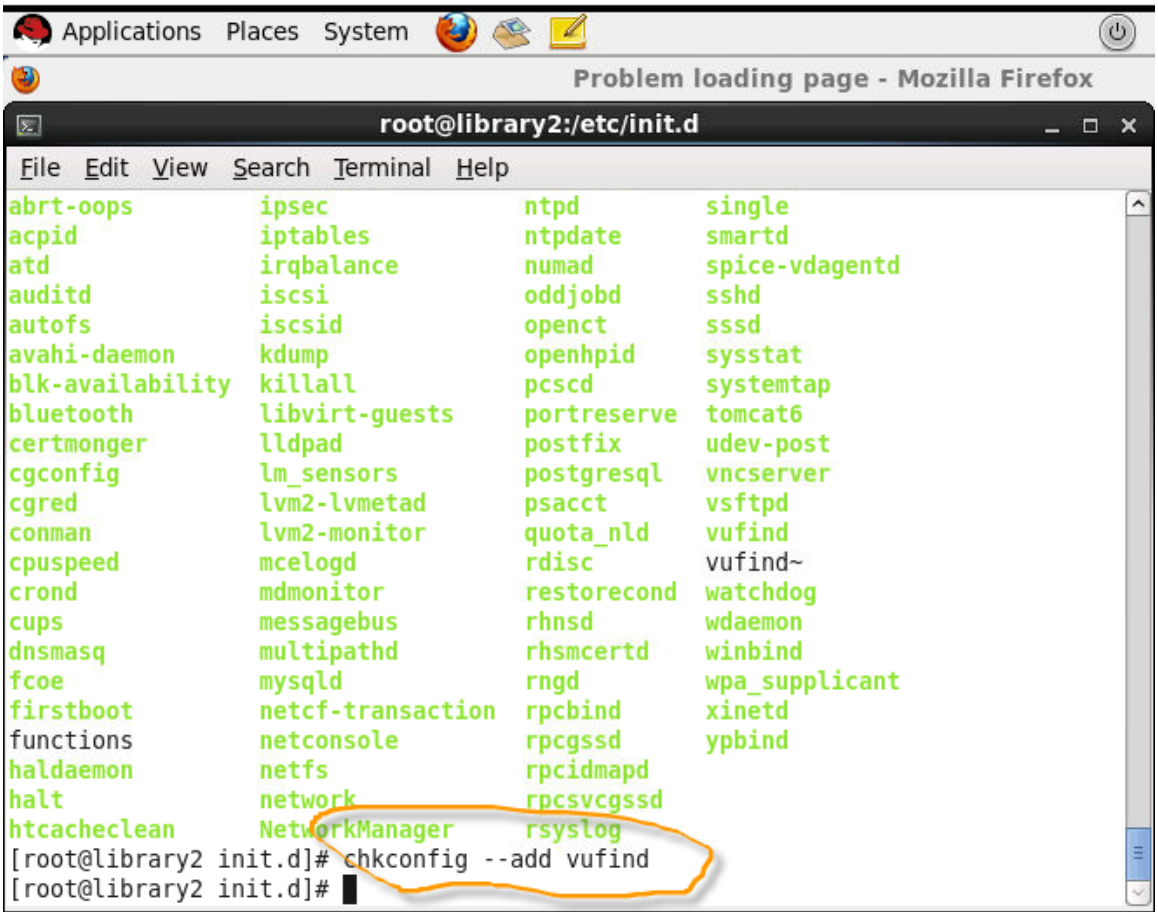
Make VuFind automatically start



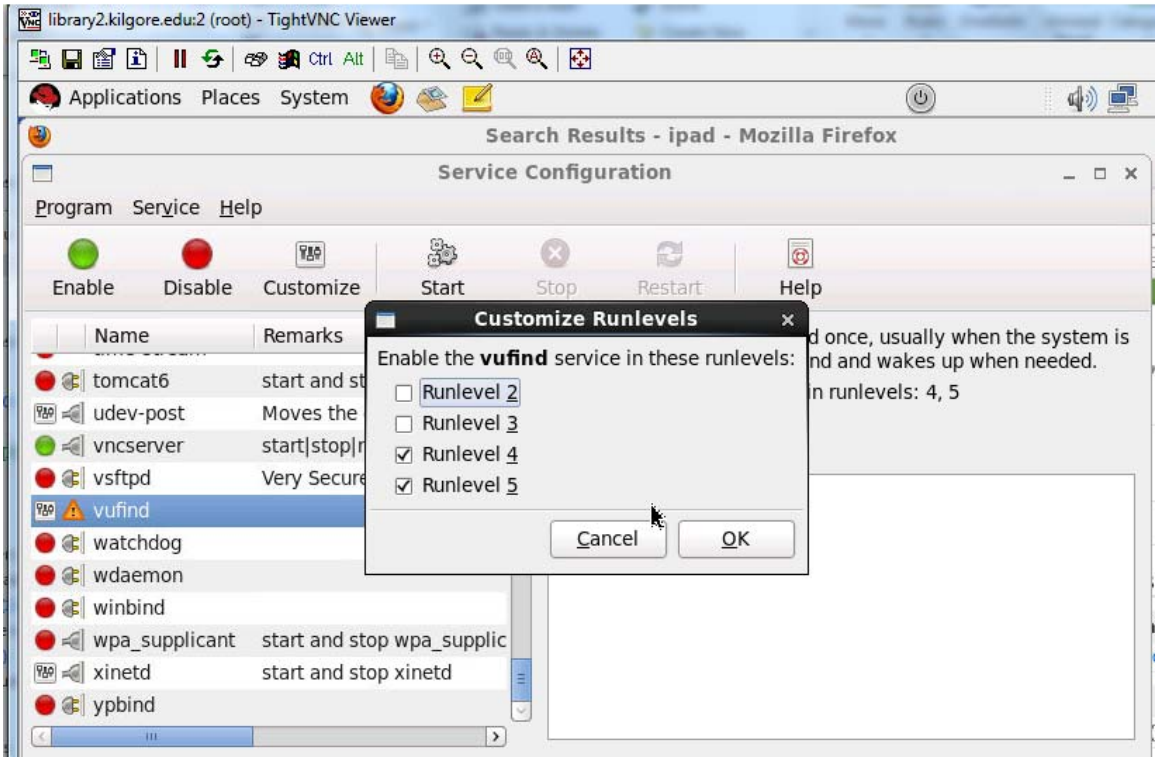
Make VuFind script executable.



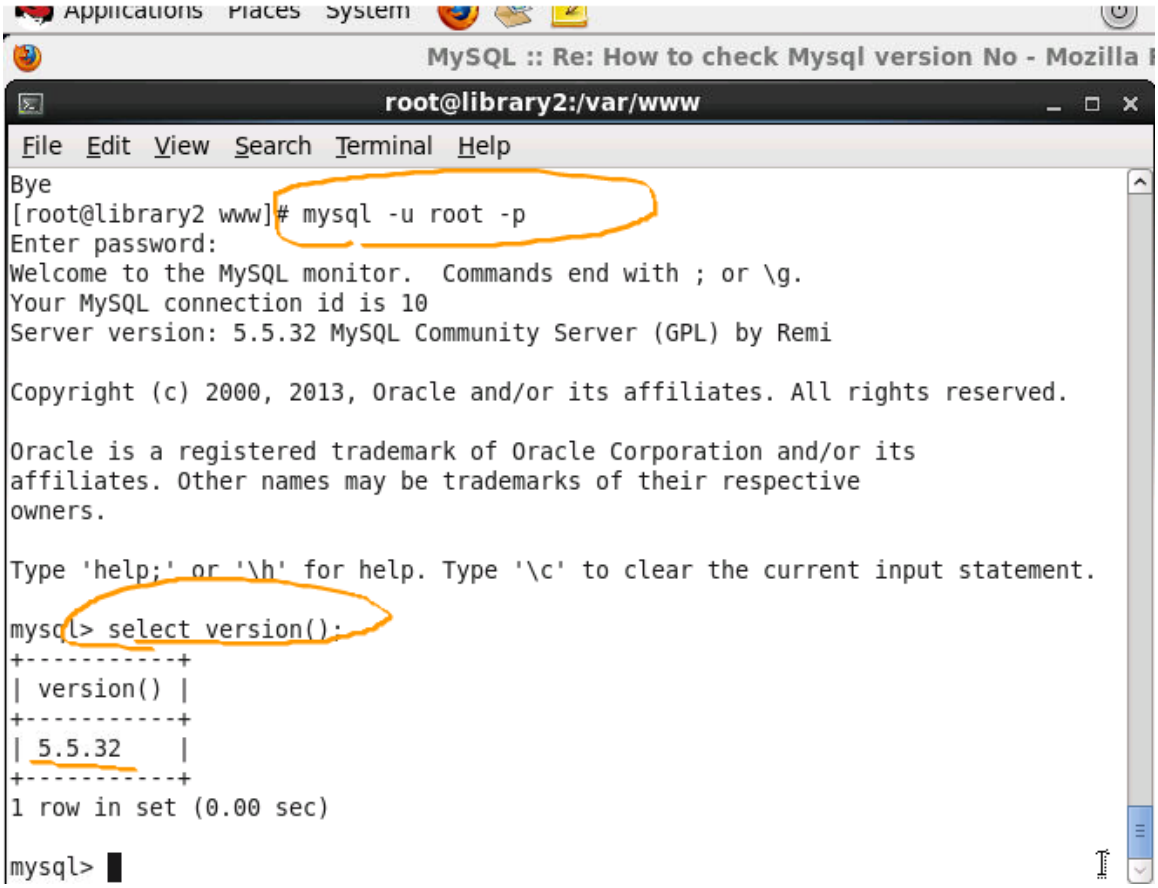
Use chkconfig command to add vufind script as automatc service



Customize VuFind runlevels as 4 and 5



MySQL is upgraded to 5.5.32 version



The end